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APUL

NEET FLIX

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The Living World

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1. What is the practical significance of taxonomy?
(1) Study of living being
(2) Identification of unknown organisms
(3) Study of interrelationship in organisms
(4) Grouping of organisms
BLW126

2. Binomial nomenclature is :
(1) Not applicable for cyanobacteria
(2) Not accepted by some countries
(3) Universal with one specific epithet for each species
(4) Universal with two specific epithet for each species
BLW127

3. Why scientific nomenclature are given in Latin or Greek languages ?
(1) Latin and Greek languages are popular languages throughout the world
(2) Most of the scientific journals are published in Latin or Greek
(3) These languages are dead languages
(4) Linnaeus who proposed scientific nomenclature was a Greek scientist
BLW128

4. Plant classification proposed by Carolus Linnaeus was artificial because it was based on:-
(1) Only a few morphological characters
(2) Economic importance of plants
(3) Only few anatomical character
(4) Only vegetative characters
BLW129

5. Read carefully and find correct statement :-
(1) All living organisms except prokaryotes show metabolism
(2) Metabolism is present in viruses
(3) Cellularity is an exceptionless property of all livings
(4) Metabolic reactions can not be demonstrated in test-tube
BLW130

6. Which of the following character forms the basis of modern taxonomic studies?
(1) Structure of cell
(2) Developmental process of organisms
(3) Ecological information of organisms
(4) All of the above
BLW131

7. Which of the following are mutually exclusive events in majority of higher plants and animals?
(1) Growth and Metabolism
(2) Growth and Reproduction
(3) Reproduction and Metabolism
(4) Consciousness and Cellular organisation
BLW132

8. Increase in mass and increase in number of individuals are twin characters of :-
(1) Growth (2) Reproduction
(3) Metabolism (4) Consciousness
BLW133

9. The brain dead coma patient who is supported by machines which replace heart and lungs has:-
(1) Only self consciousness
(2) Only consciousness
(3) Both consciousness and self consciousness
(4) Neither consciousness nor self consciousness
BLW134

10. Which classification is also known as cladistic classification?
(1) Non phylogenetic classification
(2) Phylogenetic classification
(3) Artificial classification
(4) Practical classification
BLW135

11. Which of the following plant group is most advanced according to phylogeny?
(1) Thallophyta (2) Bryophyta
(3) Angiosperm (4) Gymnosperm
BLW136

12. Which of the following is a fundamental unit of taxonomy ?
(1) Order (2) Genus
(3) Species (4) Sub-species
BLW137

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13. Which one of the following includes all the others ?
(1) Order (2) Class
(3) Family (4) Species
BLW138

14. Which of the following is not true for a species?
(1) Variations occur among members of a species
(2) Gene flow does not occur between the populations of a species
(3) Each species is reproductively isolated from the other species
(4) Members of a species can interbreed.
BLW139

15. Consider the following statements (A-D) :-
(A) Biological names are Latinised or derived from Latin irrespective of their origin.
(B) Animals, mammals and dogs are different taxa at the different level
(C) By using artificial system of classification, we can get evolutionary relations in organisms.
(D) Bacteria cannot be considered as biological species according to the biological concept of species
Which of the above statements(s) are correct?
(1) A, B & D (2) B, C & D
(3) B & C only (4) A & B only
BLW140

16. Animals, dogs and mammals represent
(1) Taxa at different levels
(2) Taxonomic hierarchy
(3) Taxa at equivalent level
(4) Categories at equivalent level
BLW141

17. Living organisms exhibit distinctive characteristics to differentiate from non living objects. Which one of the following statements can be considered as defining property of living things ?
(1) Increase in body mass as a criterion for growth
(2) Reproduction is synonymous with growth in unicellular organisms
(3) Ability of living organisms to sense their surroundings
(4) Metabolic reactions that can be demonstrated outside the body in cell-free system only
BLW142

18. First step of taxonomy is-
(1) Classification (2) Nomenclature
(3) Identification (4) Characterisation
BLW143

19. Study of diversity of organisms and evolutionary relationship amongst them on the basis of all possible characters is called as-
(1) Taxonomy (2) Systematics
(3) Metabolism (4) Phylogeny
BLW144

20. Growth can not be taken as a defining property of living organisms because -
(1) Non-living objects also exhibit growth by accumulation of material on the surface
(2) Unicellular organisms also grow by cell division
(3) Growth occurs throughout the life span in animals
(4) A dead organism does not grow
BLW145

21. Which statement is incorrect w.r.t. genus?
(1) Each genus may have one or more than one specific epithets
(2) It comprises a group of related species
(3) It has more characters in common in comparison to species of other genera
(4) Potato and brinjal belong to different genus
BLW146

22. Which of the following is not a rule for binomial nomenclature?
(1) The name of categories higher than the name of genus are not printed in italics.
(2) Binomial name consists of two Latin and one Roman word.
(3) Generic and common names can never be the same.
(4) The name of the discover is given after scientific name in abbreviated form.
BLW147

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23. Which suffix is used for 'class' according to international code of Botanical Nomenclature?
(1) - phyceae
(2) - opsida
(3) - ales
(4) More than one option is correct

BLW148

24. Find odd one out w.r.t. taxonomic categories
(1) Primata
(2) Diptera
(3) Poales
(4) Monocotyledonae

BLW149

25. Select the **incorrect** match

Taxon		Category
(1) Insecta	–	Class
(2) Tiger	–	Species
(3) Cruciferae	–	Family
(4) Orchid	–	Order

BLW150

26. Which of the following are the defining features of living organisms?
(1) Consciousness, reproduction
(2) Metabolism, cellular organisation
(3) Growth, consciousness, metabolism
(4) Cellular organisation, growth, reproduction

BLW151

27. Which of the following criteria form the basis of modern taxonomic studies?
(1) Development process
(2) Ecological information of organisms
(3) External and internal structure
(4) More than one option is correct

BLW152

28. AII given are related to the taxonomic hierarchy of mango, **except**
(1) Poales (2) Anacardiaceae
(3) Dicotyledonae (4) Angiospermae

BLW153

29. Suffixes like -phyceae, -ae and -opsida are used for the category called -
(1) Division (2) Class
(3) Order (4) Kingdom

BLW154

30. Which one of the following pair of taxa constitute or form the same taxonomic category?
(1) *Solanum* and Solanaceae
(2) Convolvulaceae and Solanaceae
(3) Felidae and carnivora
(4) *Triticum* and *Pisum*

BLW155

31. No names are recognised prior to those used by Linnaeus in 1758 in the 10th edition of
(1) Philosophia Botanica
(2) Systema Naturae
(3) Historia Naturalis
(4) Species Plantarum

BLW156

32. Process by which anything is grouped into convenient categories based on some easily observable characters called
(1) Classification (2) Nomenclature
(3) Identification (4) Characterisation

BLW157

33. Match the column I with column II.

Column I	Column II
a. Dicotyledonae	(i) Phylum
b. Arthropoda	(ii) Class
c. Diptera	(iii) Family
d. Anacardiaceae	(iv) Order

(1) a-(ii), b-(i), c-(iii), d-(iv)
(2) a-(ii), b-(i), c-(iv), d-(iii)
(3) a-(iii), b-(i), c-(iv), d-(ii)
(4) a-(i), b-(ii), c-(iii), d-(iv)

BLW158

34. Biological concept of species given by Ernst Mayr cannot be applied to -
(1) Organisms that reproduce by asexual methods only
(2) Sexually reproducing forms breeding the whole year
(3) Sexually reproducing forms breeding in a particular season
(4) Sexually reproducing organisms having external fertilization

BLW159

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35. Growth and reproduction are mutually exclusive events in
(1) Unicellular organisms only
(2) Higher animals and lower plants
(3) Majority of higher animals and plants
(4) Some bacteria and lower animals

BLW160

36. Read the following statements:
(A) All organisms are aware of their surroundings.
(B) Isolated metabolic reactions in-vitro are not living reactions.
(1) Both (A) & (B) are correct
(2) Only (B) is correct
(3) Both (A) & (B) are incorrect
(4) Only (A) is correct

BLW161

37. Which of the following are twin characteristics of growth?
(1) Increase in density and decrease in mass
(2) Increase in mass and decrease in volume
(3) Increase in volume and decrease in density
(4) Increase in mass and increase in number

BLW162

38. Taxonomy in which number and codes are assigned to all the characters and the data are processed using computers is
(1) Phenetics
(2) Biosystematics
(3) Alpha taxonomy
(4) Classical taxonomy

BLW163

39. A taxon as a group of related species with more common characters is represented by which one of the following group of organisms?
(1) Radish, cabbage and cauliflower
(2) Tomato, potato, and chilli
(3) Lion, tiger and leopard
(4) Cat, dog and mice

BLW164

40. In which one of the following options the two taxonomic categories given are correctly matched with their organism ?

Taxonomic categories organism

(1) Diptera and Felidae – *Musca*
(2) Solanaceae and polymoniales – *Datura*
(3) Canidae and Primata – Gibbon
(4) Triticum and Sapindales – Wheat

BLW165

41. External and internal structure along with the structure of cell, development process and ecological information's of organism are essential and form the basis of -
(1) Old systematic studies
(2) Classical systematic studies
(3) Old taxonomic studies
(4) Modern taxonomic studies

BLW166

42. Select odd one w.r.t. family -
(1) Muscidae (2) Felidae
(3) Anacardiaceae (4) Diptera

BLW167

43. Find odd one w.r.t. defining features of life forms.
(1) Reproduction (2) Consciousness
(3) Metabolism (4) Intrinsic growth

BLW168

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	3	3	1	3	4	2	1	2	2	3	3	2	2	1
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	1	3	4	2	1	4	3	4	4	4	2	4	1	2	2
Question	31	32	33	34	35	36	37	38	39	40	41	42	43		
Answer	2	1	2	1	3	4	4	1	3	2	4	4	1		

Biological Classification

1. The evolution of organisms in the five kingdom system is most accurately described by the following sequence :-
(1) Monera to fungi, fungi to protista, protista to plants and plants to Animals.
(2) Monera to protists, from protists to fungi, plants and animals, separately
(3) Protists to Monera, Monera to fungi, from fungi to plants and animals, separately
(4) Monera to protista and animals from protista, plants with the fungi by losing ability of photosynthesis in plants.

BBC466

2. Consider the following statements :-
I. All prokaryotic organisms were grouped together under kingdom-Monera.
II. The unicellular eukaryotic organisms were placed in kingdom-Protista.
III. *Chlorella* and *Chlamydomonas*, both have cell wall.
IV. *Paramoecium* and *Amoeba* lack cell walls.
V. Kingdom-Protista has brought together *Chlamydomonas, Chlorella* with *Paramoecium* and *Amoeba*.
Which of the statements given above are **correct**?
(1) Only I, II, III and V (2) Only II, III, IV and V
(3) Only I, II, III and IV (4) I, II, III, IV and V

BBC467

3. How many kingdoms have decomposers in Whittaker's classification ?
(1) 3 (2) 4 (3) 1 (4) 2

BBC468

4. How many kingdoms have only heterotrophic mode of nutrition in Whittaker's classification?
(1) 4 (2) 3 (3) 2 (4) 5

BBC469

5. Which of the following are not mentioned in five kingdom classification?
(1) Virus, Viroids, Prions and Lichens
(2) Only Viruses
(3) Viroids and Prions only
(4) Virus and Lichens only

BBC470

6. Insects are preserved in insect boxes. They are kept in :-
(1) Botanical gardens
(2) Herbarium
(3) Biological museum
(4) Zoological parks

BBC471

7. If mesosomes of a bacterium are removed then :-
(1) Bacteria will not be able to repair its damaged cell wall
(2) Bacteria will fail to undergo division
(3) Bacteria will fail in aerobic respiration
(4) All the above

BBC472

8. Which of the following bacteria never use oxygen in respiration?
(1) *Azotobacter*
(2) *Clostridium botulinum*
(3) *Acetobacter aceti*
(4) *Clostridium tetani*

BBC473

9. Archaebacteria which degrade the cellulose in rumen (gut) of cattles is a/an :-
(1) Actinomycetes (2) Eubacterium
(3) Methanogen (4) Halophile

BBC474

10. Read the following statements and select the correct option :-
(a) Blue green algae contain single stranded circular DNA.
(b) Gas vacuoles are present in all prokaryotes.
(c) Only 70S type of ribosomes are present in Eubacteria.
(d) Plasmid is the main DNA in bacterial cell.
(e) Cell wall is absent in Mycoplasma.
Options :-
(1) 4-correct, 1-incorrect
(2) 1-correct, 4-incorrect
(3) 3-correct, 2-incorrect
(4) 2-correct, 3-incorrect

BBC475

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11. Volutin granules in bacteria are chemically :
(1) Butyric acid granules
(2) Fatty acid granules
(3) Protein granules
(4) Phosphate granules

BBC476

12. The process by which a bacterium acquires new genes by taking up parts of a naked DNA molecule from its surroundings is called :
(1) Transformation (2) Conjugation
(3) Transduction (4) Binary fission

BBC477

13. *Nitrobacter* converts :-
(1) Ammonia into nitrite
(2) Methane into CO_2
(3) Nitrite into nitrates
(4) Nitrates into nitrite

BBC478

14. The blue green algae which has large amount of protein and is edible in nature :-
(1) *Chlorella* (2) *Spirulina*
(3) *Anabena* (4) *Oscillatoria*

BBC479

15. The bacteria that convert nitrates into nitrogen and reduce the soil fertility is :-
(1) *Nitrobacter*
(2) *Thiobacillus denitrificans*
(3) *Agrobacterium tumefaciens*
(4) *Azotobacter*

BBC480

16. Which of the following is not the activity of blue green algae ?
(1) It gives fertility to sterile alkaline soil.
(2) Responsible for water bloom.
(3) It is never used as green manure.
(4) Secrete toxin which inhibits growth of mosquito larva in water.

BBC481

17. Decomposer protists are :-
(1) Slime mould (2) Diatoms
(3) Dinoflagellates (4) Euglenoids

BBC482

18. In oogamy, fertilization involves :
(1) A large motile female gamete and a small non-motile male gamete
(2) A small non-motile female gamete and a large motile male gamete
(3) A large non-motile female gamete and a small motile male gamete
(4) A large motile female gamete and a large non-motile male gamete.

BBC483

19. Which one of the following is not a fungal disease ?
(1) White Rust of crucifers
(2) Loose smut of wheat
(3) Citrus canker
(4) Red rot of sugarcane

BBC484

20. The phycomycetes :
(1) Have hyphae without regularly occuring cross walls
(2) Have cellulosic cell wall
(3) Form fleshy fruiting bodies
(4) Are haploid throughout their life cycle.

BBC485

21. In club fungi, which stage is usually prolonged:
(1) Stage between dikaryotisation and karyogamy
(2) Stage between karyogamy and meiosis
(3) Stage between meiosis and formation of spores
(4) Stage between formation of basidia and basidiocarp

BBC486

22. Diatoms have left behind large amount of cell wall deposits in their habitat. This accumulation is referred to as ?
(1) Coral reef
(2) Paramylum
(3) Diatomite or Keiselgurh
(4) Plasmodium

BBC487

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23. Which of following option is true for below diagram?



(1) It is free living, found in fresh and stagnant water.
(2) Example of unicellular organism.
(3) Presence of two flagella, one eye spot and contractile vacuole.
(4) All the above

BBC488

24. Phycomycetes produces :-
(1) Aplanospores, Zoospores, Zygospores
(2) Oospores, Zygospores, Ascospores
(3) Ascospores, Basidiospores, Oospores
(4) Aplanospores, Zoospores, Basidiospores

BBC489

25. In which of the following, the cell wall has stiff cellulose plate on the outer surface ?
(1) Dinoflagellates (2) Desmids
(3) Diatoms (4) Euglenoids

BBC490

26. Which of the following statements is correct.
I. *Mycoplasma* has no cell wall.
II. *Mycoplasma* is the smallest living organism
III. *Mycoplasma* cannot survive without oxygen.
IV. *Mycoplasma* are pathogenic in animals and plants.
V. True sexuality is not found in bacteria.
VI. A short of sexual reproduction by adopting a primitive DNA transfer from one bacterium to the other occurs.
(1) All (2) Only III
(3) I, II, IV, V, VI (4) I, III, VI

BBC491

27. Based upon the modes of nutrition protists are grouped into :-
(1) Plant-like protists (algae) and ingestive animal-like protists (protozoa) and absorptive, fungus like protists.
(2) Chrysophytes, Dinoflagellates and euglenoids only.
(3) Slime moulds and fungi only.
(4) Flagellated protozoans and protozoans only

BBC492

28. Dinoflagellates have :-
(1) A single flagellum in the transverse groove between the cell plates.
(2) A single flagellum in the longitudinal groove between the cell plates.
(3) Two flagella one lies longitudinally and the other transversely in a furrow between the wall plates.
(4) No flagella.

BBC493

29. I. Instead of a cell wall, they have a portion rich in pellicle making their body flexible.
II. They have 2 flagella, a short and a long one.
III. They have mixotrophic nutrition.
IV. In light, they are photosynthetic, but act as heterotroph when they are in dark.
V. They are connecting link between plants and animals.
The above statements are assigned to :-
(1) Dinoflagellates
(2) Slime mould
(3) Desmids and diatoms
(4) *Euglena*

BBC494

30. Select the following statement that does not apply to diatoms: -
(1) Diatom cell wall may be impregnated with silica.
(2) Cell wall is made up of 2 half-shells fit tightly together.
(3) Diatom is a chrysophyte.
(4) Diatom is multi flagellated.

BBC495

31. Diatomaceous earth is used for all except :-
(1) Polishing.
(2) Filtration of oil and syrups.
(3) Sound and fire proof room.
(4) Biogas.

BBC496

32. Dinoflagellates are mostly :-
(1) Marine (2) Fresh water
(3) Terrestrial (4) Saprophytes

BBC497

33. Photosynthetic protists are :-
(1) Euglenoids, Diatoms and Dinoflagellates.
(2) Euglenoids and slime moulds.
(3) Diatoms and zooflagellates.
(4) Desmids and ciliates.

BBC498

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34. Red tides in warm coastal water develop due to super abundance of :-
(1) Dinoflagellates
(2) Euglenoid forms
(3) Diatoms
(4) Desmids

BBC499

35. Which of the following statements about *Euglena* is true?
(1) Euglenoids are flagellates.
(2) *Euglena* placed in continuous darkness lose their photosynthetic activity and die.
(3) The pigments of *Euglena* are quite different from those of green plants.
(4) *Euglena* is a marine protists.

BBC500

36. Slime moulds :-
(1) Are parasites.
(2) Do not produce spores.
(3) Do not produce fruiting bodies.
(4) Saprophytic protists.

BBC501

37. Which of the following statement is true about *Euglena* ?
(1) They show flagellar locomotion.
(2) They have a rigid cell wall.
(3) They have no chloroplast.
(4) They are obligate autotroph.

BBC502

38. Protozoans are not included in animal kingdom because :-
(1) Mostly asymmetrical.
(2) Unicellular eukaryotes.
(3) Heterotrophic nature.
(4) Multicellular prokaryotes.

BBC503

39. Which one is correct about *Trypanosoma*?
(1) They are flagellated protozoan.
(2) They are parasites.
(3) They cause sleeping sickness.
(4) All

BBC504

40. *Paramecium* :-
(1) It is a ciliated protozoan.
(2) Shows water current maintained by cilia which helps the food to be steered into gullet.
(3) Has a cavity (gullet) that open to the outside of the cell surface.
(4) All

BBC505

41. Select the correct statements below that correctly apply to the kingdom fungi :-
(1) Some fungi form beneficial inter-relationships with plants.
(2) Certain fungi are natural source of antibiotics.
(3) The fungal life cycle typically includes a spore stage.
(4) All

BBC506

42. When a moist bread is kept exposed in air, it becomes mouldy and black because :-
(1) Spores are present.
(2) Spores are present in the bread.
(3) Spores are in the air.
(4) The bread gets decomposed.

BBC507

43. Members of phycomycetes are found :-
I. In aquatic habitat.
II. On decaying wood.
III. On moist and damp places.
IV. As obligate parasite on plants.
(1) None of the above (2) I and IV
(3) II and III (4) All of the above

BBC508

44. Which of the following is false about Ascomycetes?
(1) Mode of nutrition saprophytic, decomposer, coprophilous and parasitic.
(2) Includes unicellular (eg. Yeast) and multicellular forms.
(3) Mycelium is coenocytic.
(4) *Aspergillus Claviceps*, *Neurospora* are important members of ascomycetes.

BBC509

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45. I. It includes unicellular as well as multicellular fungi.
II. In multicellular forms hyphae are branched and septate.
III. Conidiophore producers conidia (spores) exogenously in chain.
IV. Sexual spore are ascospores produced endogenously in ascus.
Which of the following characters are shown by :-
(1) Phycomycetes (2) Sac fungi
(3) Club fungi (4) Fungi imperfecti
BBC510

46. Basidiomycetes include :-
(1) Mushrooms, Toadstools, puffballs, and bracket fungi.
(2) Smut fungi and rust fungi.
(3) Both (1) and (2).
(4) Bread mould, sac fungi and algal fungi.
BBC511

47. Where does meiosis occur in mushroom?
(1) Basidiospore (2) Basidium
(3) Basidiocarp (4) Ascus mother cell
BBC512

48. Plasmogamy is the fusion of :-
(1) 2 haploid cells including their nuclei.
(2) 2 haploid cells with out nuclear fusion.
(3) Sperm and egg.
(4) Sperm with 2 polar nuclei.
BBC513

49. Somatogamy is the fusion of :-
(1) Two vegetative/ somatic cell, of different strain/ genotypes to form diakaryotic cell.
(2) Sperm with egg.
(3) 2 somatic cell having identical strain.
(4) Egg with another egg.
BBC514

50. Which of the following is false about Deuteromycetes?
(1) They reproduce only by asexual spores (conidia).
(2) Mycelium is branched and septate.
(3) They have only parasitic forms.
(4) They have no sexual stage (perfect stage)
BBC515

51. In the five kingdom classification of whittaker, there is :-
(1) No mention of viruses.
(2) Mention of lichens.
(3) No mention of viroids.
(4) No mention of viruses, viroids and lichens.
BBC516

52. Viruses did not find a place in classification since :-
(1) They are not truly living.
(2) They are obligate parasite.
(3) They are cellular.
(4) They are hyperparasite.
BBC517

53. The viruses are non-cellular organisms that are characterised by :-
(1) Having no nucleic acid.
(2) Having an active structure outside the living cell.
(3) Having an inert crystalline structure outside the cell.
(4) Their hyperactivity outside the cell.
BBC518

54. The Cyanobacteria of great nutritional value being marketed today is :-
(1) *Salvinia* (2) *Spirogyra*
(3) *Spirulina* (4) *Stigonema*
BBC519

55. What is true about Cyanobacteria?
(1) Oxygenic with nitrogenase.
(2) Oxygenic without nitrogenase.
(3) Non-oxygenic with nitrogenase.
(4) Non-oxygenic without nitrogenase.
BBC520

56. Select correct statement with respect to Eubacteria.
(1) Have very simple structure.
(2) Peptidoglycan nature of cell wall.
(3) Heterotrophs are most abundant in nature
(4) All
BBC521

57. Archaebacteria do not show :-
(1) Peptidoglycan in cell wall.
(2) Introns in DNA
(3) Branched chain lipids in cell membrane.
(4) Ribosomal protein with high acidic nature.
BBC522

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58. Choose the correct statement -
(1) Members of phycomycetes are facultative parasites on plants.
(2) Fusion of protoplasm between two motile or non-motile gametes is called as plasmogamy.
(3) Kingdom Plantae includes all eukaryotic chlorophyll containing organisms and non-chlorophyll organisms called plants.
(4) Trichoderma belongs to basidiomycetes.
BBC523

59. Which of the following is wrongly matched ?
(1) T.O. Diener – Viroids are found to be a free DNA.
(2) W.M. Stanley – Crystallised proteins.
(3) M.W. Beijerinck – *Contagium vivum fluidum*.
(4) D.J. Ivanowsky – Microbes smaller than a bacteria cause mosaic disease of tobacco.
BBC524

60. Which one of the following cocci appears like grapes under microscope?
(1) Streptococci (2) Diplococci
(3) Staphylococci (4) Pneumococci
BBC525

61. Nutritionally, protists are -
(1) Photoautotrophs
(2) Heterotrophs
(3) Mixotrophs
(4) Photoautotrophs, heterotrophs or autotrophs
BBC526

62. Match the following -

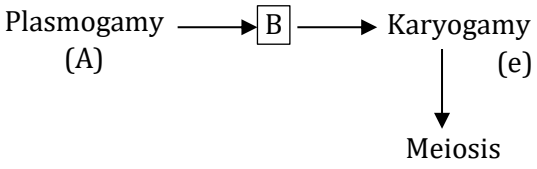
Column-I		Column-II	
A.	*Albugo*	i.	Conidia
B.	*Penicillium*	ii.	Parasitic fungi
C.	Morels	iii.	Edible fungi
D.	Mucor	iv.	Dung mould

(1) A-ii, B-i, C-iii, D-iv (2) A-i, B-iii, C-ii, D-iv
(3) A-iii, B-iv, C-ii, D-I (4) A-iv, B-ii, C-iii, D-i
BBC527

63. 'n + n' condition in fungi like Ascomycetes is known as -
(1) Dikaryophase (2) Eukaryophase
(3) Ascokaryophase (4) Prokaryophase
BBC528

64. The fungal component in lichens is known as-
(1) Phycobiont (2) Mycobiont
(3) Fungobiont (4) Fucobiont
BBC529

65. In fungi life cycle sequence in case of Ascomycetes.



B is :-
(1) Dikaryophase (2) Ascophase
(3) Mitosis (4) Fungophase
BBC530

66. Which fungus is extensively used in biochemical studies?
(1) *Neurospora* (2) *Penicillium*
(3) *Albugo* (4) Yeast
BBC531

67. Which one of the following statements about viruses is true?
(1) Viruses are the most primitive life forms.
(2) All viruses have DNA, single or double-stranded.
(3) Prions are true viruses.
(4) Viruses have a nucleic acid based genome enclosed in a protein coat.
BBC532

68. Which of the following statements which are not relevant to Archaebacteria?
(A) They live in some of the most harsh habitats.
(B) They are present in the gut of several ruminant animals.
(C) They are characterised by the presence of rigid cellulosic cell wall.
(D) They include *Mycoplasma*.
(E) They are also referred to as blue green algae.
(1) A, B, C (2) A, C, E
(3) C, D, E (4) A, C, D
BBC533

69. Which is parasitic?
(1) *Albugo* (2) *Rhizopus*
(3) *Euglena* (4) Slime mould
BBC534

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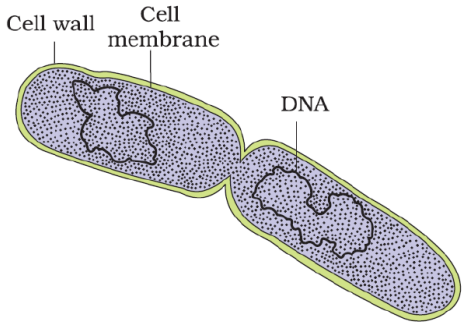
70. In 1971, T.O. diener discovered a new infections agent that was smaller than viruses -
I. It causes potato spindle tuber disease.
II. It is free RNA.
III. Mol. weight of RNA is low.
These statements are assigned to -
(1) Viruses (2) Viroids
(3) Mycoplasma (4) Prions

BBC535

71. Choose the correct statement -
(1) *Trichoderma* belong to Basidiomycetes.
(2) *Puccinia* belong to Ascomycetes.
(3) *Albugo* belongs to Basidiomycetes.
(4) *Mucor* belongs to Phycomycetes.

BBC536

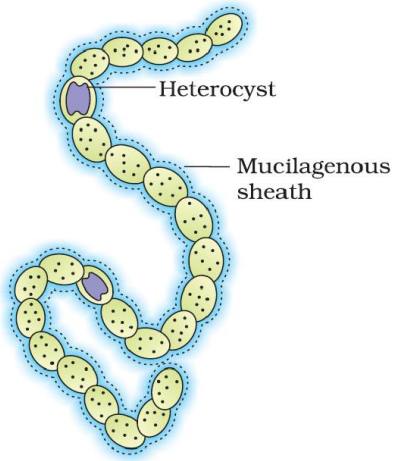
72.



This diagram represents which of the following organism?
(1) Slime mould
(2) *Euglena*
(3) Dividing fungi
(4) Dividing bacterium

BBC537

73. Identify the given figure :-



(1) *Paramecium* (2) *Gonyaulax*
(3) *Nostoc* (4) *Rhizopus*

BBC538

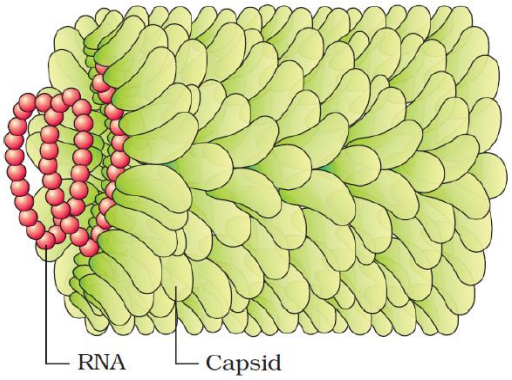
74. Identify the following in sequence :-




(1) *Mucor*, *Trypanosoma*
(2) *Mucor*, *Aspergillus*
(3) *Agaricus*, *Puccinia*
(4) *Albugo*, *Euglena*

BBC539

75.



Identify the given organism
(1) TMV (2) *Mucor*
(3) Diatoms (4) *Puccinia*

BBC540

76. Which of the following are edible Ascomycetes delicacies?
(1) Morels + Mushroom
(2) Truffles + Toadstool
(3) Morels + Truffles
(4) Puffball + Mushroom

BBC541

77. In phycomycetes, sexual reproduction occurs by
(1) Isogamy and Anisogamy.
(2) Isogamy and Oogamy.
(3) Isogamy, Anisogamy and Oogamy.
(4) Oogamy and Anisogamy.

BBC542

78. Which of the following are protozoans?
(1) Diatoms, Flagellates, Ciliates.
(2) Desmids, Flagellates, Ciliates.
(3) Amoeboids, Flagellates, Ciliates, Sporozoans.
(4) *Amoeba*, *Paramecium*, Dinoflagellates, *Plasmodium*.

BBC543

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79. Euglenoids eg. *Euglena* are found -
(1) In fresh running water.
(2) In fresh stagnant water.
(3) In marine environment.
(4) In both fresh and marine water.

BBC544

80. Which of the following statements is wrong about Archaebacteria?
(1) They live in some of the most harsh habitats.
(2) They are recently evolved group.
(3) Cell wall is peptidoglycan less.
(4) They are unlike most other bacteria.

BBC545

81. Dinoflagellates have -
(1) A single flagellum in the transverse groove between the cell plates.
(2) A single flagellum in the longitudinal groove between the cell plates.
(3) Two flagella, one lies longitudinally and the other transversely in a furrow between the wall plates.
(4) No flagella.

BBC546

82. Many fungi are _____ associating with photosynthetic organisms to form mycorrhizae or lichens -
(1) Parasitic (2) Symbiotic
(3) Photosynthetic (4) Saprobic

BBC547

83. Bacteria participate in -
(1) Many industrial and commercial processes.
(2) N_2- fixation in legume roots.
(3) Disease producing in plants, animals and human beings.
(4) All of these.

BBC548

84. _____ are small naked fragments of RNA that infect plant cells.
(1) Prions (2) Nucleons
(3) Prophages (4) Viroids

BBC549

85. _____ is a parasite on mustard.
(1) *Albugo* (2) *Puccinia*
(3) Yeast (4) *Ustilago*

BBC550

86. Match the following -

Column-I		Column-II	
A	Phycomycetes	I	Sac fungi
B	Ascomycetes	II	Algal fungi
C	Basidiomycetes	III	Fungi imperfecti
D	Deuteromycetes	IV	Club fungi

(1) A - II, B - I, C - IV, D - III
(2) A - II, B - IV, C - I, D - III
(3) A - IV, B - I, C - II, D - III
(4) A - IV, B - III, C - II, D - I

BBC551

87. Fungi show a great diversity in -
(1) Morphology
(2) Habitat
(3) Both (1) and (2)
(4) Nutrition

BBC552

88. Match the following and choose the correct combination form the options given -

Column-I		Column-II	
A	Saprophytic protists	i	*Trypanosoma*
B	Golden algae	ii	*Plasmodium*
C	Malarial parasite	iii	Desmids
D	Sleeping sickness is caused by	iv	Slime moulds

(1) A - i, B - ii, C - iii, D - iv
(2) A - ii, B - iii, C - iv, D - i
(3) A - iv, B - iii, C - ii, D - i
(4) A - iii, B - iv, C - ii, D - i

BBC553

89. All protozoans are -
(1) Saprophytes only.
(2) Parasites only.
(3) Predators only.
(4) Heterotrophs (parasite/predators) only.

BBC554

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90. Which statement is correct among the following?
(1) Cell wall is present in Mycoplasma.
(2) Spores in slime moulds have true cell wall.
(3) Bacteria have cell wall made up of cellulose.
(4) Fungi do not have cell wall.

BBC555

91. Chrysophytes group includes -
(1) *Euglena*, *Gonyaulax*
(2) *Amoeba*, *Euglena*
(3) *Paramecium*, Diatoms
(4) Diatoms, Desmids

BBC556

92. Rust disease in wheat plant is caused by -
(1) *Ustilago* (2) *Puccinia*
(3) *Agaricus* (4) *Claviceps*

BBC557

93. Find the incorrect statement about fungi -
(1) They show a great diversity in morphology and habitats.
(2) Fungi are cosmopolitan and occur in air, water, soil and as parasites too.
(3) Dikaryophase occur in Ascomycetes and Basidiomycetes.
(4) *Rhizopus* belongs to group basidiomycetes.

BBC558

94. Among the given statements which are correct ?
(A) Fusion of two nuclei called plasmogamy.
(B) In phycomycetes, mycelium is aseptate and coenocytic.
(C) In Animals, mode of nutrition is holozoic.
(D) Mad cow disease is caused by TMV virus.
(1) (A), (B), (C) and (D)
(2) (B) and (C)
(3) (C) and (D)
(4) (A), (B) and (D)

BBC559

95. Heterocystous cyanobacteria -
(1) Only Nostoc.
(2) Only Anabaena.
(3) Both Nostoc and Anabaena.
(4) Vibrio.

BBC560

96. Match the following -

Column-I		Column-II	
A	Viroids	i	Mad cow disease
B	Prions	ii	Potato spindle tuber disease
C	*Ustilago*	iii	Smut disease
D	*Colletotrichum*	iv	Red Rot of sugarcane

(1) A - i, B - ii, C - iii, D - iv
(2) A - ii, B - i, C - iii, D - iv
(3) A - ii, B - iii, C - i, D - iv
(4) A - iv, B - iii, C - i, D - ii

BBC561

97. **Assertion :** Deuteromycetes are known imperfect fungi.
Reason : Sexual reproduction is absent in deuteromycetes.
(1) Both assertion and reason are true and reason is the correct explanation of assertion.
(2) Both assertion and reason are true and reason is not the correct explanation of assertion.
(3) Assertion is true but reason is false.
(4) Both assertion and reason are false.

BBC562

98. **Assertion :** Viruses are not truly living organism.
Reason : Viruses are acellular organisms and do not show metabolism and consciousness.
(1) Both assertion and reason are true and reason is not the correct explanation of assertion.
(2) Both assertion and reason are true and reason is the correct explanation of assertion.
(3) Assertion is true but reason is false.
(4) Both assertion and reason are false.

BBC563

99. **Assertion :** Spores of slime mould can survive for many years even under adverse conditions.

Reason : Slime mould spore possess true cell wall.

(1) Both assertion and reason are true and reason is the correct explanation of assertion.

(2) Both assertion and reason are true and reason is not the correct explanation of assertion.

(3) Assertion is true but reason is false.

(4) Both assertion and reason are false.

BBC564

100. **Assertion :** Majority of Euglenoids are fresh water organisms.

Reason : All unicellular eukaryotes placed under Protista.

(1) Both assertion and reason are true and reason is the correct explanation of assertion.

(2) Both assertion and reason are true and reason is not the correct explanation of assertion.

(3) Assertion is true but reason is false.

(4) Both assertion and reason are false.

BBC565

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	4	1	3	1	3	4	2	3	4	4	1	3	2	2
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	3	1	3	3	1	1	3	4	1	1	3	1	3	4	4
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	4	1	1	1	1	4	1	2	4	4	4	3	4	3	2
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	3	2	2	1	3	4	1	3	3	1	4	1	2	1	3
Question	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Answer	4	1	1	2	1	1	4	3	1	2	4	4	3	2	1
Question	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
Answer	3	3	3	2	2	3	2	4	4	1	1	3	3	4	2
Question	91	92	93	94	95	96	97	98	99	100					
Answer	4	2	4	2	3	2	1	2	2	2					

Plant Kingdom

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1. Which of the following is not true for algae?
(1) Play role in ecology
(2) Have pigments
(3) Have thalloid body
(4) Have terrestrial nature only
BPK417

2. Which of the following contains chlorophyll?
(1) Kelps, Diatoms, Dinoflagellates
(2) Brown algae, Euglenoids, Diatoms
(3) Diatoms, Dinoflagellates, B.G.A
(4) Green algae, Brown algae, Diatoms
BPK418

3. Which of the following description does not apply to phaeophyta ?
(1) Accessory photosynthetic pigment is fucoxanthin
(2) Nearly all are fresh water algae
(3) Store food in the form of mannitol
(4) Body is filamentous or thalloid
BPK419

4. The sex organs of algae are generally :
(1) Unicellular and jacketed
(2) Unicellular and non-jacketed
(3) Multicellular and jacketed
(4) Multicellular and non-jacketed
BPK420

5. Algin and carageenin are obtained respectively from:
(1) Red algae and brown algae
(2) Red algae and green algae
(3) Brown algae and red algae
(4) Brown algae and green algae
BPK421

6. Pigments present in red algae are :
(1) chl-a, chl-d and fucoxanthin
(2) chl-a and chl-c only
(3) chl-a, chl-d and r-Phycoerythrin
(4) chl-a, chl-c and fucoxanthin
BPK422

7. Complex post fertilization development is seen in which algae :-
(1) Green algae (2) Brown algae
(3) Blue green algae (4) Red algae
BPK423

8. Presence of pyrenoids is character of algae, a pyrenoid consists of ?
(1) Core of starch surrounded by protein
(2) Core of protein surrounded by starch
(3) Core of fatty acid surrounded by protein
(4) Nucleic acids and protein
BPK424

9. Stored food of red algae and green algae are, respectively :-
(1) Starch and floridian starch
(2) Laminarin and starch
(3) Floridian starch and laminarin
(4) Floridian starch and starch
BPK425

10. Bacteriophages usually contain :-
(1) ss DNA (2) ds DNA
(3) ss RNA (4) ds RNA
BPK426

11. Viroid contains :-
(1) Low molecular weight DNA
(2) Low molecular weight RNA
(3) High molecular weight DNA
(4) High molecular weight RNA
BPK427

12. Potato spindle tuber disease is caused by :-
(1) Virus (2) Prion
(3) Viroids (4) Mycoplasma
BPK428

13. Which of the following is not found in virus?
(1) Nucleic acid (2) Capsid
(3) Cell membrane (4) Capsomeres
BPK429

14. In symbiotic association between algae and fungi which one is **not correct** about lichen :-
(1) Algae prepare food for fungi and itself also
(2) Fungi provide shelter and absorb mineral and water for its partner
(3) Lichens are very good pollution indicators because they grow in polluted areas only
(4) In lichens both partners are so close in their association that one would never imagine that they had two different organisms within them
BPK430

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15. In mycorrhiza :-
(1) Fungus provides sugars to the roots of higher plants.
(2) Roots provide only N-containing compounds to fungus.
(3) Fungus provides water and minerals to the roots.
(4) Roots provide sugars and minerals to the fungus.

BPK431

16. Bryophytes have :
(1) Independent sporophyte of long duration
(2) Dominant phase of sporophyte which is dependent
(3) Dominant phase of gametophyte which produce spores
(4) Small sporophyte phase which is generally dependent on gametophyte

BPK432

17. The leafy member of liverwort having tiny leaf-like appendages in two rows on the stem-like structure is :-
(1) *Marchantia* (Liver wort)
(2) *Porella* (Leafy liver wort)
(3) *Polytrichum* (Moss)
(4) *Riccia* (Liver wort)

BPK433

18. The plant body of Mosses is :-
(1) Less differentiated than that of liverworts
(2) More differentiated than that of algae
(3) More differentiated than that of fern
(4) Less differentiated than that of thallophytes

BPK434

19. The heart shaped prothallus in homosporous species of pteridophytes is :-
(1) Dioecious gametophyte stage
(2) Monoecious sporophytic stage
(3) Monoecious gametophytic stage
(4) Dioecious sporophytic stage

BPK435

20. Choose the incorrect statement from the following
(1) Almost all gymnosperms are xerophytes
(2) All gymnosperms show heterosporous nature
(3) All gymnosperms show seed habit
(4) All gymnosperms are monoecious

BPK436

21. The functional megaspore of gymnosperm germinate :-
(1) Inside the megaspore mother cell
(2) Inside the megasporangium
(3) Inside the soil
(4) Into endosperm after fertilization

BPK437

22. Which of the following is not a character of all divisions of vascular plants ?
(1) Sporophytic main plant body
(2) Differentiation of plant body into root, stem and leaves
(3) Xylem and phloem for transporting materials between roots and leaves
(4) Formation of seeds

BPK438

23. Choose the **correct** statement from the following :-
(1) *Pinus* have vessel in Xylem
(2) *Cycas* have persistent leaves
(3) *Alsophila* is a smallest Pteridophyte
(4) All gymnosperms are homosporous

BPK439

24. Choose the **correct** statement
(1) Both pteridophytes and gymnosperms are completely successful land plants
(2) Sporophylls of pteridophytes are photosynthetic and bear sex organs
(3) All gametophytes are non-vascular
(4) All sporophytes are diploid and differentiated into root, stem and leaf.

BPK440

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25. Match the column I with column II regarding to angiosperms :-

Column-I	Column-II
(A) Megasporangium	(i) Pollen sac
(B) Microsporophyll	(ii) Pistil
(C) Microsporangium	(iii) Ovule
(D) Megasporophyll	(iv) Stamen

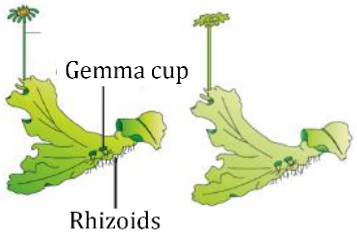
(1) A-ii, B-i, C-iv, D-iii
(2) A-i, B-ii, C-iii, D-iv
(3) A-iii, B-i, C-iv, D-ii
(4) A-iii, B-iv, C-i, D-ii

BPK441

26. Choose the **correct** one :-
(1) Male gamete + Egg cell + Synergids → Embryo
(2) Male gamete + Egg cell + Antipodals → Embryo
(3) Male gamete + Egg cell + Sec. Nucleus → Endosperm
(4) Male gamete + Secondary Nucleus → Endosperm

BPK442

27. Identify the bryophytes marked as A and B:



(A) **(B)**

(1) (A) = Male *Sphagnum*, (B) = Female *Sphagnum*
(2) (A) = Female *Marchantia*, (B) = Male *Marchantia*
(3) (A) = Male *Marchantia*, (B) = Female *Marchantia*
(4) (A) = Female *Sphagnum*, (B) = Male *Sphagnum*

BPK443

28. Which of the following is not correctly matched?
(1) Gemma cups - *Marchantia*
(2) Unicellular organism - *Chlorella*
(3) Biflagellate zoospores - Brown *algae*
(4) Uniflagellate gametes -*Polysiphonia*

BPK444

29. (A) Main plant body is Gametophyte.
(B) The gametophyte is differentiated into true roots, stems and leaves.
(C) Their sporophyte lacks vascular bundles.
(D) Macrophyllous plant is selaginella.
How many statements from A to D are correct about pteridophyta :-
(1) 0 (2) 1 (3) 2 (4) 3

BPK445

30. Which one of the following is not found in fern gametophyte?
(1) Chlorophyll (2) Rhizoids
(3) Sex organs (4) Vascular tissue

BPK446

31. Which one of the following is a correct statement
(1) Protonemal and leafy stage occur in gametophyte of pteridophyte.
(2) In gymnosperms, female gametophyte is free living
(3) Antheridia and archegonia absent in pteridophytes
(4) Precursor of seed habit occurs in pteridophytes.

BPK447

32. Identify the diagram given below?



(1) *Equisetum* (2) *Dryopteris*
(3) *Salvia* (4) *Salvinia*

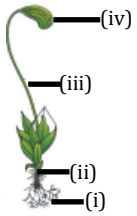
BPK448

33. Thallus of brown algae is divided into three parts as
(1) Foot, Seta and Capsule
(2) Root, Stem and Leaf
(3) Frond, Stipe and Holdfast
(4) Rhizoid, Gemma cup and Archaegonia

BPK449

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34. (A) Name of the plant ?
(B) Name of parts labelled as (i), (ii), (iii) and (iv) ?



	(A)	(B)
(1)	*Funaria*	(i) Rhizome, (ii) Main Axis (iii) Seta, (iv) Capsule
(2)	*Sphagnum*	(i) Capsule, (ii) Seta (iii) Main axis, (iv) Rhizome
(3)	*Marchantia*	(i) Adventitious roots (ii) Stem (iii) Seta, (iv) Capsule
(4)	*Funaria*	(i) Rhizoids, (ii) Main Axis (iii) Seta, (iv) Capsule

(1) 1 (2) 2 (3) 3 (4) 4

BPK450

35. Select the correct statement :-
(1) Gymnosperms are both homosporous and heterosporous
(2) *Salvinia*, *Ginkgo* and *Pinus* all are gymnosperms
(3) *Sequoia* is one of the tallest tree in gymnosperm
(4) The leaves of gymnosperms are not well adapted to exteremes of climate

BPK451

36. Presence of archegonium is a common feature of :-
(1) Algae, Fungi and Bryophytes
(2) Fungi, Bryophytes and Pteridophytes
(3) Bryophytes, Pteridophytes and Gymnosperm
(4) Pteridophytes, Gymnosperm and Angiosperm

BPK452

37. In which of the following option all listed genera belongs to the same class of Algae ?
(1) *Chara*, *Fucus*, *Polysiphonia*
(2) *Volvox*, *Spirogyra*, *Chlamydomonas*
(3) *Porphyra*, *Ectocarpus*, *Ulothrix*
(4) *Sargassum*, *Laminaria*, *Gracillaria*

BPK453

38. Identify and count the number of members of green algae from given box.

Chara, Volvox, Spirogyra, Albugo, Azotobacter, Fucus, Marchantia, Ulothrix

(1) 2 (2) 3 (3) 4 (4) 5

BPK454

39. How many plants in the given below list are the vascular embryophytes?
Spirogyra, Volvox, Fucus, Polysiphonia, Polytrichum, Sphagnum, Marchantia, Funaria, Selaginella, Equisetum.
(1) Six (2) Three (3) Four (4) Two

BPK455

40. Which one of the following sentence is not true for Bryophytes?
(1) Amphibians of the plant kingdom.
(2) Non vascular embryophytes
(3) Zygotic meiosis
(4) Gametophyte develops from germinate spores.

BPK456

41. Mention the ploidy level of followings :-
(a) Protonemal cell of moss.
(b) Rhizoidal cell of prothallus.
(c) Leaf cell of moss.
(d) Gemma cell in *Marchantia*.
(e) Endosperm of *Pinus*.
(f) Leaf cell of fern.
Options :-
(1) n, 2n, n, n, 3n, 2n respectively
(2) 2n, n, n, n, n, 2n respectively
(3) n, n, n, n, n, 2n respectively
(4) n, n, 2n, n, 3n, 2n respectively

BPK457

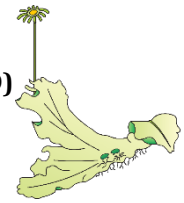
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42. Observe the following diagrams of plants carefully and identify the heterosporous plant.

(A) 

(B) 

(C) 

(D) 

(1) A and C only (2) A and D only
(3) only A (4) A, B and D only

BPK458

43. Match the following Column-I and Column-II

Column-I		Column-II	
A.	*Gelidium*	i	Complex post fertilisation development
B.	*Fucus*	ii	Fucoxanthin
C.	*Polysiphonia*	iii	Agar-Agar

(1) A-i, B-ii, C-iii (2) A-i, B-iii, C-ii
(3) A-ii, B-iii, C-i (4) A-iii, B-ii, C-i

BPK459

44. The life cycle of Bryophyta is haplo - diplontic type, it means that -

(1) Haploid generation is multicellular and diploid generation is single celled.
(2) Haploid generation is dominant and multicellular, diploid generation is also multicellular but comparatively less developed.
(3) Diploid generations are dominant.
(4) Haploid generation is highly reduced.

BPK460

45. Recognise the figure and find out the correct matching :-



(A)



(B)

(1) A- *Polysiphonia*, B-*Dictyota*
(2) A-*Chara*, B-*Fucus*
(3) A-*Laminaria*, B-*Fucus*
(4) A-*Polysiphonia*, B-*Chara*

BPK461

46. Which option is not related with phaeophyceae?

(1) Gelatinous coating of algin.
(2) Gametes are pear shaped and bear two laterally attached flagella.
(3) The food is stored as floridean starch.
(4) They possess chlorophyll a,c, carotenoids and xanthophylls.

BPK462

47. Match the column given below :

Column-I Genus		Column-II Class	
a.	*Adiantum*	i.	*Sphenopsida*
b.	*Lycopodium*	ii.	*Pteropsida*
c.	*Equisetum*	iii.	*Psilopsida*
d.	*Psilotum*	iv.	*Lycopsida*

correct option is :

(1) a-ii b-i c-iii d-iv (2) a-ii b-iv c-i d-iii
(3) a-ii b-iii c-iv d-I (4) a-iii b-iv c-i d-ii

BPK463

48. Pteridophyta have advancement over bryophyta in :-

(1) Presence of vascular tissues
(2) Presence of roots
(3) Necessity of water for fertilisation
(4) (1) and (2) both

BPK464

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49. The spread of living pteridophytes is limited and restricted to narrow geographical region, because :-
(1) Their gametophytes require cool, damp, shady places to grow
(2) They need water for fertilization
(3) Their main plant body does not possess well developed root, stem and leaves
(4) Both (1) & (2)

BPK465

50. Angiosperm range in size from the smallest _____ to tall tree of _____.
(1) *Eucalyptus*, *Wolffia* (2) *Zamia*, *Sequoia*
(3) *Wolffia*, *Eucalyptus* (4) *Cynodon*, Bamboo

BPK466

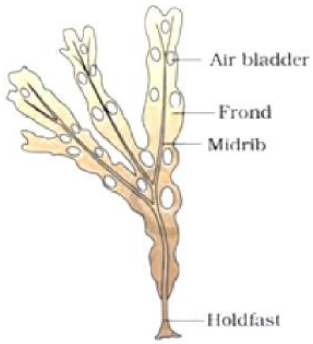
51. Among the following, which is not characteristic feature of bryophytes :
(1) Motile sperms
(2) Presence of archegonium
(3) Water essential for fertilization
(4) Photosynthetically independent sporophyte

BPK467

52. The main plant body of bryophytes is ...A... . It produces ...B..., hence is called a ...C... . Fill in the blanks by choosing the correct option for A, B and C.
(1) A-diploid, B-spores, C-sporophyte
(2) A-haploid, B-gametes, C-gametophyte
(3) A-diploid, B-endospores, C-sporophyte
(4) A-haploid, B-conidia, C-gametophyte

BPK468

53. Study the given algae and answer the following question.



(a) Name of algae
(b) Name of division
(c) Stored food
(1) (a)-*Laminaria* (b)-Chlorophyceae (c)- Starch
(2) (a)-*Fucus* : (b)-Chlorophyceae (c)- Starch
(3) (a)-*Fucus* (b)-Phaeophyceae (c)- Laminarin
(4) (a)-*Laminaria* (b)-Rhodophyceae (c)- Glycogen

BPK469

54. Which one of the following is not an ecological importance of moss ?
(1) Provides food for herbaceous mammals birds and other animals
(2) Use for trans-shipment of living materials due to high water holding capacity.
(3) Pioneering organisms to colonise on rocks with lichens
(4) form dense mats on the soil and reduce the impact of falling rain.

BPK470

55. Match the following (Column-I with Column-II)

Column-I		Column-II	
(a)	*Chlamydomonas*	(i)	Moss
(b)	*Sequoia*	(ii)	Pteridophyte
(c)	*Selaginella*	(iii)	Algae
(d)	*Sphagnum*	(iv)	Gymnosperm

(1) a-iii, b-i, c-iv, d-ii
(2) a-iii, b-iv, c-iii, d-i
(3) a-i, b-iii, c-iv, d-ii
(4) a-iv, b-ii, c-iii, d-i

BPK471

56. Choose the incorrect match
(1) *Cycas* - Dioecious
(2) *Equisetum* - Heterosporous
(3) *Funaria*- Free living gametophyte
(4) *Selaginella*- Precursor to the seed habit

BPK472

57. Find the incorrect from the following :-
(1) Sporophyte of moss is more elaborate than that of liver worts
(2) Spores of bryophytes germinate to form free living sporophytes
(3) Funaria and Sphagnum are mosses
(4) After meiosis spores are produced within the capsule of bryophytes

BPK473

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58. Sexual reproduction is oogamous and accompanied by complex post fertilisation development found in:-
(1) Brown algae (2) Red algae
(3) Blue green algae (4) Green algae
BPK474

59. Phycobilins are characteristic pigments of :-
(1) Rhodophyceae and phaeophyceae
(2) Rhodophyceae and Pyrophyceae
(3) Pyrophyceae and Cyanophyceae
(4) Rhodophyceae and Cyanophyceae
BPK475

60. Unbranched stem is found in -
(1) *Cycas*, *Pinus* (2) *Cycas*, *Cedrus*
(3) *Pinus*, *Cedrus* (4) *Cycas*
BPK476

61. Unlike Bryophytes & Pteridophytes, in Gymnosperms the male & female gametophyte do not have an independent free living existence because .....
(1) Gametophyte is the main body.
(2) Gametophyte retained on sporophytes.
(3) both Gametophyte & sporophyte are independent.
(4) They do not have double fertilization.
BPK477

62. Alga rich in protein is
(1) *Chlorella* (2) *Spirogyra*
(3) *Oscillatoria* (4) *Ulothrix*
BPK478

63. Which is the non embryonic plant group ?
(1) Gymnosperms (2) Bryophytes
(3) Algae (4) Pteridophytes
BPK479

64. "Gemmae" are multicellular green structures for vegetative propagation. These are found in gemma cups present on -
(1) Prothallus of *Salvinia*
(2) Thallus of *Marchantia*
(3) Protonema of *Funaria*
(4) Prothallus of Fern
BPK480

65. Find the mismatched pair w.r.t. plants
(1) Bryophyta- Embryophytes
(2) Pteridophyta-First tracheophytes
(3) Gymnosperms-Naked seeded plants
(4) Angiosperms- Archegoniate spermatophyte
BPK481

66. Bryophytes are also called Amphibians of plant kingdom because .....
(1) They requires wind to complete their life cycle
(2) They requires land to complete their life cycle
(3) They requires both water and land for their survival
(4) They requires only marine water to complete life cycle
BPK482

67. The members of rhodophyceae are commonly called red algae because they are
(1) Greater concentrations in warmer area of ocean
(2) Floridean starch as a stored food
(3) Predominance of r-phycoerythrin in their body
(4) Complex post fertilization development
BPK483

68. _____ the first terrestrial plants to possess vascular tissue:-
(1) Bryophyta (2) Angiosperms
(3) Pteridophyta (4) Gymnosperms
BPK484

69. Read the following paragraph and fill in the blanks with appropriate words :-
In pteridophytes the main plant body is ___(A)___ which bears sporangia that are subtended by leaf-like appendages called ___(B)___. The sporangia produce spores by ___(C)___ and after germination of spores, they give rise to photosynthetic gametophytes called ___(D)___.
Choose the correct option :-
(1) **A-**Sporophyte, **B-**Sporangia, **C-**Meiosis **D-**Protonema
(2) **A-**Gametophyte, **B-**Sporophylls, **C-**Mitosis, **D-**Sporophyte
(3) **A-**Sporophyte, **B-**Sporophylls, **C-**Meiosis, **D-**Prothallus
(4) **A-**Gametophyte, **B-**Sporophylls, **C-**Mitosis, **D-**Protonema
BPK485

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70. Fusion between one large, nonmotile (static) female gamete and a smaller, motile male gamete such type of fertilization occurs in -
(1) *Volvox*, *Fucus* (2) *Ulothrix*
(3) *Eudornia* (4) *Spirogyra*
BPK486

71. In pteridophytes, spores germinate to give rise to inconspicuous, small but multicellular, free living, mostly photosynthetic thalloid gametophytes, called :-
(1) Protonema (2) Prothallus
(3) Endosperm (4) Embryo sac
BPK487

72. "Horse tail" is a -
(1) Bryophyte (2) Pteridophyte
(3) Fungi (4) Mosses
BPK488

73. Xylem and phloem first appeared in
(1) Pteridophyte (2) Bryophyte
(3) Algae (4) Gymnosperm
BPK489

74. *Selaginella* and *Salvinia* are _____ Pteridophyte.
(1) Homosporous (2) Heterosporous
(3) Aquatic (4) Marine
BPK490

75. Branched stem is the property of :
(1) *Cycas*, *Pinus* (2) *Cycas*, *Cedrus*
(3) *Pinus*, *Cedrus* (4) *Cycas*
BPK491

76. Flower, fruit and seeds are produced in -
(1) Pteridophyta (2) Gymnosperm
(3) Angiosperm (4) Bryophyta
BPK492

77. Which pigment is responsible for the characteristic brown coloration of Phaeophyceae?
(1) Chlorophyll a (2) Chlorophyll b
(3) Carotene (4) Fucoxanthin
BPK493

78. Brown algae are mostly found in:
(1) Freshwater habitats
(2) Terrestrial environments
(3) Marine environments
(4) Desert regions
BPK494

79. Which of the following is a common example of brown algae?
(1) *Ulva* (2) *Spirogyra*
(3) *Fucus* (4) *Chlamydomonas*
BPK495

80. Brown algae have a unique structure called a holdfast, which is used for:
(1) Photosynthesis
(2) Reproduction
(3) Attachment to substrates
(4) Nutrient absorption
BPK496

81. The primary storage polysaccharide in brown algae is:
(1) Starch (2) Cellulose
(3) Glycogen (4) Mannitol
BPK497

82. The air bladders found in brown algae serve the function of:
(1) Buoyancy to keep the algae floating
(2) Nutrient storage
(3) Protection from predators
(4) Gas exchange during photosynthesis
BPK498

83. The cell walls of brown algae contain a unique polysaccharide known as:
(1) Chitin (2) Cellulose
(3) Laminarin (4) Algin
BPK499

84. Brown algae belong to class?
(1) Chlorophyceae
(2) Rhodophyceae
(3) Phaeophyceae
(4) Bacillariophyceae
BPK500

85. Which of the following statements about brown algae is false?
(1) They are primarily marine.
(2) They have chlorophyll a and c as primary pigments.
(3) They are mostly unicellular.
(4) They are structurally complex
BPK501

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86. The unique reproductive structures of brown algae contain:
(1) Sporangia
(2) Antheridia and oogonia
(3) Zoospores
(4) Alternation of generations

BPK502

87. **Assertion :** The members of phaeophyceae or brown algae are found primarily in marine habitats.
Reason : Brown algae possess chlorophyll a, c, carotenoids and xanthophylls.
(1) Both A and R are true and R is the correct explanation of A
(2) Both A and R are true but R is not the correct explanation of A
(3) A is true but R is false
(4) Both A and R are false

BPK503

88. **Assertion :** The sporophyte in mosses is more elaborate than that in liverworts.
Reason : The predominant stage of the life cycle of a moss have single set of chromosome.
(1) Both A and R are true and R is the correct explanation of A
(2) Both A and R are true but R is not the correct explanation of A
(3) A is true but R is false
(4) Both A and R are false

BPK504

89. **Assertion** : Free-living, mostly photosynthetic thalloid gametophytes called prothallus in bryophytes.
Reason : *Sequoia* is one of the tallest tree species belong to angiosperm
(1) Both A and R are true and R is the correct explanation of A
(2) Both A and R are true but R is not the correct explanation of A
(3) A is true but R is false
(4) Both A and R are false

BPK505

90. **Assertion:** The gymnosperms are heterosporous.
Reason : The two kinds of spores are produced on different types of sporangia that are borne on different types of sporophylls.
(1) Both A and R are true and R is the correct explanation of A
(2) Both A and R are true but R is not the correct explanation of A
(3) A is true but R is false
(4) Both A and R are false

BPK506

91. The members of rhodophyceae are commonly called red algae because of the predominance of the-
(1) 'r'- phycocyanin
(2) 'r'-phycoerythrin
(3) carotene
(4) Both (1) and (2)

BPK507

92. Fusion between one large, nonmotile (static) female gamete and a smaller, motile male gamete is termed as _____.
(1) Isogamy (2) Anisogamy
(3) Oogamy (4) Zoospory

BPK508

93. Bryophytes are plants which can live in soil but are dependent on water for_____.
(1) Asexual reproduction
(2) Fusion
(3) Sexual reproduction
(4) Pollination

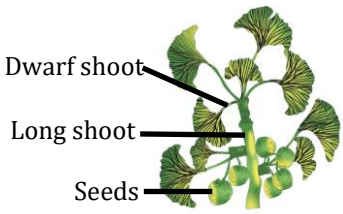
BPK509

94. Roots in some genera of gymnosperm have fungal association in the form of _____.
(1) Mycobiont
(2) Lichen
(3) Phycobiont
(4) Mycorrhiza

BPK510

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95. Select the correct option about given diagram:-



(1) The male and the female gametophytes do not have an independent free-living existence.
(2) The pollen grains and ovules are developed in specialised structures called flowers.
(3) This is the giant redwood tree *Sequoia* is one of the tallest tree species in gymnosperms
(4) Both (1) and (3)

BPK511

96. Which of the following diagrams represents flowering plant :-

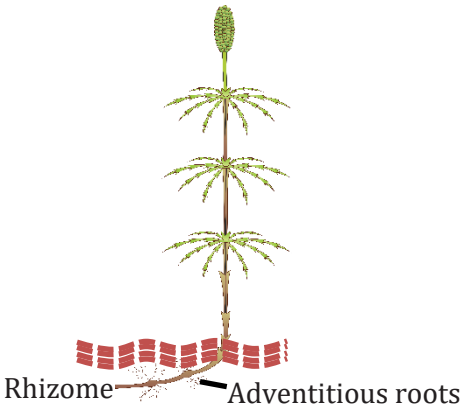


(A) **(B)** **(C)** **(D)**

(1) A, B, D (2) A, B, C
(3) B, D (4) Only D

BPK512

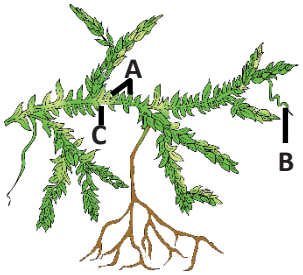
97. Which of the following is not correct about given diagram?



(1) Sporophylls may form distinct compact structures called strobili.
(2) Its prothallus is mostly photosynthetic thalloid gametophytes.
(3) Its possess vascular tissues – xylem and phloem.
(4) Its needle-like leaves reduce the surface area.

BPK513

98. Identify A, B and C in given below diagram.



(1) A–leaves, B–stems, C–roots
(2) A–stems, B–leaves C–roots
(3) A–stems, B–branches, C–leaves
(4) A–leaves, B–roots, C–stem

BPK514

99. Unbranched stem is characteristic of which gymnosperm ?

(1) *Pinus* (2) *Cycas*
(3) *Cedrus* (4) *Lycopodium*

BPK515

100. Both thallophyta and bryophyta are most correctly termed as :-

(1) Phanerogames
(2) Cryptogames
(3) Non-vascular cryptogames
(4) Vascular cryptogames

BPK516

101. Non motile gametes formation has been observed in the algal members belonging to:-

(1) Chlorophyceae
(2) Brown algae
(3) Phaeophyceae
(4) Rhodophyceae

BPK517

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	4	1	2	2	3	3	4	2	4	2	2	3	3	3	3
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	4	2	2	3	4	2	4	2	3	4	4	2	4	1	4
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	4	4	3	4	3	3	2	3	4	3	3	1	4	2	3
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	3	2	4	4	3	4	2	3	2	2	2	2	2	4	4
Question	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Answer	2	1	3	2	4	3	3	3	3	1	2	2	1	2	3
Question	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
Answer	3	4	3	3	3	4	1	4	3	3	2	2	2	4	1
Question	91	92	93	94	95	96	97	98	99	100	101				
Answer	2	3	3	4	1	4	4	4	2	3	4				

Morphology of Flowering Plants

1. Which monocot plant shows exceptionally reticulate venation and bears tendrillar stipules?
(1) *Gloriossa* (2) *Calophyllum*
(3) *Smilax* (4) *Dracaena*
BMP384

2. In which plant, the leaves are small and short lived, the petioles expand and become photosynthetic:-
(1) *Opuntia* (2) *Asparagus*
(3) *Australian acacia* (4) *Ruscus*
BMP385

3. In *Chrysanthemum*, the lateral branches originate from the basal and underground portion of the main stem grow horizontally beneath the soil and then come out obliquely upward giving rise to leafy shoot. These branches are called :-
(1) Stolon (2) Sucker (3) Runner (4) Offset
BMP386

4. Match the column-I with column-II

	Column I		Column II
a.	Runner	i	Opuntia
b.	Aquatic runner	ii.	Oxalis
c.	Flat phylloclade	iii.	Euphorbia
d.	Cylindrical phylloclade	iv.	Eichhornia

Choose the correct answer from the following options :-
(1) a-ii, b-iv, c-iii, d-i (2) a-ii, b-iv, c-i, d-iii
(3) a-iv, b-i, c-ii, d-iii (4) a-iv, b-ii, c-i, d-iii
BMP387

5. In which plants primary root is short lived and replaced by a large number of fibrous roots?
(1) Gymnosperm plants
(2) Monocotyledonous plants
(3) Dicotyledonous plants
(4) All angiosperm plants
BMP388

6. Thorns are developed from
(1) Floral buds (2) Axillary buds
(3) Epidermis (4) All of the above
BMP389

7. Hanging roots from branch of banyan tree are called :-
(1) Stilt roots (2) Prop roots
(3) Fibrous roots (4) Pneumatophores
BMP390

8. When the incision of the lamina reaches up to the midrib breaking it into a number of leaflets, then the leaf is called :-
(1) Compound leaf (2) Simple leaf
(3) Alternate leaf (4) Opposite leaf
BMP391

9. Underground stems of potato, ginger, turmeric, zaminkand and Colocasia are modified to :-
(1) Protection from browsing animals
(2) Perform perennation and storage
(3) Water and mineral absorption
(4) Increase the surface area
BMP392

10. Which of the following is incorrect ?
(1) Roots help in water and mineral absorption from soil.
(2) Roots provide a proper anchorage.
(3) Roots store food material and synthesize plant growth regulators.
(4) Roots lack meristematic activity
BMP393

11. Which one of the following option gives correct categorisation of flowers according to the position of gynoecium on the thalamus :-

	Hypogynous	Perigynous	Epigynous
(1)	Chinarose, Mustard	Plum, Guava	Peach, Cucumber
(2)	Chinarose, Rose,Apple	Plum, Peach	Guava, Cucumber
(3)	Chinarose, Brinjal	Rose, Plum	Cucumber, Ray florets of funflower
(4)	Chinarose, Apple	Rose, Peach	Cucumber, Ray florest of sunflower

BMP394

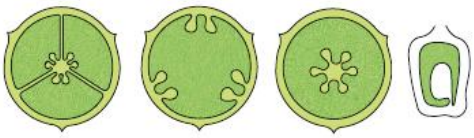
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12. Select the incorrectly matched pair :-
(1) Persistent sepal — Lily
(2) Polypetalous petals — Mustard
(3) Spathe — Maize
(4) Pappus — Sunflower

BMP395

13. The given diagrams show different type of placentation. Choose **correct** option with respect to figure :-



(A) (B) (C) (D)

	Diagram	Type of placentation	Example
(1)	A	Basal	Sunflower, Marigold
(2)	B	Parietal	Mustard, Argemone
(3)	C	Free central	China rose, Prime rose
(4)	D	Axile	Tomato, Lemon

BMP396

14. What is perianth ?
(1) Part of flower in which calyx and corolla are not always distinguishable.
(2) When stamens and gynoecium are not distinguishable.
(3) When only calyx is present
(4) When calyx and corolla both are absent in a flower.

BMP397

15. A sterile stamen is called :-
(1) Phyllode (2) Staminode
(3) Pappus (4) Spathe

BMP398

16. When sepals or petals in a whorl just touch one another at the margin, without overlapping, then it is said to be ?
(1) Twisted (2) Imbricate
(3) Valvate (4) Vexillary

BMP399

17. In which type of inflorescence the main axis continues to grow & the flowers borne laterally in acropetal manner on it :-
(1) Racemose (2) Cymose
(3) Cyathium (4) Hypanthodium

BMP400

18. Find the incorrect statement for hypanthodium inflorescence :-
(1) It has flask shaped fleshy receptacle
(2) It has three types of flowers
(3) It has bisexual flowers
(4) Present in Ficus

BMP401

19. When the margins of petals overlap one another without any particular direction, this condition is found in :-
(1) Gulmohar (2) Cotton
(3) Mustard (4) China Rose

BMP402

20. When a flower can be divided into two equal radial halves in any radial plane passing through the centre, then it is said to be :-
(1) Zygomorphic (2) Actinomorphic
(3) Bilateral (4) Asymmetric

BMP403

21. Consider following statements :-
(A) In caryopsis fruits pericarp is fused with the seed coat and found in wheat and rice.
(B) Litchi is a nut fruit which is indehiscent fruit in which fleshy aril is edible.
Which of the above statements is/are **correct**?
(1) A only (2) B only
(3) Both A and B (4) Neither A nor B

BMP404

22. What are the aggregate fruits :-
(1) Fruit which develop from monocarpellary gynoecium or multicarpellary syncarpous gynoecium.
(2) Whole inflorescence is modified into fruit
(3) Fruit which develop from multicarpellary, apocasssrpous gynoecium.
(4) Fruit which develops without fertilization

BMP405

23. Presence of which of the following character is salient feature of family brassicaceae :-
(1) Axile placentation
(2) Replum (false septum)
(3) Adventitous root system
(4) Didynamous androecium

BMP406

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24. In tomato plant the ovary possesses :-
(1) Swollen placenta (2) Oblique septum
(3) Axile placentation (4) All of the above

BMP407

25. Floral formula $Br\oplus\male P_{(3+3)}\ A_{3+3}\underline{G}_{(3)}$ represents which one of the following group of plants of a specific family:-
(1) Mustard and Chinarose
(2) *Allium* and *Asparagus*
(3) Lily and *Petunia*
(4) Tulip and Tobacco

BMP408

26. Ebr $\oplus\male K_{2+2}C_{4x}A_{2+4}\underline{G}_{(2)}$ is the floral formula of:-
(1) Mustard (2) Brinjal
(3) Sesbania (4) *Colchicum*

BMP409

27. Perennial herbs with underground bulbs or corms or rhizomes are common in the family:-
(1) Liliaceae (2) Fabaceae
(3) Solanaceae (4) Brassicaceae

BMP410

28. Exstipulate, simple and alternate with reticulate venation types of leaves present in family :-
(1) Liliaceae (2) Fabaceae
(3) Solanaceae (4) (1) and (2) Both

BMP411

29. Composite fruit develops from :-
(1) Multicarpellary, syncarpous gynoecium
(2) Multicarpellary, apocarpous gynoecium
(3) Multicarpellary, superior ovary
(4) Complete inflorescence

BMP412

30. The floral formula of *Petunia* is :-
(1) $\oplus\male K_{(5)}C_{(5)}A_5\underline{G}_{(2)}$
(2) $\oplus\male K_5\,C_5\ A_{(5)}\underline{G}_{(2)}$
(3) $\oplus\male K_{(5)}\ C_{(5)}\ A_5\underline{G}_{(2)}$
(4) $\oplus\male K_{(5)}C_{(5)}A_{(5)}\underline{G}_{(2)}$

BMP413

31. Consider the following four statements (a-d) and select the option which includes all the correct ones:-
(a) In *Rhizophora* pneumatophores are present which help to get oxygen for respiration.
(b) Prop roots are present in banyan tree for support.
(c) Stilt roots are present in maize and sugarcane.
(d) Adventitious roots are develop from radicle.
(1) Statements (a), (b) and (c)
(2) Statements (a) and (b)
(3) Statements (c) and (d)
(4) Statements (a), (c) and (d)

BMP414

32. In how many given plants stem is modified for storage of food ?
Potato, Sweet potato, Turmeric, Turnip, Ginger, *Opuntia*, *Pistia*, Zaminkand, *Colocasia*
(1) Seven (2) Five
(3) Eight (4) Six

BMP415

33. Which of the following option give the correct categorisation of six plants on the basis of arrangement of leaves (A, B & C) on the stem or branch ?

	(A) Alternate phyllotaxy	**(B) Opposite phyllotaxy**	**(C) Whorl phyllotaxy**
(1)	*Nerium* & *Calotropis*	China rose & *Alstonia*	*Ocimum* & mustard
(2)	*Ocimum* & *Alstonia*	*Nerium* & mustard	China rose & *Calotropis*
(3)	Mustard & china rose	Guava & *Calotropis*	*Nerium* & *Alstonia*
(4)	*Nerium* & *Alstonia*	Mustard & Chinarose	*Ocimum* & *Calotropis*

BMP416

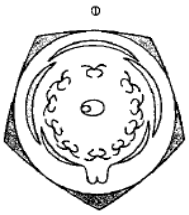
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34. Consider the following four statements (a-d) and select the option which includes all the correct ones :
(a) In some leguminous plants, the leaf base may become swollen, which is known as pulvinus.
(b) Long thin flexible petioles allow leaf blades to flutter in wind, there by cooling the leaf and bringing fresh air to leaf surface.
(c) A single leaf arises at each node in mustard and china rose.
(d) Leaves originate from shoot apical meristem and are arranged in basipetal manner.
(1) a, b and c (2) b, c and d
(3) a and b only (4) c and d only
BMP417

35. Observe the following examples. How many plants show hypogynous condition?
Mustard, China rose, Rose, Plum, Peach, Guava Apple, Cucumber, Brinjal
(1) One (2) Three (3) Seven (4) Five
BMP418

36. "Long filament is attached to the back of the anther at a point only and anther can swing freely in any direction.
This condition is not found in :-
(1) Wheat (2) Maize
(3) Rice (4) Sunflower
BMP419

37. Which one of the following is incorrect about given below diagram.

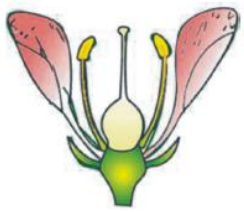


(1) $\oplus \male K_{(5)}\, C_{1+2+(2)}\, A_{(9)+1}\, G_1$
(2) Legume fruit with one to many non endospermic seed.
(3) Five Petals, with vexillary aestivation.
(4) Found in *Crotalaria*, *Sesbania*, gram and lupin.
BMP420

38. How many plants have zygomorphic flower from given below ?
Mustard, Datura, Gulmohar, *Cassia*, Chilli, Pea, Potato, Onion, Soyabean, Bean
(1) 5 (2) 9 (3) 1 (4) 3
BMP421

39. How many of the following plants show axile placentation :-
China rose, Lemon, Pea, Mustard, *Argemone*, Tomato, *Primrose*, *Dianthus*
(1) Two (2) Three (3) Five (4) Six
BMP422

40. In the given following diagram, how many plants not belong to this condition ?



Mustard, Plum, Cucumber, *Hibiscus*, pumpkin, tobacco, lily, onion, tulip, apple, peach, lupin, gram, radish, Capsicum
(1) Nine (2) Four (3) Eleven (4) Eight
BMP423

41. Which of the following characters are found in Asteraceae family ?
(a) Capitulum inflorescence
(b) Presence of pappus
(c) Syngenesious stamen
(d) Gynoecium bicarpellary, syncarpous and superior ovary
(e) Basal placentation
(1) a, b and c (2) a, b and e
(3) b and c (4) a, b, c and e
BMP424

42. Match the column-I with column-II :-

	Column I		Column II
I.	Sunflower	a.	Lodicules
II.	Wheat	b.	pappus
III.	*Gossypium*	c.	Monoadelphous
IV.	China rose	d.	Staminal tube

(1) I-b, II-a, III-c, IV-d
(2) I-b, II-a, III-d, IV-c
(3) I-a, II-b, III-d, IV-c
(4) I-c, II-d, III-a, IV-b
BMP425

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43. How many plants in the list given below have axile placentation ?
Mustard, gram, *tulip*, *Asparagus*, Arhar, Sunhemp, Chilli, *Colchicum*, Onion, Pea, *Trifolium*, *Atropa*, Potato.
(1) Six (2) Seven
(3) Five (4) Eight

BMP426

44. How many plants in the list give below belong to floral formula $\oplus\,\male\,K_{(5)}\overset{\frown}{C_{(5)}A_{(5)}}\underline{G_2}$
Lupin, Datura, Turnip, *Gloriosa*, Brinjal, Chilli, *Aloe*, Raddish, Gram, *Indigofera*.
(1) Four (2) Three (3) Two (4) Five

BMP427

45. Match of the following column-I and II, and choose correct option :-

	Column I		Column II
I.	Pinnately compound leaf	A.	Leguminus
II.	Pinnately compound leaf	B.	Banyan
III.	Simple leaf	C.	Neem
IV.	Pulvinus	D.	Silk cotton

(1) A-IV, B-III, C-I, D-II
(2) A-II, B-III, C-IV, D-I
(3) A-III, B-II, C-IV, D-I
(4) A-III, B-IV, C-II, D-I

BMP428

46. How many characters in the list given below belongs to family solanaceae?
Parianth, Epipetalous, Trimerous, Swollen Placenta, Epiphyllous, Pentamerous, Oblique septum, Diadelphous, Persistent calyx.
(1) Four (2) Five
(3) Six (4) Three

BMP429

47. Which of the following is correct for the diagram given below :



(1) It is found in lady's finger and cotton flowers
(2) It is found in Gulmohar which also possess zygomorphic flowers
(3) It is found in Bean which also possess zygomorphic flower
(4) It is found in mustard which also possess actinomorphic flower.

BMP430

48. Study the following statements carefully and give the answer.
(A) Petioles allow leaf blade to flutter in wind.
(B) Veins provide rigidity to the leaf blade and act as channels of transport.
(C) A bud is present in the axil of both leaf and leaflet.
(D) In pinnately compound leaf, all leaflets are attached to a common point petiole.
(1) Both A and B are correct
(2) Both A and D are correct
(3) Both B, D are wrong
(4) Both B and C are wrong

BMP431

49. Identify the given floral diagram and choose suitable option from the following characteristics:-



(1) Epitepalous and parietal placentation
(2) Oblique septum and swollen placenta
(3) Stamens five and epipetalous
(4) Both (2) and (3)

BMP432

50. Which one of the following options gives the correct categorisation of six plants according to the type of aestivation (X, Y, Z) they give out.

	(X) Valvate	(Y) Twisted	(Z) Imbricate
(1)	Mustard, *Cassia*	Gulmohar, China rose	Cotton, Potato
(2)	Cotton, China rose	*Cassia*, Mustard	Gulmohar, Potato
(3)	Mustard, Cotton	Potato, China rose	Gulmohar, *Cassia*
(4)	Mustard, Potato	Cotton, China rose	Gulmohar, *Cassia*

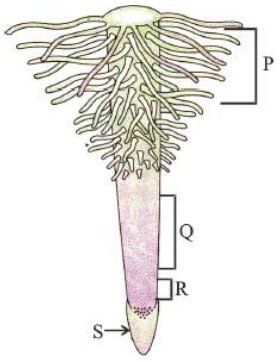
BMP433

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51. How many examples represent Actinomorphic symmetry ?
Mustard, *Datura*, Chilli, Pea, Gulmohar, Bean, *Cassia*, *Canna*
(1) Three (2) Four (3) Five (4) Two

BMP434

52. Given below is the diagrammatic sketch of root. Identify the parts labelled P,Q,R and S, and select the right option about them :-



	P	Q	R	S
(1)	Region of Maturation	Region of meristematic activity	Region of elongation	Root cap
(2)	Root cap	Region of elongation	Region of meristematic activity	Region of Maturation
(3)	Root cap	Region of meristematic activity	Region of elongation	Region of Maturation
(4)	Region of Maturation	Region of elongation	Region of meristematic activity	Root cap

BMP435

53. Identify the following figures and choose the correct option.

(P) 
(Q) 
(R) 
(S) 

(1) **P**-Palmately compound leaf,
Q-Cymose inflorescence,
R-Perigynous flower,
S- Twisted aestivation

(2) **P**-Pinnately compound leaf,
Q-Racemose, inflorescence,
R-Epigynous flower,
S-Imbricate aestivation

(3) **P**-Pinnately compound leaf,
Q-Cymose inflorescence,
R-Epigynous flower,
S-Vexillary aestivation

(4) **P**-Palmately compound leaf,
Q-Cymose inflorescence,
R-Perigynous flower,
S-Vexillary aestivation

BMP436

54. Aestivation of petals in the flower of cotton is correctly shown in :

(1) 
(2) 
(3) 
(4) 

BMP437

55. Which of the following floral formula is not occur in Asteraceae.

(1) Br% ♀$K_pA_0\overline{G}_{(2)}$

(2) Br⊕♂$K_P\overset{\frown}{C}_{(5)}A_{(5)}\overline{G}_{(2)}$

(3) Br%$K_pC_{(5)}A_0G_0$

(4) EBr%$K_pC_{(5)}A_0G_0$

BMP438

56. Identify A and B inflorescence :-


(A)

(B)

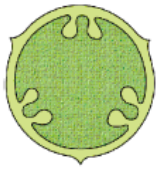
(1) A-Cymose, B-Racemose
(2) A-Racemose, B-Cymose
(3) A-Racemose, B-Racemose
(4) A-Cymose, B-Cymose

BMP439

57. Br %♂P_2 $A_{3 \text{ or } 6}$ $\underline{G}_1$ or $\underline{G}_{(3)}$
Give floral formula is related to :-
(1) Liliaceae (2) Malvaceae
(3) Asteraceae (4) Poaceae
BMP440

58. "The filaments are united together to form a long staminal canal or staminal tube around style"
This condition is related to :-
(1) Malvaceae (2) Asteraceae
(3) Cruciferae (4) Liliaceae
BMP441

59. Given figure is a type of placenta of a flowering plant. Choose correct statement related to below given placenta.



(a) Found in unilocular syncarpous ovary.
(b) Ovules develops on the outer wall of the ovary on peripheral part.
(c) Presence of false septum.
(d) Shown by Primrose & *Argemone*.
(1) a, b
(2) b, d
(3) a, c
(4) All are correct statement
BMP442

60. Monocot plants are characterised by the presence of -
(1) Tap roots (2) Fibrous roots
(3) Pneumatophores (4) Stilt roots
BMP443

61. Stem originates from :-
(1) Radicle (2) Plumule
(3) Cotyledon (4) Leaf
BMP444

62. In which type of inflorescence, the flowers are borne in a basipetal order :-
(1) Racemose (2) Cymose
(3) Spike (4) Catkin
BMP445

63. When gynoecium occupies the highest position and other parts are situated below it, then flower is called as:-
(1) Epigynous (2) Perigynous
(3) Hypogynous (4) Actinomorphic
BMP446

64. Flowers of peach, plum and rose are :
(1) Epigynous (2) Perigynous
(3) Hypogynous (4) Polygamous
BMP447

65. Replum is found in :-
(1) Pea (2) China rose
(3) Mustard (4) *Citrus*
BMP448

66. Floral formula of ashwagandha is :-
(1) Br or Ebr ⊕♀$\underline{K}_{2+2}C_4A_{2+4}\underline{G}_{(2)}$
(2) Br or Ebr ⊕♀$\underline{K}_{(5)}\overset{\frown}{C_{(5)}A_5}\underline{G}_{(2)}$
(3) Br or Ebr ⊕♀$\underline{K}_{(5)}C_{(5)}A_5\underline{G}_{(2)}$
(4) Br or Ebr ⊕♀$\underline{K}_{(5)}\overset{\frown}{C_{(5)}A_5}\underline{G}_{(5)}$
BMP449

67. What type of phyllotaxy is found in Chinarose and Mustard ?
(1) Opposite (2) Alternate
(3) Whorled (4) None of above
BMP450

68. Many pulses of daily use belong to one of the families following :-
(1) Solanaceae (2) Fabaceae
(3) Liliaceae (4) Poceae
BMP451

69. The wheat grain has an embryo with one large shield shaped cotyledon known as :-
(1) Coleoptile (2) Epiblast
(3) Coleorrhiza (4) Scutellum
BMP452

70. *Gloriosa* and *Tulip* are the members of family
(1) Liliaceae (2) Malvaceae
(3) Mimosoideae (4) Solanaceae
BMP453

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71. Swollen placenta and oblique septum is characteristic feature of :-
(1) Cruciferae (2) Solanaceae
(3) Liliaceae (4) Fabaceae
BMP454

72. Arrangement of ovules within the ovary is known as:
(1) Placentation (2) Aestivation
(3) Fragmentation (4) Hibernation
BMP455

73. The cells of which region undergo rapid elongation, enlargement and are responsible for the growth of the root in length ?
(1) Region of meristematic acitivity
(2) Region of elongation
(3) Region of maturation
(4) Root cap
BMP456

74. When the anthers of stamens are free but the filaments are united into single bundle. Such a condition, termed as:-
(1) Polyadelphous (2) Tetradynamous
(3) Monoadelphous (4) Syngenesious
BMP457

75. Syngenesious condition is found in :-
(1) Solanaceae (2) Fabaceae
(3) Asteraceae (4) Liliaceae
BMP458

76. Smallest anterior petals in papilionaceous corolla known as :
(1) Perianth (2) Wings
(3) Standard (4) Keel
BMP459

77. Replum is formed due to the growth of :
(1) Placenta (2) Thalamus
(3) Ovule (4) Style
BMP460

78. The term 'Monoadelphous' is related to
(1) Corolla (2) Calyx
(3) Gynoecium (4) Androecium
BMP461

79. Feathery stigma is present in :-
(1) Wheat (2) Pea
(3) Gram (4) Datura
BMP462

80. 3-7 epicalyx, 5 petals which are free with twisted aestivation are feature of :-
(1) Malvaceae (2) Asteraceae
(3) Solanaceae (4) Febaceae
BMP463

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	3	2	2	2	2	2	1	2	4	3	1	2	1	2
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	3	1	3	1	2	3	3	2	4	2	1	1	3	4	3
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	1	2	3	1	2	4	1	1	2	2	2	2	4	2	2
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	2	3	1	4	4	1	4	3	3	4	2	4	1	3	2
Question	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Answer	2	2	3	2	3	2	2	2	4	1	2	1	2	3	3
Question	76	77	78	79	80										
Answer	4	2	4	1	1										

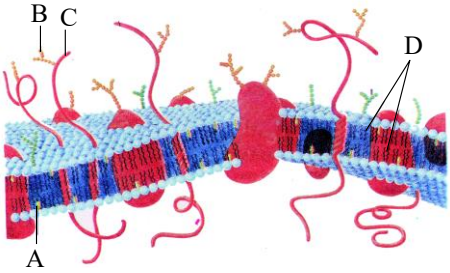
Cell : The Unit of Life

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1. Hypothesis that the bodies of animals and plants are composed of cells and products of cells was proposed by :-
(1) Schleiden (2) Schwann
(3) Rudolf Virchow (4) Robert Brown
BCL274

2. To keep the cell in the 'Living state', various chemical reactions occur in :-
(1) Nucleus (2) Cell membrane
(3) Cytoplasm (4) Cell wall
BCL275

3. Choose correct option regarding labelled parts A or B or C or D :-



(1) A → Cholestrol B → Protein
(2) B → Protein C → Sugar
(3) C→ Protein D → Lipid bilayer
(4) D → Lipid bilayer A → Sugar
BCL276

4. According to fluid-mosaic model, the quasi-fluid nature of _____ enables lateral movement of _____ within the overall bilayer. This ability to move within the membrane is measured as its _____.
(i) Carbohydrates (ii) Lipids
(iii) Proteins (iv) Fluidity
(v) Selective permeability
Correct sequence to fill the blanks :-
(1) ii, iii, iv (2) iii, i, iv
(3) iii, ii, v (4) i, ii, iv
BCL277

5. Which of the following is not dependent upon fluidity of cell membrane ?
(1) Selective permeability
(2) Endocytosis
(3) Secretion
(4) Cell growth
BCL278

6. Transport of molecules across the membrane against their concentration gradient is :-
(1) An energy independent process
(2) Osmosis
(3) Diffusion
(4) An energy dependent process
BCL279

7. Mark the correct statements related to cell wall.
(a) Non-living rigid structure
(b) Present in fungi and plants
(c) Provides barriers to undesirable macromolecules
(d) Primary cell wall is capable of growth
(e) Primary wall gradually diminishes as the cell matures
(1) a, b, d (2) d only
(3) c, d (4) All are correct
BCL280

8. Oligosacchrides present on the outer surface of plasma membrane are important for :-
(1) cell to cell recognition
(2) fluidity of plasma membrane
(3) stability of plasma membrane
(4) selective permeability of plasma membrane
BCL281

9. The fluid nature of plasma membrane is important from the point of view of how many function in the list given below ?
Cell division, endocytosis, secretion, cell growth, formation of intercellular junctions.
(1) Four (2) Three
(3) Two (4) Five
BCL282

10. Presence of which of the following in an eukaryotic cell differentiate it from a prokaryotic cell?
(1) Cell membrane
(2) Cytoplasm
(3) Membrane bound cell organelles
(4) Ribosomes
BCL283

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11. Mitochondria and chloroplasts have membranes yet they are not the part of endomembrane system because :-
(1) Their membranes are thin as compare to the membranes of ER & Golgi body.
(2) They have two membranes each where as endomembrane system has one membrane
(3) Their functions are not coordinated with the ER, golgi body and Lysosomes
(4) They originated from prokaryotic cells

BCL284

12. ER is concerned with the synthesis of :-
(1) Lipid (2) Steroid hormone
(3) Proteins (4) All of the above

BCL285

13. RER (Rough Endoplasmic Reticulum) :-
(a) Bears 80S ribosomes on their outer surface
(b) Frequently observed in the cells actively involved in protein synthesis and secretion
(c) Are extensive and continuous with outer nuclear membrane
(d) Synthesizes steroid hormones in animal cells
Find out the incorrect statement(s) :-
(1) a and c (2) b and d
(3) b only (4) d only

BCL286

14. A Golgi complex has :-
(1) Fixed number of cisternae
(2) Varied number of cisternae
(3) One cisternae in higher plants
(4) Convex trans face and concave cis face cisternae

BCL287

15. The cis and the trans faces of golgi body are :-
(1) Completely similar but not inter-connected
(2) Entirely different and not interconnected
(3) Entirely different but interconnected
(4) Completely similar and interconnected

BCL288

16. Materials to be packaged in the form of vesicles from the.....(i).....fuse with the.....(ii).....of the golgi apparatus and move towards the....(iii)....
Choose the **correct** terms for blanks :-
(1) (i) - lysosome, (ii) - maturing face, (iii) - trans face
(2) (i) - ER, (ii) - cis face, (iii) - forming face
(3) (i) - vacuole, (ii) - concave face, (iii) - convex face
(4) (i) - ER, (ii) - forming face, (iii) - trans face

BCL289

17. Digestion of the tail of frog tadpole larva during metamorphosis is :-
(1) Autophagy (2) Heterophagy
(3) Pinocytosis (4) Autolysis

BCL290

18. Cell sap contains :-
(1) Water + Reserve food
(2) Water + Excretory products
(3) Water + Secretory materials
(4) All of the above

BCL291

19. In plant cells the vacuole can occupies upto..... of the volume of the cell.
(1) 10 percent (2) 90 percent
(3) 50 percent (4) 70 percent

BCL292

20. In which of the following options function is correctly matched with their particular cellular structure ?

	Functions	Cellular structure
(1)	Protein synthesis	SFR
(2)	Formation of glycoprotein	Golgi body
(3)	Antolysis	RER
(4)	Photosynthesis	Vacuole

BCL293

21. The number of mitochondria per cell is variable depending on the :-
(1) Size of the cells
(2) Physiological activity of the cells
(3) Shape of the cells
(4) Position of the cells

BCL294

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22. Mitochondria :-
(a) are easily visible under the microscope (without specifically stained)
(b)are typically sausage-shaped or cylindrical
(c) are double membrane bound structure
(d) have two aqueous compartments
Option :
(1) a, d correct and b, c incorrect
(2) a, b correct and c, d incorrect
(3) a incorrect and b, c, d correct
(4) a, d incorrect and b, c correct
BCL295

23. Site of ATP formation in mitochondria is :-
(1) Matrix
(2) Cristae
(3) F_1 particle
(4) Outer compartment
BCL296

24. Mitochondria are believed to be of symbiotic origin from bacteria because they have similarity with bacteria in :-
(1) DNA
(2) Ribosomes
(3) Division method
(4) All of the above
BCL297

25. Mitochondria are the sites of :-
(1) Protein synthesis
(2) Cellular digestion
(3) Anaerobic respiration
(4) Aerobic respiration
BCL298

26. In the stem of potato, presence of which of the following is significant ?
(1) Amyloplasts (2) Elaioplasts
(3) Chromoplasts (4) Aleuroplasts
BCL299

27. The chloroplasts contain :-
(1) Fat soluble pigments anthocyanin and chlorophylls.
(2) Water soluble pigments carotenoids and chlorophylls.
(3) Water soluble pigments chlorophylls and anthocyanin.
(4) Fat soluble pigments carotenoids and chlorophylls.
BCL300

28. Outer membrane of chloroplast is comparatively more permeable than inner membrane :-
(1) Because it is comparatively thin layer
(2) Due to the absence of proteins
(3) Due to the presence of porins
(4) Because it has more cholesterol
BCL301

29. The stroma of the chloroplast contains enzymes for the synthesis of :-
(1) Lipids
(2) Carbohydrates
(3) Proteins
(4) Carbohydrates and proteins
BCL302

30. Chloroplasts are called semiautonomous organelles of the cell because they have :-
(1) DNA and RNA
(2) Ribosomes
(3) Enzymes for protein synthesis
(4) All of the above
BCL303

31. Ribosomes are known as :-
(1) Cell within cell
(2) Largest cell organelle
(3) Universal cell organelle
(4) Autonomous cell organelle
BCL304

32. Ribosomes are composed of :-
(1) Deoxyribonucleic acid and ribonucleic acid.
(2) Deoxyribonucleic acid and proteins.
(3) Ribonucleic acid and proteins.
(4) Proteins, ribonucleic acid and deoxyribonucleic acid.
BCL305

33. If the concentration of magnesium ions in the cell is less than 0.001 M, then it will affect the synthesis of :-
(1) Carbohydrates (2) Proteins
(3) Lipids (4) Vitamines
BCL306

34. In the animal cell, 70 S ribsomes are found in:-
(1) Mitochondria and plastid
(2) Mitochondria
(3) Mitochondria, plastid and ER
(4) ER and freely in cytoplasm
BCL307

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35. An elaborate network of filamentous proteinaceous structures present in the cytoplasm is involved in :-
(1) Intracellular digestion
(2) Maintenance of the shape of the cell
(3) ATP formation
(4) Carbohydrates synthesis

BCL308

36. Both the cilium and flagellum emerge from ..... like structure called the basal body.
(1) Mitochondria (2) Centriole
(3) Nucleus (4) Golgi body

BCL309

37. Cilium and flagellum are :-
(1) Without covering
(2) Covered with two membranes
(3) Out growths of cell wall
(4) Covered with plasma membrane

BCL310

38. Arms of 'A' tubules in cilium and flagellum, consist of an enzymatic protein :-
(1) Actin (2) Dynein
(3) Myosin (4) Tubulin

BCL311

39. Which of the following is correct regarding arrangement of microtubules in centrioles ?
(1) 9 (doublet) + 2
(2) 9 (Triplet) + 2
(3) 9 (Triplet) + 0
(4) 9 (doublet) + 0

BCL312

40. Radial spokes in centrioles connect the peripheral microtubules :-
(1) With each other
(2) With central tubules
(3) With central hub
(4) With endoplasmic reticulum

BCL313

41. Centrioles are :-
(1) Double membrane bound
(2) Membrane less
(3) Triple membrane bound
(4) Single membrane bound

BCL314

42. Nucleus as a "cell organelle" was first described by ..... as early as .....
(1) Robert Hooke, 1665
(2) Robert Brown, 1831
(3) Flemming, 1931
(4) Strasburger, 1831

BCL315

43. Nuclear envelope consists of parallel membranes with a space called perinuclear space which is about :-
(1) 10 to 50 Å (2) 100 to 500 Å
(3) 1000 to 5000 Å (4) 1 to 5 Å

BCL316

44. Large and more numerous nucleoli are present in cells actively carrying out -
(1) Lipid synthesis
(2) Glycogen synthesis
(3) Protein synthesis
(4) Nucleic acid synthesis

BCL317

45. (a) Spherical structure
(b) Present in the nucleoplasm
(c) Membraneless
(d) Site for ribosomal RNA synthesis
Above given informations are true for which cell structure ?
(1) Nucleolus (2) Ribosome
(3) Centriole (4) Centromere

BCL318

46. Read the following statements (A–D)
(A) Nuclear matrix or nucleoplasm contains nucleolus and chromatin
(B) Outer nuclear membrane usually remains continuous with the ER and also bears ribosomes on it
(C) Nuclear pores are the passages through which movement of protein molecules take place in only one direction from nucleus to the cytoplasm.
(D) Nucleous act as ribosome factory
Which of the above statements are correct?
(1) A, B, C, D (2) Only C
(3) A, C, D (4) A, B, D

BCL319

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47. Interphase nucleus has loose and indistinct network of nucleoprotein fibres called chromatin, but during different stages of cell division, cells show 'structured chromosomes' in place of the :-
(1) Nucleus (2) Cytoplasm
(3) Mitochondria (4) Nucleolus
BCL320

48. Histone proteins are rich in which of the following amino acids ?
(1) Arginine and Histidine
(2) Arginine and Lysine
(3) Histidine and Lysine
(4) Tryptophan and Lysine
BCL321

49. Every chromosome essentially has the _____ on the sides of which disc shaped structures called _____ are present.
Choose the correct words for the blanks respectively
(1) Centrosome, centrioles
(2) Kinetochore, centrosome
(3) Centromere, Kinetochores
(4) DNA, RNA
BCL322

50. Chromosome, which has centromere slightly away from the middle of the chromosome resulting into one shorter arm and one longer arm, is called -
(1) Submetacentric (2) Acrocentric
(3) Metacentric (4) Telocentric
BCL323

51. Read the following statements :-
(A) Both the chromatids of chromosome are joined by a structure called centromere
(B) During cell division spindle fibres are attached to satellite
(C) Telomeres are rich in Guanine base.
(D) Every chromosome essentially has a secondary constriction
Find out correct sequence of true (T) and false (F) statements
(1) T F T T (2) F T F T
(3) T F T F (4) T T F F
BCL324

52. A single human cell contain approximately ..... long thread of DNA distributed among its ..... chromosomes
(1) 2 cm, 46 (2) 2 meter, 46
(3) 2 meter, 23 (4) 2 cm, 23
BCL325

53. Which one of the following nuclear parts is correctly described ?
(1) Nucleolus = It is a site for protein synthesis
(2) Heterochromatin = This is dark stained, thick part of chromatin
(3) Perinuclear space = Contain nucleolus and chromatin
(4) Centriole = Play an important role in cell division in plants.
BCL326

54. If volume of the cell is filled with semi-fluid matrix called cytoplasm, what kind of cell is it?
(1) Eukaryotic cells (2) Prokaryotic cells
(3) Both (1) and (2) (4) None of these
BCL327

55. Which of the following is incorrect matching?
(1) Round and biconcave – RBC
(2) Amoeboid – WBC
(3) Elongated – Tracheid
(4) Long and narrow - Nerve cells
BCL328

56. Which of the following statements is not true?
(1) Response to gram stain is due to the cell envelope.
(2) The cell envelope acts as a single protective unit.
(3) The glycocalyx layer is similar in all the bacteria.
(4) Slime layer is loose sheath of glycocalyx.
BCL329

57. Select the incorrect statement:
(1) Glycocalyx differs in composition and thickness among different bacteria.
(2) All organisms are made of cells or aggregates of cells.
(3) ER helps in synthesis of proteins, lipoproteins, and glycogen.
(4) Cells of all living organisms have nucleus.
BCL330

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58. Select the incorrect statement:
(1) The chloroplast contains chlorophyll and carotenoid pigments.
(2) Chromoplast contains water soluble carotenoid pigments like carotene and xanthophylls.
(3) Plastid is easily observed under microscope.
(4) Chloroplast is a double membrane bound organelle.

BCL331

59. Find out the incorrect statement:
(1) Centrioles form basal bodies of cilia or flagella.
(2) Centriole gives rise to spindle fibres at the time of cell division in animal cells.
(3) Centrosome is an organelle usually containing two cylindrical structures called centrioles.
(4) Peripheral fibril of centriole is doublet.

BCL332

60. Which of the following is true about the internal structure of axoneme?
(1) Central sheath is connected to one of the tubules of each peripheral doublet by a radial spoke.
(2) Axoneme is not covered by plasma membrane at all.
(3) There are only radial spokes and peripheral doublets found in axoneme.
(4) Peripheral doublets are not connected with each other.

BCL333

61. **Assertion:** Genetic material in prokaryotic is said to be naked.
Reason: Genetic material in prokaryotic is not associated with histone proteins and absence of nuclear envelope around it.
(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.

BCL334

62. **Assertion:** In lipid bilayer in membrane, lipid is arranged with polar head towards outer side and the hydrophobic tail towards the inner part
Reason: This ensures that the nonpolar tail of saturated hydrocarbon is protected from aqueous environment.
(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.

BCL335

63. Which of the following is not function of endoplasmic reticulum?
(1) Formation of acrosome
(2) Detoxification
(3) Mechanical support
(4) Formation of nuclear membrane

BCL336

64. Peri-plastidial space is found in:
(1) Between two membranes of a chloroplast
(2) Between two thylakoids
(3) Between two inter-grana lamellae
(4) Between two chloroplasts of a cell

BCL337

65. Read following statements:
(A) Larger and more numerous nucleoli are present in cells actively carrying out protein synthesis.
(B) The content of nucleolus is continuous with the rest of the nucleoplasm as it is a membrane bound structure.
(C) Nucleolus is main site for synthesis of ribosomal protein.
(D) Nuclear envelope consists of two parallel membranes with a space between called peri-mitochondrial space.
(E) The centrioles form the basal body of cilia or flagella.
How many above statements are not correct?
(1) 5 (2) 3 (3) 4 (4) 1

BCL338

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66. Which one of the following is correct?
(a) Nuclear matrix or nucleoplasm contains nucleolus and chromatin.
(b) Outer nuclear membrane usually remains continuous with the ER and also bears ribosomes on it.
(c) Small and lesser number of nucleoli are present in cells actively carrying out protein synthesis.
(d) Perinuclear space forms a barrier between the materials present inside the nucleus and that of the cytoplasm.
(1) a and b (2) a, b and d
(3) a, b, c and d (4) b, c and d

BCL339

67. On the basis of ease of extraction membrane proteins are of two type - extrinsic and intrinsic. How can intrinsic protein be separated from membrane?
(1) By treating with mild detergent
(2) By using gram stain
(3) By using light microscope
(4) By using electron microscope

BCL340

68. Enzymes of Calvin cycle are located in:
(1) Mitochondria matrix
(2) Stroma of chloroplast
(3) Granum of Leucoplast
(4) All the above

BCL341

69. Match column I with column II & select the correct option: -

Column I	Column II
(a) Contractile vacuole	(i) Osmo regulation
(b) Mitochondria	(ii) ATP synthesis
(c) Lysosomes	(iii) Glycosylation
(d) Golgi body	(iv) Hydrolytic enzymes

(1) (a) - (i), (b) - (ii), (c) - (iii), (d) - (iv)
(2) (a) - (iv), (b) - (iii), (c) - (ii), (d) - (i)
(3) (a) - (i), (b) - (ii), (c) - (iv), (d) - (iii)
(4) (a) - (iii), (b) - (iv), (c) - (i), (d) - (ii)

BCL342

70. How many cell organelles in the list given below are surrounded by single membrane?
Mitochondria, Lysosome, Chloroplast, Golgi apparatus, Nucleus, Nucleolus, Endoplasmic reticulum, Ribosome, Peroxisome, Centriole
(1) Four (2) Six
(3) Nine (4) Three

BCL343

71. Regarding differences between animal & plant cells, which of the following statement is correct?
(1) Animal cells contain centrioles & cell wall, plant cells do not.
(2) Animal cells contain centrioles & plant cells contain plastids, cell wall and large central vacuole.
(3) Animal cells do not contain centrioles, but plant cells contain centrioles.
(4) Animal cells contain plastids, while plants cells do not contain plastids.

BCL344

72. Read the following statements with respect to mitochondria:-
(i) It divides and increases its number by binary fission.
(ii) During the early course of evolution these were prokaryotes.
(iii) Its matrix performs electron transport chain, and the inner membrane performs Kreb's cycle.
(iv) These are fully autonomous organelles.
(v) Its matrix possess 70S ribosomes, RNA and linear dsDNA.
How many of the above statements are correct?
(1) Three (2) Two
(3) Four (4) Five

BCL345

73. What is the correct arrangement of Lipid molecules in the cell membrane

(1) Polar tail - inner side,
Non-polar head - outer side

(2) Non-polar head – Outside,
Polar tail - Inner side

(3) Polar tail – Outside,
Non-polar head - Inner side

(4) Polar head - Outside,
Non-polar tails - Inner side

BCL346

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	3	3	1	1	4	4	1	4	3	3	4	4	2	3
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	4	4	4	2	2	2	3	3	4	4	1	4	3	4	4
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	3	3	2	2	2	2	4	2	3	3	2	2	2	3	1
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	4	1	2	3	1	3	2	3	3	4	3	4	2	4	1
Question	61	62	63	64	65	66	67	68	69	70	71	72	73		
Answer	1	1	1	1	2	1	1	2	3	1	2	2	4		

Cell Cycle and Cell Division

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1. Which of the following is the most active phase of cell cycle ?
(1) Metaphase (2) Interphase
(3) Anaphase (4) Prophase
BCD183

2. In plant cells, during the S-phase -
(1) DNA replication occurs in nucleus and centrioles duplicates in the cytoplasm
(2) DNA replication occurs in nucleus
(3) Centrioles duplicates in the cytoplasm
(4) Centrioles duplicates in the nucleus and DNA replication occurs in the cytoplasm.
BCD184

3. If the cell had 2n number of chromosomes at G_1, after S phase the number of chromosome will be-
(1) 2n (2) 4n (3) n (4) 8n
BCD185

4. During cell division, in cell -
(1) Only DNA replication takes place
(2) Only cell growth takes place
(3) Both DNA replication and cell growth takes place.
(4) Neither DNA replication nor cell growth takes place
BCD186

5. Mitosis occurs -
(1) Only in diploid somatic cells
(2) Only in diploid reproductive cells
(3) In both diploid and haploid somatic cells
(4) Only in haploid somatic and reproductive cells
BCD187

6. Cells at the end of prophase, when viewed under the microscope do not show -
(1) Nucleolus
(2) Golgi complex
(3) Endoplasmic reticulum
(4) All of the above
BCD188

7. During cytokinesis in plant cells, plate formation:-
(1) Starts at periphery of the cell and grows inward
(2) Starts at centre of the cell and grows inward
(3) Starts at centre of the cell and grows outward
(4) Starts at periphery of the cell and grows outward
BCD189

8. If the volume of cell is 110 cc and volume of cytoplasm is 100 cc then what will be the KI (Karyoplasmic index) of this cell ?
(1) 10 (2) 1
(3) 0.1 (4) 100
BCD190

9. During cell cycle (choose correct one) :
(1) Cytoplasmic increase is a continuous process and nuclear DNA replication occurs in specific stage.
(2) Nuclear DNA replication is a continuous process and cytoplasmic increase occurs in specific stage
(3) Both cytoplasmic increase and nuclear DNA replication are continuous process
(4) Both cytoplasmic increase and nuclear DNA replication occurs in specific stage
BCD191

10. Which of the following statements is not correct?
(1) Prophase is marked by the intiation of condensation of chromosomal material.
(2) Spindle fibres attach to kinetochores of chromosomes in metaphase.
(3) In anaphase when daughter chromosomes move towards opposite poles, their centromeres remain directed towards spindle equator.
(4) Chromosomes lost their identity as discrete element in telophase.
BCD192

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11. Meiosis involves -
(1) Two cycles of each, nuclear division and DNA replication.
(2) One cycle of each, nuclear division and DNA replication.
(3) One cycle of nuclear division and two cycles of DNA replication
(4) Two cycles of nuclear division and one cycle of DNA replication.

BCD193

12. Both the chromatids of each chromosome become distinct in which phase of meiosis-I ?
(1) Metaphase-I (2) Pachytene
(3) Diakinesis (4) Anaphase-I

BCD194

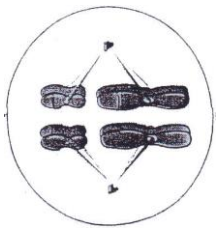
13. Which of the following stages is characterised by the appearance of the sites at which crossing over occurs ?
(1) Zygotene (2) Leptotene
(3) Metaphase-I (4) Pachytene

BCD195

14. Which of the following stages represents the transition to metaphase ?
(1) Diakinesis (2) Diplotene
(3) Prophase-I (4) Leptotene

BCD196

15. Identify the given diagram and answer the question given below :-



Which one statement is true for above diagram?
(1) The bivalent (homologous chromosomes) align on the equatorial plate
(2) Sister chromatids separate
(3) Nuclear membrane reappear
(4) Homologous chromosomes separate

BCD197

16. Anaphase-I is characterised by -
(1) Disjunction of homologous chromosomes
(2) Division of centromere
(3) Doubling the number of chromosomes in cell
(4) Doubling the amount of DNA in cell.

BCD198

17. Meiosis-I results in formation of _____ of cells with _____ of chromosomes in each cell as compare to mother cell.
(1) Tetrad, half number
(2) Dyad, half number
(3) Tetrad, same number
(4) Dyad, double number

BCD199

18. If a cell has 10 pg (pico gram) DNA in G_1 stage, what will be the amount of DNA in each daughter cell after meiosis-I ?
(1) 10 pg (2) 20 pg
(3) 40 pg (4) 5 pg

BCD200

19. Meiosis maintains the specific number of chromosomes from -
(1) Cell to cell
(2) Organism to organism
(3) Generation to generation
(4) Organ to organ

BCD201

20. After meiosis-I if daughter cell has 6 double chromatid chromosomes then what will be the number of chromosomes in each daugther cell after meiosis-II ?
(1) 6 (each with two chromatids)
(2) 12 (each with single chromatid)
(3) 6 (each with single chromatid)
(4) 12 (each with two chromatids)

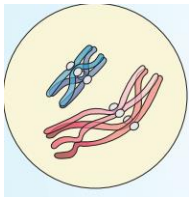
BCD202

21. **Assertion:** DNA synthesis occurs in G_1 and G_2 periods of cell cycle.
Reason: During G_1 and G_2 phases, the DNA content becomes double.
(1) Both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) Both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) Assertion is true but the reason is false.
(4) Both the assertion and reason are false.

BCD203

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22. Find the correct statement w.r.t the given phase.



(1) Leptotene – chromosome compaction continues.
(2) Zygotene - bivalent becomes distinct and clearly appears as tetrads.
(3) Diplotene - terminalization of chiasmata.
(4) Pachytene – Recombination between sister chromatids.

BCD204

23. Mitosis accomplishes not only the segregation of ___A___ chromosomes into ___B____ nuclei (karyokinesis), but the cell itself is divided into two daughter cells by the separation of cytoplasm called cytokinesis at the end of which cell ___C___ gets completed.
Fill in the blanks with appropriate words.
(1) B – duplicated, C – division
(2) A – duplicated, B – parent
(3) A – duplicated, C – division
(4) A – daughter, C – cycle

BCD205

24. Match the phases in column I with the respective characters in column II.

Phases	Characters of phases
a. Interkinesis	i. Bivalents on equatorial plate
b. Anaphase II	ii. Centromere division
c. Metaphase I	iii. Homologous chromosomes separate
d. Anaphase I	iv. Followed by prophase II

(1) a – iv, b – ii, c – i, d - iii
(2) a – iv, b – iii, c – ii, d – i
(3) a – iii, b – iv, c – i, d – ii
(4) a – iv, b – i, c – iii, d – ii

BCD206

25. Cells enter G_0 phase when they:
(1) Exit the G_2 phase.
(2) Exit the G_1 phase.
(3) Suspend the G_1 phase.
(4) Suspend the G_1 phase.

BCD207

26. All are correct for Metaphase I except:
(1) The bivalent chromosomes align on the equatorial plate.
(2) The microtubules from opposite poles of the spindle get attached to the kinetochores of sister chromatids.
(3) The microtubules from the opposite poles of the spindle attach to the kinetochore of homologous chromosomes.
(4) No statement is incorrect.

BCD208

27. **Statement A -** Four haploid cells are formed at the end of meiosis I.
Statement B - Nucleolus, Golgi complex and ER disintegrate during telophase.
(1) Only statement A is correct.
(2) Only statement B is correct.
(3) Both statement A and B are correct.
(4) Both statement A and B are incorrect.

BCD209

28. Select the incorrect statement from the following:
(1) In animals, mitotic cell division is only seen in the diploid somatic cells.
(2) Plants can show mitotic division in both haploid and diploid cells.
(3) In an adult's heart, the cells do not divide.
(4) All organisms start their life cycle from multiple cell.

BCD210

29. Select the total number of correct statements:
I. Cell-plate formation occurs in plant cell during cytokinesis.
II. During cytokinesis mitochondria and plastid gets distributed between two daughter cells in mitosis.
III. Liquid endosperm in coconut is syncytium.
IV. Furrow formation occurs in Animal cell during cytokinesis.
(1) 1 (2) 2 (3) 3 (4) 4

BCD211

30. Cross-like configuration when non-sister chromatids of a bivalent comes in contact during the first meiotic division are:

(1) Chiasmata

(2) Bivalents

(3) Chromomeres

(4) Centromeres

BCD212

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	2	1	3	3	4	3	3	1	3	4	2	4	1	1
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	1	2	1	3	3	4	1	3	1	2	2	4	4	4	1

Photosynthesis in Higher Plants

1. Photosynthesis is a :-
(1) Photochemical oxidation
(2) Physicochemical reduction
(3) Physicochemical oxidation reduction
(4) Photobiochemical oxidation
BPP258

2. When KOH is added in an experiment showing photosynthesis, what will be effect on it ?
(1) Rate will increase
(2) Rate will decrease
(3) Rate will remains same
(4) Rate will enhance
BPP259

3. Concept of gaseous exchange during photosynthesis was established by the experiment using :-
(1) O^{18} (2) Cladophora
(3) Flash light (4) Bell jar
BPP260

4. Which one carries out photosynthesis without evolution of oxygen ?
(1) Green plants
(2) Purple and green sulphur bacteria
(3) BGA
(4) Fungi
BPP261

5. $2H_2A + CO_2 \xrightarrow{\text{Light}} 2A + CH_2O + H_2O$
In above equation if we replace 'A' with oxygen [O], then this equation will correct for which of the following organisms ?
(1) Purple sulphur bacteria
(2) Green sulphur bacteria
(3) Cyanobacteria
(4) All of the above
BPP262

6. According to division of labour, the membrane system in chloroplast is responsible for :-
(1) Trapping the light energy
(2) Synthesis of ATP
(3) Synthesis of NADPH + H^+
(4) All of the above
BPP263

7. Chlorophyll 'b', as compared to chlorophyll 'a', has :-
(1) One more hydrogen and two less oxygens
(2) Two more hydrogens and two more oxygens
(3) Two less hydrogens and one more oxygen
(4) One less hydrogen and one less oxygen
BPP264

8. The precursor substance for chlorophyll synthesis in plants, is :-
(1) Succinyl CoA (2) CO_2
(3) Phycobilin (4) Lutein
BPP265

9. We can conclude that chlorophyll 'a' is the chief pigment of photosynthesis, because :-
(1) It is most abundant photosynthetic pigment
(2) Action spectrum of photosynthesis resembles maximum with the absorption spectrum of chlorophyll 'a'
(3) Chlorophyll 'a' has unique structure to absorb UV
(4) No other pigment can absorb light
BPP266

10. In experiment, by the help of which first action spectrum of photosynthesis was described, bacteria were used :-
(1) to absorb most of the blue light
(2) to detect the sites of O_2 evolution
(3) to absorb most of the red light
(4) to absorb most of the blue and red light
BPP267

11. Each photosystem has all the pigments (except one molecule of ____) forming a light harvesting system also called antennae.
Choose correct word from the following for above blank :-
(1) Xanthophyll (2) Carotene
(3) Chlorophyll 'a' (4) Chlorophyll 'b'
BPP268

12. In photosynthesis, the light energy is utilized in:
(1) Converting ATP into ADP
(2) Converting CO_2 into O_2
(3) Converting $C_6H_{12}O_6$ into CO_2
(4) Converting ADP into ATP
BPP269

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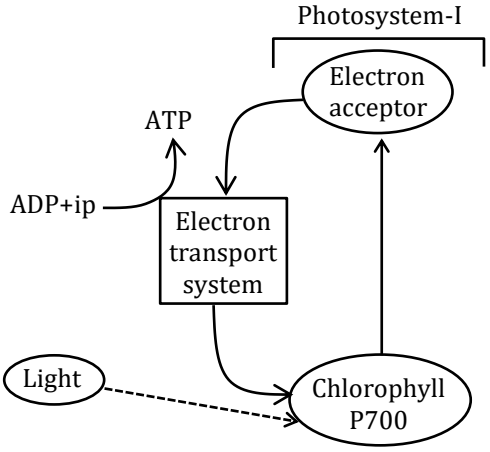
13. Splitting of water during photosynthesis occurs on the :-
(1) inner side of stroma lamellae
(2) outer side of stroma lamellae
(3) outer side of grana lamellae
(4) inner side of grana lamellae

BPP270

14. Which of the following does not participate when the light reaction synthesize only ATP?
(1) PS-II (2) PS-I
(3) Ferredoxin (4) Plastocyanin

BPP271

15. Which of the following locations shows only the process showed in above diagram ?



(1) Periplastidial space
(2) Lamellae which form stacks in the chloroplast
(3) The fluid part of the chloroplast
(4) Lamellae which connects the grana

BPP272

16. Which of the following is hydrogen carrier in both cyclic and non cyclic photo-phosphorylation?
(1) Pheophytin (2) Plastoquinone
(3) Cytochrome f (4) FRS

BPP273

17. During photosynthesis :-
(A) ATP is formed by only cyclic photophos-phorylation
(B) NADPH + H^+ are formed by only noncyclic photophosphorylation
(C) PQ is helpful in creating proton gradient
From the above statements :-
(1) A and B are correct
(2) A and C are correct
(3) B and C are correct
(4) All statements are correct

BPP274

18. During creation of proton gradient in chloroplast:-
(1) concentration of protons increases in stroma
(2) concentration of protons decreases in thylakoid lumen
(3) pH decreases in thylakoid lumen
(4) pH decreases in stroma

BPP275

19. Which of the following components are required for chemiosmosis ?
(1) A membrane, a proton pump, an electron gradient and ATPase
(2) F_0–F_1 particle, ATP, a membrane and proton gradient
(3) A proton pump, a membrane, a proton gradient and ATP synthase
(4) A proton pump, a membrane, a electron gradient and PS-II

BPP276

20. Proton gradient across the thylakoid membrane is broken down due to the movement of protons :-
(1) from stroma to thylakoid lumen with the help of plastoquinone
(2) from thylakoid lumen to stroma with the help of plastoquinone
(3) from thylakoid lumen to stroma through the transmembrane channel of the CF_0 of the ATP synthase
(4) from stroma to thylakoid lumen through the transmembrane channel of the CF_0 of the ATP synthase

BPP277

21. By the use of C^{14} radioisotope, which fact of photosynthesis was established ?
(1) Carbohydrates are synthesised from CO_2
(2) Green colour is essential in photosynthesis
(3) Dark reaction is associated with light reaction
(4) O_2 is released from hydrogen donor

BPP278

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22. Which of the following is present in Calvin cycle?
(1) Photophosphorylation
(2) Oxidative carboxylation
(3) Oxidative phosphorylation
(4) Reductive carboxylation

BPP279

23. Number of NADPH + H^+ molecules used during the conversion of CO_2 into one molecule of glucose through C_3 cycle, is :-
(1) 4 (2) 6 (3) 12 (4) 18

BPP280

24. For the synthesis of two molecules of PGAL, how many ATP and NADPH + H^+ molecules are required in Calvin cycle ?
(1) 18 ATP and 12 NADPH + H^+
(2) 3 ATP and 2 NADPH + H^+
(3) 2 ATP and 2 NADPH + H^-
(4) 12 ATP and 12 NADPH + H^-

BPP281

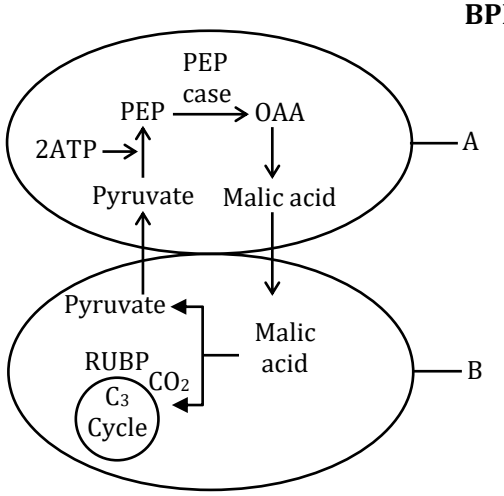
25. C_4 cycle takes place in :-
(1) Wheat (2) Sugarcane
(3) *Chlorella* (4) Barley

BPP282

26. In Hatch & Slack pathway decarboxylation of C_4 acid and regeneration of phosophoenol pyruvate occurs respectively in :-
(1) mesophyll cell and bundle sheath cell
(2) bundle sheath cell and mesophyll cell
(3) bundle sheath cell and bundle sheath cell
(4) mesophyll cell and mesophyll cell

BPP283

27.



Select correct option regarding Hatch and Slack pathway shown in above figure :-
(1) A = Impermeable to gaseous exchange
B = Permeable to gaseous exchange
(2) A = Impermeable to gaseous exchange
B = Impermeable to gaseous exchange
(3) A = Permeable to gaseous exchange
B = Impermeable to gaseous exchange
(4) A = Permeable to gaseous exchange
B = Permeable to gaseous exchange

BPP284

28. Mesophyll chloroplast of which plant synthesizes hexose by C_3 cycle ?
(1) C_3 plant
(2) Wheat
(3) Both (1) and (2)
(4) C_4 plant

BPP285

29. For the fixation of 12 molecules of CO_2 in Rice and Maize, how many molecules of ATP are required respectively ?
(1) 18 and 30 (2) 36 and 60
(3) 3 and 5 (4) 18 and 60

BPP286

30. Which of the following is not correct regarding differences between C_3 & C_4 plants?

	Characteristic	C_3 Plant	C_4 Plant
(1)	Does plant have RuBisCO	yes	yes
(2)	Number of carbons in the primary CO_2 acceptor	3	4
(3)	Cell type in which calvin cycle takes place	Mesophyll	Bundle sheath
(4)	Cell type in which the initial carboxylation occurs	Mesophyll	Mesophyll

BPP287

31. Which of the following factors favour the oxygenase activity of RuBisCO ?
(1) $O_2\downarrow$, $CO_2\uparrow$, Light intensity$\downarrow$, temperature$\uparrow$
(2) $CO_2\uparrow$, $O_2\downarrow$, Light intensity$\uparrow$, temperature$\downarrow$
(3) $O_2\downarrow$, $CO_2\downarrow$, Light intensity$\downarrow$, temperature$\downarrow$
(4) $CO_2\downarrow$, $O_2\uparrow$, Light intensity$\uparrow$, temperature$\uparrow$

BPP288

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32. Which of the following is correct for photorespiration ?
(1) RuBP + $O_2 \rightarrow$ 2, Phosphoglycerate
(2) RuBP + $CO_2 \rightarrow$ 2, Phosphoglycerate
(3) RuBP + $CO_2 \rightarrow$ Phosphoglycerate + Phosphoglycolate
(4) RuBP + $O_2 \rightarrow$ Phosphoglycerate + Phosphoglycolate

BPP289

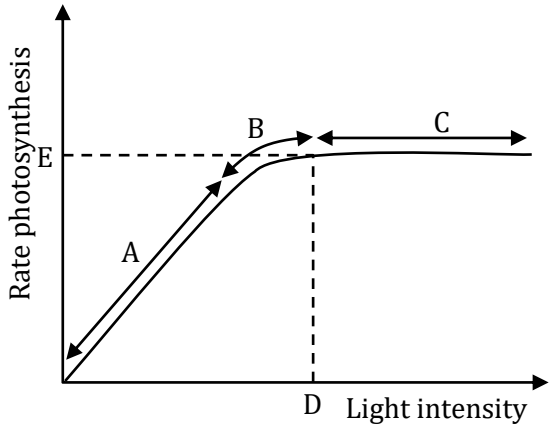
33. Which of the following is correct for photorespiration?
(1) Synthesis of ATP with release of CO_2
(2) Synthesis of sugar and NADPH both
(3) Release of CO_2 and utilisation of ATP
(4) Synthesis of NADPH but not of sugar

BPP290

34. Which of the following factors affecting photosynthesis is dependent on the genetic predisposition of the plant?
(1) Orientation of leaves
(2) Light
(3) Temperature
(4) Carbondioxide concentration

BPP291

35. Identify the incorrect statement regarding the following graph :-



(1) B $\rightarrow$ Here light Intensity is not a limiting factor
(2) C $\rightarrow$ Here light intensity is a major limiting factor
(3) E $\rightarrow$ maximum rate of photosynthesis
(4) A $\rightarrow$ In this region light intensity is a limiting factor

BPP292

36. Under the normal condition, which one is the major limiting factor for photosynthesis ?
(1) CO_2 concentration
(2) Light
(3) Temperature
(4) Chlorophyll concentration

BPP293

37. CO_2 fertilising effect can be shown by :-
(1) Tomato (2) Sorghum
(3) Maize (4) Sugar cane

BPP294

38. Which of the following are affected by temperature?
(1) Light reactions of photosynthesis
(2) Dark reactions of photosynthesis
(3) Both light reactions and dark reactions of photosynthesis
(4) Neither light reactions nor dark reactions of photosynthesis

BPP295

39. The C_4 plants respond to ____ temperatures and show higher rate of photosynthesis and C_3 plants have a much ____ temperature optimum. Choose the correct words respectively for above blanks from the following options :-
(1) lower, lower (2) lower, higher
(3) higher, lower (4) higher, higher

BPP296

40. Water stress can affect the photosynthesis by:-
(1) reducing the CO_2 availability
(2) reducing the surface area of the leaves
(3) reducing the metabolic activities of leaves
(4) all of the above

BPP297

41. Who demonstrated the first action spectrum of photosynthesis ?
(1) Julius von Sachs
(2) Cornelius van Niel
(3) T.W. Engelmann
(4) Jan Ingenhousz

BPP298

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42. Identify the incorrect statement :-
(1) Plants restore to the air whatever breathing animals and burning candle remove.
(2) Photosynthesis only occur in blue and red wavelength of light.
(3) Oxygen released during photosynthesis comes from H_2O.
(4) Sunlight is essential to the plant process that somehow purifies the air.

BPP299

43. How many of the given statements is/are correct for chloroplast?
A. Shows division of labour
B. Can arrange themselves according to incident light
C. Has membrane system for trapping sunlight
D. Has light dependent, enzymatic reactions for CO_2 assimilation
(1) Four (2) Three (3) Two (4) One

BPP300

44. Chloroplast shows division of labour and it cannot be verified by the fact that :-
(1) There are thylakoids for light reaction.
(2) Light reaction is performed in grana and dark reaction in stroma.
(3) Light reaction is performed in stroma and dark reaction in grana.
(4) There are two photosystem to perform non-cyclic photophosphorylation.

BPP301

45. Accessory pigments of photosynthesis includes:
(1) Chl b (2) Xanthophyll
(3) Carotenoids (4) All are correct

BPP302

46. Match the column-I with column-II.

Pigment		Colour (in chromatogram)	
(A)	Chlorophyll a	(i)	Reaction centre
(B)	Chlorophyll b	(ii)	Accessory pigments
(C)	Xanthophylls	(iii)	Yellow-green

Choose the correct answer from the options given below :
(1) A-i, B-ii, C-iii (2) A-i, B-iii, C-ii
(3) A-ii, B-i, C-iii (4) A-iii, B-ii, C-i

BPP303

47. Along with chlorophyll-a, accessory pigments are also involved in photosynthesis. This can be concluded by the fact that :-
(1) There is no one to one overlap between the absorption spectrum of chlorophyll-a and the action spectrum of photosynthesis.
(2) The reaction centre is different in photosystem-I and photosystem-II.
(3) Most of the photosynthesis takes place in the blue and red regions of the spectrum
(4) Accessory pigments protect chl-a from photooxidation.

BPP304

48. Which of the following is not the raw material for dark reaction ?
(1) ATP (2) NADPH
(3) CO_2 (4) Glucose

BPP305

49. In PS I, the reaction centre of Chl a has absorption peak at ____; whereas in PS II, the reaction centre of Chl a has absorption maxima at ___ :-
(1) 700 nm, 680 nm (2) 680 nm, 700 nm
(3) 400 nm, 500 nm (4) 700 nm, 800 nm

BPP306

50. Light reaction/photochemical phase includes
(1) Absorption of light, water splitting, and O_2 release
(2) ATP and $NADPH_2$ formation
(3) CO_2 fixation and glucose formation
(4) Both (1) and (2)

BPP307

51. How many O_2 is evolved for the production of 1 molecule of glucose :-
(1) 6 (2) 8
(3) 10 (4) 12

BPP308

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52. The electron flow from PSII to NADP via ETS and PSI follows :-
(1) Uphill → Downhill → Uphill → Downhill
(2) Downhill → Uphill → Downhill → Uphill
(3) Downhill → Uphill → Uphill → Downhill
(4) Uphill → Downhill → Downhill → Uphill

BPP309

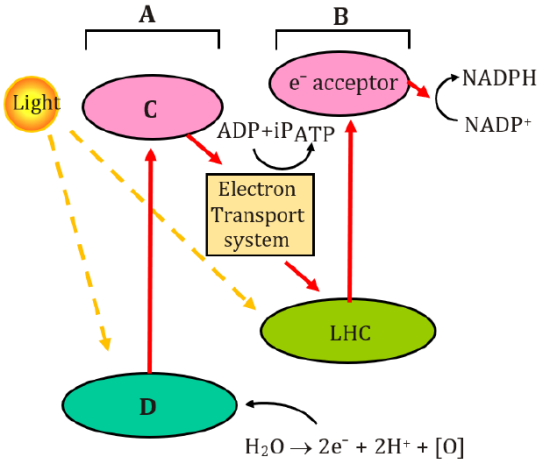
53. Water harvesting complex in PS II is located on :-
(1) Inner side of thylakoid membrane
(2) Outer side of thylakoid membrane
(3) Lumen of thylakoid membrane
(4) Stroma lamellae

BPP310

54. Which of the following conditions not supports in the photorespiration :
(1) More oxygen and less carbon dioxide
(2) Less oxygen and more carbon dioxide
(3) More temperature
(4) More light intensity

BPP311

55. Which of the following is not correct information for the given figure :-



(1) D is made up of hundreds of pigment molecule bound to protein.
(2) C is a part of photosystem I.
(3) B excited when receive red light of 700 nm wavelength.
(4) Whole scheme of transfer of electrons starting from (A).

BPP312

56. Photolysis of each water molecule in light reaction will yield :-
(1) 2 electrons and 2 protons, O_2
(2) 4 electrons and 4 protons, ½ O_2
(3) 4 electrons and 4 protons, O_2
(4) 2 electrons and 2 protons, ½ O_2

BPP313

57. Select the odd one with respect to requirements of chemiosmosis :-
(1) Electron pump (2) ATP synthase
(3) Proton gradient (4) Membrane

BPP314

58. Assume that thylakoid membrane is punctured so that the interior of the thylakoid is no longer separated from the stroma. This damage will have the most direct effect on which of the following process?
(1) Synthesis of ATP
(2) The absorption of light energy by chlorophyll
(3) Flow of electrons
(4) Oxidation of NADPH + H^+

BPP315

59. In which of the following cells of a C_4 leaf would you find the highest concentration of enzyme RuBisCO?
(1) Epidermal cell
(2) Mesophyll cell
(3) Bundle sheath cell
(4) Both (2) and (3)

BPP316

60. Connecting link between light phase and dark phase of photosynthesis.
(1) Only ATP
(2) Only NADPH(H^+)
(3) Only NADH + H^+
(4) ATP & NADPH(H^+)

BPP317

61. **A** + $CO_2 \xrightarrow{B}$ 2x(3-PGA):
Here A and B respectively?
(1) Malic acid, PEP carboxylase
(2) RuBisCO, PEP
(3) RuBP, RuBisCO
(4) Glucose, Hexokinase

BPP318

62. Which one is correct for C_4 Plants?
(1) PEP, 4 carbon molecule presents in mesophyll cells.
(2) Mesophyll cells lack RuBisCO enzyme.
(3) Aspartic acid or malic acid is transported to the mesophyll cells.
(4) CO_2 of mesophyll cells from bundle sheath enter the Calvin cycle.

BPP319

63. What does not occur in photorespiration?
(1) Phosphoglycolate formation
(2) Decrease CO_2 fixation
(3) Synthesis of ATP
(4) All the above

BPP320

64. Which statement is **incorrect** ?
(1) C_3 plants respond to higher temperature show higher photosynthesis rate while C_4 plants have lower optimum temperature.
(2) Tropical plants have higher temperature optimum than the plants adapted to temperate climate.
(3) Light reaction is less temperature sensitive than dark reaction.
(4) The effect of water as a factor is more through its effect on plant, rather than directly on photosynthesis.

BPP321

65. Identify the incorrect statement :-
(1) Increase in CO_2 concentration up to 0.05 percent can cause on increase in CO_2 fixation rates.
(2) C_4 plants shows saturation at 360 ppm and C_3 at 450 ppm.
(3) C_3 plants respond to higher CO_2 concentration (more than 0.05) showing increased rates of photosynthesis.
(4) Green house crops allowed to grow in CO_2 enriched atmosphere that leads to higher yield.

BPP322

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	2	4	2	3	4	3	1	2	2	3	4	4	1	4
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	2	3	3	3	3	1	4	3	3	2	2	3	3	2	2
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	4	4	3	1	2	1	1	3	3	4	3	2	2	3	4
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	2	1	4	1	4	1	1	1	2	2	4	1	1	3	4
Question	61	62	63	64	65										
Answer	3	2	3	1	3										

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Respiration in Plants

1. Select the correct sequence of steps in glycolysis:-
(1) PGAL → 3-PGA → 1,3-BisPGA → PEP
(2) PGAL → 1,3 – Bis PGA → PEP → 3–PGA
(3) 1,3–Bis PGA → 3–PGA → 2–PGA → PEP
(4) PGAL → PEP → 1,3–Bis PGA → 2–PGA

BRP168

2. Which of the following is not an oxidation reaction ?
(1) 1, 3 Bis PGAL to 1, 3 Bis PGA
(2) 1, 3 Bis PGA to 3-PGA
(3) Pyruvic acid to acetyl CoA
(4) Malic acid to oxaloacetic acid

BRP169

3. In some biochemical reactions, two redox equivalents in the form of hydrogen atoms comes out from PGAL and transferred to NAD^+, site of this process is :-
(1) Stroma of chloroplast
(2) Cytoplasm
(3) Mitochondrial matrix
(4) Outer membrane of mitochondria

BRP170

4. Pace maker enzyme of EMP pathway is :-
(1) Hexokinase
(2) Pyruvate kinase
(3) Enolase
(4) Phosphofructokinase

BRP171

5. Before entering the TCA cycle, pyruvic acid converted into acetyl CoA. This process is :-
(1) Oxidative carboxylation
(2) Oxidative dehydration
(3) Oxidative decarboxylation
(4) Reductive phosphorylation

BRP172

6. Number of CO_2 molecules liberated during EMP pathway of respiration is :-
(1) Less than the CO_2 molecules liberated during link reaction
(2) More than the CO_2 molecules liberated during link reaction
(3) Equal to the CO_2 molecules liberated during link reaction
(4) Sometime less and sometime more than the CO_2 molecules liberated during link reaction

BRP173

7. Which one of the following is odd ?
(1) EMP pathway
(2) TCA cycle
(3) CA cycle
(4) Organic acid cycle

BRP174

8. The net yield of ATP, except substrate level phosphorylation, from Krebs cycle per glucose molecule is :-
(1) 12 (2) 24
(3) 22 (4) 36

BRP175

9. During the complete oxidation of one sucrose molecule, how many ATP are produced by $FADH_2$ of Krebs cycle ?
(1) 24 (2) 8
(3) 3 (4) 4

BRP176

10. By the aerobic respiration of one molecule of glucose, how many CO_2 molecules produced in the mitochondria but outside the TCA cycle ?
(1) 2 (2) 4
(3) 6 (4) 12

BRP177

11. Which of the following complexes of mitochondrial ETS contains two copper centres?
(1) Cytochrome C oxidase
(2) Succinate dehydrogenase
(3) NADH dehydrogenase
(4) Cytochrome bc_1

BRP178

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12. Choose correct answers regarding mitochondrial ETS from the following options for question (a) mobile electron carrier between complex-III & IV and (b) complex which does not pump proton (H^+) in intermembrane space:-
(1) a = UQ, b = complex-II
(2) a = Cytc, b = complex-IV
(3) a = Cytc, b = complex-II
(4) a = UQ, b = complex-III

BRP179

13. Among the 12 molecules of H_2O produced through ETS in respiration during oxidation of 1 glucose molecule, how many can be produced due to EMP pathway ?
(1) 2 (2) 6 (3) 4 (4) 12

BRP180

14. Which one of the following electron carrier of ETS is/are mobile in nature?
(1) UQ and cyt. b (2) only cyt. c
(3) cyt. c and UQ (4) only cyt. a_3

BRP181

15. Amino acids after deamination depending on their structure would enter the respiratory pathway :-
(1) at some stage within the Krebs cycle
(2) as pyruvate
(3) as acetyl CoA
(4) All of the above

BRP182

16. Pyruvic acid $\xrightarrow[\text{in cytoplasm}]{NADH_2,\ 'C'}$ _A'_'_+ B
Choose the correct combination :-
(1) A = Lactic acid, B = CO_2, C = NAD
(2) A = Ethanol, B = H_2O, C = NAD
(3) A = Ethanol, B = CO_2, C = NAD
(4) A = Acetyl CoA, B = CO_2, C = NAD

BRP183

17. Enzymes ____ and _____ catalyse the reactions in which pyruvic acid is converted to CO_2 and ethanol.
Choose correct option for above blanks respectively :-
(1) alcohol decarboxylase, alcohol dehydrogenase
(2) pyruvic acid decarboxylase, pyruvic acid dehydrogenase
(3) alcohol decarboxylase, pyruvic acid dehydrogenase
(4) pyruvic acid decarboxylase, alcohol dehydrogenase

BRP184

18. When a molecule of pyruvic acid is subjected to anaerobic oxidation and form alcohol then indirectly it can be considered that there is :-
(1) Gain of 4 ATP (2) Loss of 3 ATP
(3) Loss of 6 ATP (4) Gain of 8 ATP

BRP185

19. When tripalmitin is the respiratory substrate:
(1) The amount of CO_2 released is less than the O_2 consumed
(2) The amount of CO_2 released is more than the O_2 consumed
(3) The amount of CO_2 released is equal to the O_2 consumed
(4) CO_2 is not released

BRP186

20. If 1 glucose = 600 Kcal, 1 ATP = 12 Kcal and ATP gain during complete oxidation of this glucose molecule is 30 ATP, than the efficiency of respiration is :-
(1) 40% (2) 50% (3) 60% (4) 100%

BRP187

21. Respiration is :-
(1) Anabolic + exergonic
(2) Amphibolic + exergonic
(3) Catabolic + endergonic
(4) Amphibolic + endergonic

BRP188

22. Plant can go along without respiratory organs because :-
(1) Stem respire at higher rate than leaf
(2) Large transport of gases is required from one part to another
(3) Do not have great demands for gaseous exchange
(4) More than one option is correct

BRP189

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23. Which one of the following is a common pathway in breakdown of glucose, fat and protein ?
(1) Glycolysis (2) Link reaction
(3) Krebs cycle (4) β-oxidation
BRP190

24. What is the incorrect statement regarding respiration :-
(1) During oxidation all the energy is released free into the cell in single step.
(2) Carbon skeleton producing during respiration is used as a precursors for biosynthesis.
(3) Energy released in respiration trapped as chemical energy in the form of ATP.
(4) The breaking of the C-C bond of complex compounds through oxidation leading to release of energy.
BRP191

25. Which enzymes are/is used in fermentation:-
(1) Pyruvic acid dehydrogenase
(2) Alcohol decarboxylase
(3) Lactate dehydrogenase
(4) Both (1) and (3)
BRP192

26. During glycolysis the number of ATP molecules utilised to change glucose into 2 phosphoglycerate are :-
(1) 4 (2) 3 (3) 2 (4) 1
BRP193

27. Phosphofructokinase (PFK) is an allosteric enzyme that catalyses the conversion of Fructose-6-phosphate to Fructose 1, 6-bisphosphate. In the presence of O_2, an decrease in the amount of ATP in a cell would be expected to :-
(1) Inhibit the enzyme, thus slow the rate of glycolysis
(2) Activate the enzyme, thus slow the rate of glycolysis
(3) Inhibit the enzyme, thus increase the rate of glycolysis
(4) Activate the enzyme, thus increase the rate of glycolysis
BRP194

28. End product of link reaction is
(1) Citric acid (2) Glyceraldehyde
(3) Acetyl CoA (4) Pyruvic acid
BRP195

29. Net gain ATP from one molecule of glucose during glycolysis or EMP pathway :-
(1) 38 ATP (2) 4 ATP
(3) 8 ATP (4) 30 ATP
BRP196

30. In eukaryotic respiration, ATP synthase complex is present in :-
(1) Mitochondrial matrix
(2) Inner mitochondrial membrane
(3) Cytoplasm
(4) Peri mitochondrial space
BRP197

31. How many ATP will produced during the production of 1 molecule of acetyl CoA from one molecule of glucose ?
(1) 11 ATP (2) 15 ATP
(3) 13 ATP (4) 6 ATP
BRP198

32. Four carbon intermediate molecule of TCA cycle is :-
(1) Citric acid
(2) Succinyl CoA
(3) α–Ketoglutaric acid
(4) Ribulose bis phosphate
BRP199

33. How many molecules of CO_2 are released in respiration per glucose molecule?
(1) Two (2) Three (3) Four (4) Six
BRP200

34. Although the aerobic process of respiration takes place only in the presence of oxygen, the role of oxygen is limited to the.....**A**.... of the process. yet, the presence of oxygen is vital, since it drives the whole process by.....**B**.... from the system :-
(1) A = ETS, B = removing electron
(2) A = Krebs cycle, B = Removing Hydrogen
(3) A = Terminal stage,
B = removing hydrogen
(4) A = Link reaction & Krebs cycle,
B = Removing electrons
BRP201

35. Match the column-I and column-II.

Column-I		Column-II	
(A)	Complex-I	(i)	Cytochrome bc_1
(B)	Complex-II	(ii)	Succinate dehydrogenase
(C)	Complex-III	(iii)	Cytochrome-c-oxidase
(D)	Complex-IV	(iv)	NADH-dehydrogenase

Choose the correct answer from the options given below :

(1) A - (ii), B - (iv), C - (iii), D - (i)

(2) A - (i), B - (iv), C - (ii), D - (iii)

(3) A - (iv), B - (ii), C - (i), D - (iii)

(4) A - (iv), B - (ii), C - (iii), D - (i)

BRP202

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	2	2	4	3	1	1	3	2	1	1	3	1	3	4
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	3	4	2	1	3	2	3	3	1	3	3	4	3	3	2
Question	31	32	33	34	35										
Answer	1	2	4	3	3										

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Plant Growth and Development

1. Which of the following is odd with respect to the parameters of growth measurements?
(1) Dry weight (2) Number of cells
(3) Fresh weight (4) Types of cells

BPD195

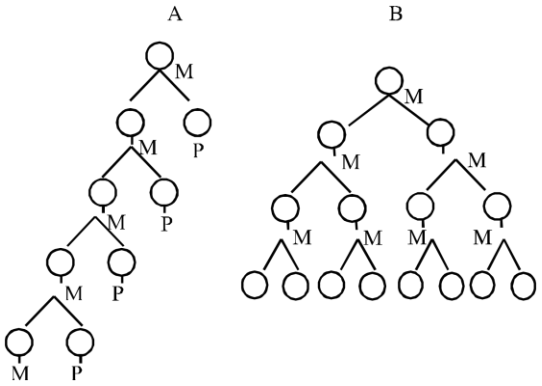
2. Which of the following is a correct match regarding growth measuring parameters?
(1) Maize root apical meristem - surface area
(2) Pollen tube - number of cells
(3) Watermelon - size of cells
(4) Dorsiventral leaf - length

BPD196

3. In which phase of growth thickening of cell-wall and modification of protoplasm takes place ?
(1) Meristematic phase
(2) Elongation phase
(3) Maturation phase
(4) Both (1) & (3)

BPD197

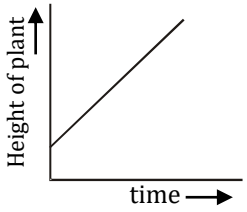
4.



P = Permanent cell, M = Meristematic cell
Which growth pattern is shown by above diagram ?
(1) A = Arithmetic ; B = Arithmetic
(2) A = Geometric ; B = Arithmetic
(3) A = Arithmetic ; B = Geometric
(4) A = Geometric ; B = Geometric

BPD198

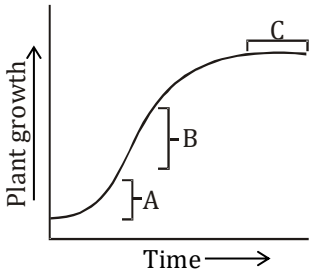
5.



Which is a correct mathematical expression of above graph ?
(1) $L_0 = L_t + rt$ (2) $W_1 = W_0e^{rt}$
(3) $W_0 = W_1e^{rt}$ (4) $L_t = L_0+rt$

BPD199

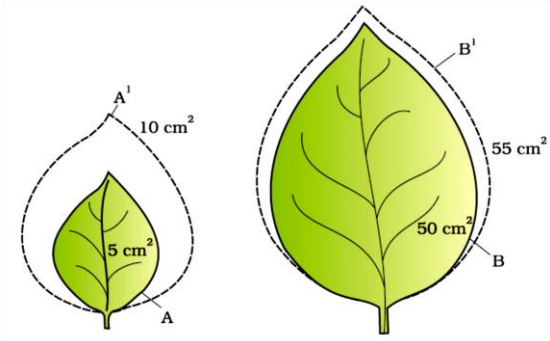
6.



Select correct option with respect to A, B and C for given graph :-
(1) A = lag phase ; B = stationary phase ; C = log phase
(2) A = stationary phase ; B = log phase ; C = lag phase
(3) A = log phase ; B = lag phase ; C = stationary phase
(4) A = lag phase ; B = log phase ; C = stationary phase

BPD200

7.



Find the correct statement with respect to above diagram:-
(1) Absolute growth rate in leaf A & leaf B is different
(2) Relative growth rate in leaf B is more than leaf A
(3) Relative growth rate in leaf A is more than leaf B
(4) Relative growth rate in leaf A and leaf B is equal

BPD201

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8. Which of the following is a correct sequence of the developmental process in a plant cell ?
(1) Differentiation → Senescence → Expansion → Division
(2) Division → Senescence → Differentiation → Expansion
(3) Division → Expansion → Differentiation → Senescence
(4) Expansion → Differentiation → Division → Senescence

BPD202

9. Which of the following is odd with respect to cause of plasticity ?
(1) Cotton (2) Coriander
(3) Buttercup (4) *Larkspur*

BPD203

10. Which of the following is most appropriate regarding plant growth or development?
(1) Growth = Development + Differentiation
(2) Developement = Growth – Differentiation
(3) Development = Differentiation – Growth
(4) Development = Growth + Differentiation

BPD204

11. Which of the following plant hormones helps to prevent leaf drop at early stages?
(1) Gibberellin (2) ABA
(3) Ethylene (4) Auxin

BPD205

12. Which of the following is odd with respect to the site of cytokinin synthesis ?
(1) Ripening fruits
(2) Root apices
(3) Young fruits
(4) Developing shoot buds

BPD206

13. Which of the following is not a correct match?
(1) $\frac{\text{Low CK}}{\text{High Auxin}}$ = root differentiation
(2) Chloroethylphosphonic acid = Source of ethylene
(3) Pomalin = CK + GA
(4) $\frac{\text{High CK}}{\text{Low Auxin}}$ = root differentiation

BPD207

14. Precursor for ethylene synthesis is ___'A'__ and for auxin synthesis is __'B'___ .
(1) A = tryptophan amino acid, B = Methionine amino acid
(2) A = Chlorophyll, B = Xanthophyll
(3) A = Adenine, B = Terpene
(4) A = Methionine amino acid, B = Tryptophan amino acid

BPD208

15. How many plant growth regulators, in the list given below, are synthetic (not naturally found in plants)?
Dormin, Naphthalene acetic acid, Zeatin, Kinetin, Indole-3-acetic acid, 2,4-dichloro-phenoxy acetic acid, Gibberellic acid
(1) Three (2) Four
(3) One (4) Five

BPD209

16. Match the following :-

Phytohormone		Function	
A	Auxin	a	Thinning in cotton
B	Gibberellin	b	Stress hormone
C	Cytokinin	c	Lateral growth of shoot
D	ABA	d	Bolting
E	Ethylene	e	Root initiation in stem cuttings

(1) A-a, B-b, C-c, D-d, E-e
(2) A-e, B-d, C-c, D-b, E-a
(3) A-c, B-a, C-d, D-e, E-b
(4) A-b, B-c, C-a, D-e, E-d

BPD210

17. Which of the following events in plants is/are affected by interaction of more than one plant growth regulators ?
(1) Dormancy in seeds / buds
(2) Abscission and senescence
(3) Apical dominance
(4) All of the above

BPD211

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18. The flowering in short day plants, does not take place, when :-
(1) The critical photoperiod is interrupted by dark
(2) An uninterrupted critical dark period is present
(3) The critical dark period is interrupted by red light
(4) The critical dark period is interrupted by far red light

BPD212

19. Which of the following is not correct ?
(1) Biennial plants are monocarpic plants.
(2) Biennial plants normally flower and die in the second season.
(3) Wheat, barley, rye are some of the common biennial plants
(4) Vernalisation prevents precocious reproductive development

BPD213

20. Which of the following can cause seed dormancy?
(1) Immature embryo
(2) Presence of chemical inhibitors
(3) Impermeable and hard seed coat
(4) All of the above

BPD214

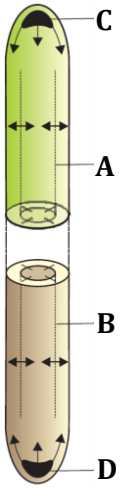
21. **Assertion (A) :** Primary growth of the plants contributes to the elongation of the plants along their axis.
Reason (R) : Root apical meristem and a shoot apical meristem are responsible for primary growth of the plants.
(1) Both Assertion and Reason are true.
(2) Both Assertion and Reason are false.
(3) Assertion is true but Reason is false.
(4) Assertion is false but Reason is true.

BPD215

22. **Assertion (A) :** The constantly dividing cells both at the root apex and the shoot apex. Show the meristematic phase of growth.
Reason (R) : The cells of this region are rich in protoplasm and are without nuclei.
(1) Both Assertion and Reason are true.
(2) Assertion is true but Reason is false.
(3) Assertion is false but Reason is true.
(4) Both Assertion and Reason are false.

BPD216

23. Choose the correct option for A, B, C or D :-



(1) A–Shoot apical meristem
(2) B–Vascular cambium
(3) C–Vascular cambium
(4) D–Lateral meristem

BPD217

24. Which of the following is correct about growth :-
(1) Most fundamental and inconspicuous character of living beings.
(2) Defined as reversible increase in size.
(3) Occur at expense of energy.
(4) Only present in multicellular organism.

BPD218

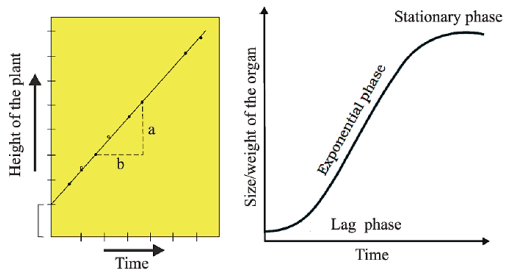
25. The stress hormone that helps in regulating abscission :-
(1) Auxins (2) Abscisic acid
(3) Ethylene (4) Cytokinin

BPD219

26. N^6- furfurylamino purine is :-
(1) Terpenes
(2) Adenine derivates
(3) Carotenoids
(4) Indole compound

BPD220

27. Recognise the figure and find out the correct labelling :-



(A) (B)

(1) A–Constant linear growth is geometric growth,
B–Relative growth rate is arithmetic growth

(2) A– Constant linear growth is arithmetic growth,
B– Relative growth rate is geometric growth

(3) A– Relative growth rate is geometric growth,
B– Constant linear growth is arithmetic growth

(4) A– Relative growth rate is arithmetic growth,
B– Constant linear growth is geometric growth

BPD221

28. During differentiation, cell undergoes structural changes in their :-

(1) Cell wall only

(2) Protoplasm only

(3) Protoplast only

(4) Cell wall and protoplasm

BPD222

29. Arithmetic growth includes :-

(1) Constant growth rate

(2) It is found in organism growth in natural environment.

(3) Efficiency index

(4) Relative growth rate

BPD223

30. (I) O_2 helps in releasing metabolic energy essential for growth activities

(II) Nutrients (micro and macro elements) are required by plants for the synthesis of protoplasm.

(III) Optimum temperature range for plant growth is needed

(IV) Environmental signal like light and gravity affect all phases of growth

Choose the **correct** option.

(1) All are correct

(2) I, II and III are correct

(3) I, II and IV are correct

(4) I and IV are correct

BPD224

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	4	3	3	3	4	4	3	3	3	4	4	1	4	4	1
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	2	4	3	3	4	1	2	2	3	2	2	2	4	1	2

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Sexual Reproduction in Flowering Plants

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1. Which of the following statement is correct ?
(1) *Papaver* is syncarpous
(2) *Papaver* is apocarpous
(3) Michelia is syncarpous
(4) Both (2) and (3)
BSR315

2. Anther of angiosperm generally composed of
(1) One sporangium (2) Two sporangia
(3) Three sporangia (4) Four sporangia
BSR316

3. Main function of middle layer is
(1) Mechanical support (2) Protection
(3) Storage of food (4) Dehiscence
BSR317

4. How many microspores are formed from 10 microspore mother cells ?
(1) 10 (2) 20 (3) 40 (4) 80
BSR318

5. Which type of dehiscence of anther is found in most of the angiosperms ?
(1) Transverse (2) Longitudinal
(3) Valvular (4) By apical pores
BSR319

6. Microscopic structure of the flower that contains vegetative cell is?
(1) Embryo sac
(2) Egg apparatus
(3) Mature pollen grain
(4) Nucellus
BSR320

7. Pollen grains of which plant cause allergy :-
(1) *Chenopodium* (2) *Parthenium*
(3) *Sorghum* (4) *All of these*
BSR321

8. Which pollen grain lose their viability within 30 minutes of their release.
(1) Rose (2) Wheat (3) Pea (4) Potato
BSR322

9. Progeny resulting in cross pollination :-
(1) Are always sterile
(2) Always show recessive character
(3) Show greater genetic variability
(4) Are homozygous
BSR323

10. In monocots male gametophyte is represented by :-
(1) Ovule (2) Nucellus
(3) Pollen grain (4) Megaspore
BSR324

11. Generally how many megaspores take part in the development of female gametophyte of angiosperms ?
(1) One (2) Two
(3) Three (4) Four
BSR325

12. Reduction division in angiosperm occurs at the time of ?
(1) Asexual reproduction
(2) Apomixis reproduction
(3) Vegetative reproduction
(4) Microspore and megaspore formation
BSR326

13. Which structure of ovule, takes part in storage of food ?
(1) Integuments (2) Nucellus
(3) Embryosac (4) Micropyle
BSR327

14. When ovule is straight so the funiculus, chalaza and micropyle lie in one vertical line, the ovule is said to be ?
(1) Anatropous ovule
(2) Amphitropous ovule
(3) Orthotropous ovule
(4) Hemitropous ovule
BSR328

15. Most common type of ovule of angiosperms is
(1) Orthotropous (2) Anatropous
(3) Hemitropous (4) Amphitropous
BSR329

16. *Polygonum* type of embryosac is -
(1) 8-celled (2) 4-celled
(3) 5-celled (4) 7-celled
BSR330

17. The ploidy level of secondary nucleus of embryosac is :-
(1) 2n (2) 3n (3) 4n (4) 5n
BSR331

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18. In which of the following plant, pollination takes place due to secretion of foul odour by the flowers?
(1) *Maize* (2) *Pinus*
(3) *Vallisneria* (4) *Rafflesia*
BSR332

19. Pericarp develops from -
(1) Septum (2) Ovary wall
(3) Funiculus (4) Integument
BSR333

20. Syngamy means.
(1) The fusion of cytoplasm of spores
(2) Delayed meiosis
(3) Delayed mitosis
(4) The fusion of male and female gametes
BSR334

21. The double fertilization is the only characteristic of :-
(1) Bryophytes (2) Pteridophytes
(3) Gymnosperm (4) Angiosperm
BSR335

22. The pollen tube was first discovered by :-
(1) P. Maheshwari (2) Strasburger
(3) Nawaschin (4) Amici
BSR336

23. The pollen tube enters the synergid through :
(1) Lateral walls
(2) Filiform apparatus
(3) Chalazal end
(4) Any of these regions
BSR337

24. Many pollen tubes are formed during the germination of pollen grains in -
(1) *Solanum* (2) *Salvia*
(3) *Capsella* (4) *Cucurbita*
BSR338

25. Polyembryony was first observed in ?
(1) *Mangifera* (2) *Cucurbita*
(3) *Citrus* (Orange) (4) *Euphorbia*
BSR339

26. The outermost layer of endosperm of maize grain is called -
(1) Scutellum (2) Perisperm
(3) Aleurone layer (4) Tunica
BSR340

27. Non endospermic seeds are present in :-
(1) Castor (2) Pea (3) Maize (4) Wheat
BSR341

28. When cells of nucellus in ovule make the embryo directly then it is ?
(1) Amphimictic (2) Parthenogenetic
(3) Both (1) & (2) (4) Apomictic
BSR342

29. The function of innermost layer of anther is:
(1) Dehiscence (2) Mechanical
(3) Protective (4) Nutritive
BSR343

30. In flowering plants, sporogenous tissue gives rise to –
(1) Only the wall of the sporangium
(2) Both wall and the sporogenous cells
(3) Microspore tetrad
(4) Only tapetum and sporogenous cells
BSR344

31. Which of the following statements are correct?
(1) Pollen grains are rich in nutrients.
(2) In some cereals like rice and wheat pollen grains lose viability within 30 minutes of their release
(3) In some members of rosaceae, Leguminosae and solanaceae, pollen grains maintain viability for months
(4) All of the above
BSR345

32. What would be the ploidy of cells of tetrad ?
(1) n (2) 2n (3) 3n (4) 4n
BSR346

33. Each lobe of a typical anther in angiosperm having two theca it is called :
(1) Monothecous (2) Dithecous
(3) Monosporangiate (4) Tetrasporangiate
BSR347

34. From outer to inner, what is the sequence of wall layers in anther lobes?
(1) Epidermis, middle layers, tapetum, endothecium
(2) Epidermis, endothecium, tapetum, middle layers
(3) Epidermis, endothecium, middle layers, tapetum
(4) Tapetum, middle layers, endothecium, epidermis
BSR348

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35. Which statement is correct.
(1) China rose plant have unisexual flowers
(2) China rose plant have bisexual flowers
(3) Maize have bisexual flowers
(4) Both (1) and (3)

BSR349

36. Sporopollenin is a :-
(1) Polymer of amines acids
(2) Polymer of carotenoids
(3) Polymer of NAG
(4) Polymer of ribose sugar

BSR350

37. Which plant is monoecious?
(1) *Marchantia* (2) Papaya
(3) *Chara* (4) Both (1) and (2)

BSR351

38. Read the following four statements (A-D) :-
(A) Both wind and water pollinated flowers are not very colourful and do not produce nectar.
(B) Wind pollinated flowers often have a single ovule in each ovary
(C) Wind pollinated pollens are light and non sticky
(D) *Vallisneria* is an aquatic plant but it is pollinated by insect
How many of the above statements are correct?
(1) One (2) Two (3) Three (4) Four

BSR352

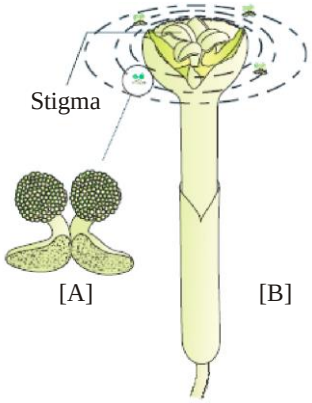
39. If pollen release and stigma receptivity within a flower is non synchronised then it is known as :-
(1) Dichogamy (2) Homogamy
(3) Heterostyly (4) Cleistogamy

BSR353

40. Which of the following statements are correct?
(1) The body of ovule fuses with funicle in the region called chalaza
(2) Polar nuclei are situated in the central cell above the egg apparatus
(3) Cleistogamous flowers are invariably autogamous
(4) Antipodal cells are present towards the micropylar end

BSR354

41. Following diagrams (A) & (B) represent ?



(1) (A) Female flower of *Hydrilla*
(B) Male flower of *Hydrilla*
(2) (A) Male flower of *Hydrilla*
(B) Female flower of *Hydrilla*
(3) (A) Female flower of *Vallisneria*
(B) Male flower of *Vallisneria*
(4) (A) Male flower of *Vallisneria*
(B) Female flower of *Vallisneria*

BSR355

42. In which set of plants an ovary contains one ovule.
(1) Papaya, water melon, orchids
(2) Wheat, paddy, mango
(3) Papaya, water melon, mango
(4) Papaya, water melon, wheat

BSR356

43. The functional megaspore generally undergoes:
(1) 2 meiotic divisions to form mature embryo sac
(2) 3 mitotic divisions to form mature embryo sac
(3) 2 mitotic divisions to form mature embryo sac
(4) 3 meiotic divisions to form mature embryo sac

BSR357

44. The number of ovules in an ovary may be
(1) One (2) Many
(3) Two (4) One to many

BSR358

45. Each ovule has one or two protective envelopes called
(1) Micropyle (2) Integuments
(3) Hilum (4) Chalaza

BSR359

46. Chalaza represents the
(1) Tip of the ovule (2) Base of the ovule
(3) Both (1) & (2) (4) Stalk of the ovule

BSR360

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47. Polar nuclei are situated in the central cell
(1) Below the egg apparatus
(2) Above the egg apparatus
(3) Below the antipodals
(4) Both (2) and (3)
BSR361

48. Pollen tube releases :-
(1) One male gamete in the cytoplasm of synergid & other in the cytoplasm of central cell
(2) Two male gametes in the cytoplasm of central cell
(3) Two male gametes in the cytoplasm of antipodal cell
(4) Two male gametes in the cytoplasm of a synergid
BSR362

49. In a flowering plant, how many division/s is/ are required for the formation of mature male gametophyte from mature pollen grain (having two cells) –
(1) 1 meiotic division is required
(2) 1 meiotic and 1 mitotic divisions are required
(3) 1 mitotic division is required
(4) 1 meiotic and 2 mitotic divisions are required
BSR363

50. The first cell of male gametophyte in flowering plant is ?
(1) Microspore mother cell
(2) Megaspore mother cell
(3) Male gamete
(4) Microspore
BSR364

51. How many meiosis are required for the formation of 80 male gametes in a typical flowering plant?
(1) 20 (2) 10 (3) 80 (4) 64
BSR365

52. How many in the list given below are diploid in a typical angiosperm plant.
Endothecium, Middle layer, Perisperm, Egg cell, Antipodals, Synergids, Nucellus, Integuments, Pollen grain, Endosperm, Embryosac.
(1) four (2) five (3) six (4) seven
BSR366

53. Read the following four statements (A-D) with respect to flowering plants :-
(A) Ploidy of the cells of the tetrad is haploid.
(B) The stigma serves as a landing platform for pollen grains.
(C) Pollen grains are generally spherical measuring about 100-200 micrometers in diameter.
(D) Exine of pollen grain has prominent apertures, called germ pores, where sporopollenin is absent.
How many of the above statements are correct?
(1) One (2) Two (3) Three (4) Four
BSR367

54. In which of the following plants, the nucellar cells start dividing and develop into the embryos ?
(1) *Citrus* and mango
(2) *Parthenium* and grasses
(3) Pea and gram
(4) Wheat and rice
BSR368

55. Fruit production without fertilisation is called?
(1) Parthenogenesis (2) Parthenocarpy
(3) Polyembryony (4) Amphimixis
BSR369

56. The zygote gives rise to the ..... and subsequently to the globular, heart shaped and mature embryo.
(1) Torpedo embryo (2) Endosperm
(3) Proembryo (4) All of the above
BSR370

57. In which of the following, endosperm persists in the mature seed ?
(1) Pea (2) Bean
(3) Groundnut (4) Castor
BSR371

58. Both autogamy and geitonogamy is prevented in :
(1) Papaya and date palm.
(2) Castor and maize
(3) Mango and papaya.
(4) Castor and date palm.
BSR372

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59. An example of syncarpous ovary is :
(1) *Hibiscus*
(2) *Papaver*
(3) *Michelia*;
(4) Both (1) and (2)

BSR373

60. A typical angiosperm anther is :
(1) Bilobed, dithecous and tetrasporangiate.
(2) Bilobed, monothecous and bisporangiate.
(3) Bilobed, monothecous and tetrasporangiate.
(4) Bilobed, dithecous and bisporangiate.

BSR374

61. In which set of plants, an ovary contains many ovule ?
(1) Wheat and paddy.
(2) Papaya and watermelon.
(3) Paddy and mango.
(4) Wheat and orchids.

BSR375

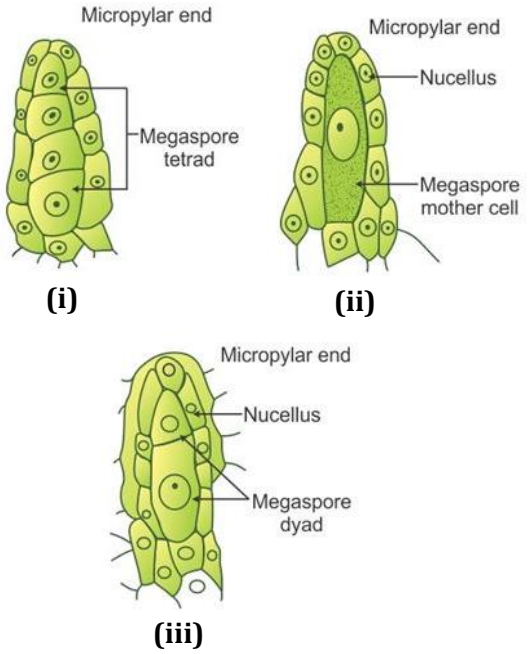
62. Choose the correct set of statements from the option given below:
(A) The microsporangia develop further and become pollen sacs.
(B) Longitudinal groove runs lengthwise separating the theca.
(C) Sporogenous tissue occupies the periphery of each microsporangium.
(D) Cells of the tapetum do not have dense cytoplasm
(1) A and B
(2) B and C
(3) A and C
(4) All of the above

BSR376

63. Which one of the following represents megasporangium ?
(1) Ovule
(1) Embryo sac
(3) Ovary
(4) Nucellus

BSR377

64. Few events of development of megaspore from megaspore mother cell is shown in figure below. Arrange the diagram in progressive order of their occurrence and select the correct order from options given below.



Megasporogenesis

Options :
(1) iii, ii, i
(2) i, ii, iii
(3) ii, iii, i
(4) ii, iii, i

BSR378

65. Choose the correct set of statements from the option given below:
(A) Hilum represents the junction between ovule and funicle.
(B) Chalaza, represents the tip of the of the ovule.
(C) Cells of the nucellus have abundant reserve food materials.
(D) Micropyle is surrounded by integuments.
(1) A, B and C
(2) B and C
(3) A and C
(4) All of the above

BSR379

66. Characteristic feature of angiosperm is :-
(1) Syngamy
(2) Triple fusion
(3) Double fertilization
(4) Seed formation

BSR380

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67. Read the following 4 statements (A-D) with respect to flowering plants :
(A) Vegetative cell of pollen grain is smaller than generative cell.
(B) Exine is made up of sporopollenin that can be degraded.
(C) The period for which pollen grains remain viable is highly variable.
(D) *Parthenium* or carrot grass, causes pollen allergy.
How many of the above statements are correct?
(1) Two (2) Three (3) One (4) Four
BSR381

68. Pollen grains of a large number of species can be stored for years in liquid nitrogen at :
(1) –180°C (2) –135°C
(3) –148°C (4) –196°C
BSR382

69. What will be the ploidy of the cells of the nucellus, MMC, the functional megaspore and female gametophyte?
(1) Diploid, haploid, diploid and haploid.
(2) Diploid, diploid, haploid and haploid.
(3) Haploid, haploid, diploid and diploid.
(4) Haploid, diploid, diploid and haploid.
BSR383

70. Which layer helps in the protection of anther?
(1) Epidermis, endothecium and tapetum.
(2) Endothecium, tapetum and middle layer.
(3) Epidermis, endothecium and middle layer.
(4) Epidermis, middle layer and tapetum.
BSR384

71. How many megaspores remain functional after meiosis of a megaspore mother cell ?
(1) One (2) Three (3) Four (4) Two
BSR385

72. How many nuclei and gametes participates in double fertilization ?
(1) 5 nuclei and 2 gametes
(2) 5 nuclei and 3 gametes
(3) 4 nuclei and 2 gametes
(4) 4 nuclei and 3 gametes
BSR386

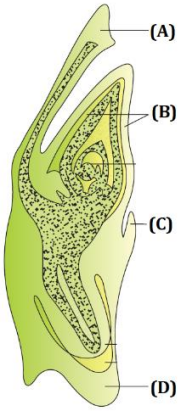
73. Which of the following statements are correct?
(1) Egg apparatus includes 2 egg cell and 2 synergid.
(2) Antipodal cells are situated at chalazal end.
(3) Typical embryo sac is 7 celled and 9 nucleate.
(4) Egg cell is present in the smaller central cell.
BSR387

74. How many meiosis are required for the formation of 100 functional megaspores and 80 functional microspores respectively.
(1) 100 and 20 (2) 80 and 70
(3) 50 and 70 (4) 80 and 100
BSR388

75. Chasmogamous and cleistogamous flowers are produced by :
(1) *Commelina* (2) *Oxalis*
(3) Both (1) and (2) (4) Papaya
BSR389

76. Which type of pollination is seen in cleistogamous flower ?
(1) Autogamy and geitonogamy
(2) Only autogamy
(3) Only Xenogamy
(4) Both (1) and (3)
BSR390

77. Identify A, B, C, D and D in the following structures -



(1) A–Scutellum, B–Coleoptile, C–Epiblast, D–Coleorhiza
(2) A–Coleoptile, B–Epiblast, C–Scutellum, D–Coleorhiza
(3) A–Coleorhiza, B–Scutellum, C–Epiblast, D–Coleoptile
(4) A–Epiblast, B–Coleorhiza, C–Scutellum, D–Coleoptile,
BSR391

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78. Which type of pollination brings genetically different pollen grains on the stigma ?
(1) Autogamy and geitonogamy
(2) Only autogamy
(3) Xenogamy
(4) Both (1) and (3)

BSR392

79. Which of the following is **incorrect** for insect pollinated flower ?
(1) The flowers pollinated by flies and beetles secrete foul odour to attract these animal
(2) Flowers are large and brightly coloured.
(3) Single ovule in each ovary.
(4) Nectar and pollen grains are the usual floral rewards.

BSR393

80. Choose the correct set of statements :
(A) *Zostera* has long ribbon like pollen grains.
(B) Water lily and water hyacinth are pollinated by water.
(C) Anemophilous flowers usually have single ovule and produce large amount of pollen grains.
(D) Pollen grains of *Hydrilla* are covered by mucilaginous covering.
(1) A, C and D
(2) B and C
(3) A and C
(4) All of the above

BSR394

81. Pollination by water in flowering plants is limited to :
(1) 20 genera, mostly monocots.
(2) 30 genera, mostly dicots.
(3) 50 genera, mostly monocots.
(4) 30 genera, mostly monocots.

BSR395

82. The central cell after triple fusion forms :
(1) Primary endosperm cell
(2) Embryo
(3) Zygote
(4) Both (1) and (2)

BSR396

83. Which of the following are endospermic seeds?
(1) Maize
(2) Castor
(3) Gram
(4) Both (1) and (2)

BSR397

84. Choose the correct set of statements :
(A) Artificial hybridisation is one of the major approaches of crop improvement programme.
(B) Removal of anthers from the flower bud before the anther dehisces is called bagging.
(C) Filiform apparatus present at the micropylar part of the synergids guides the entry of pollen tube.
(D) Primary endosperm nucleus (PEN) is diploid in nature.
(1) A, C and D (2) B and C
(3) A and C (4) All of the above

BSR398

85. Which of the following are parthenocarpic fruits ?
(1) Banana (2) Grapes
(3) Apple (4) Both (1) and (2)

BSR399

86. Read the following four statements (A-D):-
(A) Pericarp is fleshy in mango, orange and guava.
(B) Apple and pear are example of parthenocarpic fruits.
(C) In dicot seed, root cap enclosed in an undifferentiated sheath called coleorhiza.
(D) In dicot seed, the zygote gives rise to the proembryo and subsequently to the globular, heart-shaped and mature embryo.
How many of the above statements are correct?
(1) One (2) Three
(3) Two (4) Four

BSR400

87. Polyembryony is seen in :
(1) *Citrus*
(2) Mango
(3) Papaya
(4) Both (1) and (2)

BSR401

88. The arrangement of nuclei in a normal embryo sac in the dicot plant is :
(1) 2+4+3 (2) 3+2+3
(3) 2+3+3 (4) 3+3+2

BSR402

89. In some plants, embryo develops without fertilization. This phenomenon is known as :
(1) Parthenogenesis (2) Parthenocarpy
(3) Apomixis (4) Polyembryony

BSR403

90. Continued self-pollination result in:
(1) High vigour
(2) Variations
(3) Heterozygosity
(4) Inbreeding depression

BSR404

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	1	4	3	3	2	3	4	2	3	3	1	4	2	3	2
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	4	1	4	2	4	4	4	2	4	3	3	2	4	4	3
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	4	1	2	3	2	2	3	3	1	3	2	2	2	4	2
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	2	1	4	3	4	2	2	3	1	2	3	4	1	4	1
Question	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Answer	2	1	1	3	3	3	1	4	2	3	1	2	2	1	3
Question	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
Answer	2	1	3	3	1	4	1	4	3	4	2	4	2	1	4

Sexual Reproduction in Flowering Plants

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1. Which of the following statement is correct ?
(1) *Papaver* is syncarpous
(2) *Papaver* is apocarpous
(3) Michelia is syncarpous
(4) Both (2) and (3)
BSR315

2. Anther of angiosperm generally composed of
(1) One sporangium (2) Two sporangia
(3) Three sporangia (4) Four sporangia
BSR316

3. Main function of middle layer is
(1) Mechanical support (2) Protection
(3) Storage of food (4) Dehiscence
BSR317

4. How many microspores are formed from 10 microspore mother cells ?
(1) 10 (2) 20 (3) 40 (4) 80
BSR318

5. Which type of dehiscence of anther is found in most of the angiosperms ?
(1) Transverse (2) Longitudinal
(3) Valvular (4) By apical pores
BSR319

6. Microscopic structure of the flower that contains vegetative cell is?
(1) Embryo sac
(2) Egg apparatus
(3) Mature pollen grain
(4) Nucellus
BSR320

7. Pollen grains of which plant cause allergy :-
(1) *Chenopodium* (2) *Parthenium*
(3) *Sorghum* (4) *All of these*
BSR321

8. Which pollen grain lose their viability within 30 minutes of their release.
(1) Rose (2) Wheat (3) Pea (4) Potato
BSR322

9. Progeny resulting in cross pollination :-
(1) Are always sterile
(2) Always show recessive character
(3) Show greater genetic variability
(4) Are homozygous
BSR323

10. In monocots male gametophyte is represented by :-
(1) Ovule (2) Nucellus
(3) Pollen grain (4) Megaspore
BSR324

11. Generally how many megaspores take part in the development of female gametophyte of angiosperms ?
(1) One (2) Two
(3) Three (4) Four
BSR325

12. Reduction division in angiosperm occurs at the time of ?
(1) Asexual reproduction
(2) Apomixis reproduction
(3) Vegetative reproduction
(4) Microspore and megaspore formation
BSR326

13. Which structure of ovule, takes part in storage of food ?
(1) Integuments (2) Nucellus
(3) Embryosac (4) Micropyle
BSR327

14. When ovule is straight so the funiculus, chalaza and micropyle lie in one vertical line, the ovule is said to be ?
(1) Anatropous ovule
(2) Amphitropous ovule
(3) Orthotropous ovule
(4) Hemitropous ovule
BSR328

15. Most common type of ovule of angiosperms is
(1) Orthotropous (2) Anatropous
(3) Hemitropous (4) Amphitropous
BSR329

16. *Polygonum* type of embryosac is -
(1) 8-celled (2) 4-celled
(3) 5-celled (4) 7-celled
BSR330

17. The ploidy level of secondary nucleus of embryosac is :-
(1) 2n (2) 3n (3) 4n (4) 5n
BSR331

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18. In which of the following plant, pollination takes place due to secretion of foul odour by the flowers?
(1) *Maize* (2) *Pinus*
(3) *Vallisneria* (4) *Rafflesia*
BSR332

19. Pericarp develops from -
(1) Septum (2) Ovary wall
(3) Funiculus (4) Integument
BSR333

20. Syngamy means.
(1) The fusion of cytoplasm of spores
(2) Delayed meiosis
(3) Delayed mitosis
(4) The fusion of male and female gametes
BSR334

21. The double fertilization is the only characteristic of :-
(1) Bryophytes (2) Pteridophytes
(3) Gymnosperm (4) Angiosperm
BSR335

22. The pollen tube was first discovered by :-
(1) P. Maheshwari (2) Strasburger
(3) Nawaschin (4) Amici
BSR336

23. The pollen tube enters the synergid through :
(1) Lateral walls
(2) Filiform apparatus
(3) Chalazal end
(4) Any of these regions
BSR337

24. Many pollen tubes are formed during the germination of pollen grains in -
(1) *Solanum* (2) *Salvia*
(3) *Capsella* (4) *Cucurbita*
BSR338

25. Polyembryony was first observed in ?
(1) *Mangifera* (2) *Cucurbita*
(3) *Citrus* (Orange) (4) *Euphorbia*
BSR339

26. The outermost layer of endosperm of maize grain is called -
(1) Scutellum (2) Perisperm
(3) Aleurone layer (4) Tunica
BSR340

27. Non endospermic seeds are present in :-
(1) Castor (2) Pea (3) Maize (4) Wheat
BSR341

28. When cells of nucellus in ovule make the embryo directly then it is ?
(1) Amphimictic (2) Parthenogenetic
(3) Both (1) & (2) (4) Apomictic
BSR342

29. The function of innermost layer of anther is:
(1) Dehiscence (2) Mechanical
(3) Protective (4) Nutritive
BSR343

30. In flowering plants, sporogenous tissue gives rise to –
(1) Only the wall of the sporangium
(2) Both wall and the sporogenous cells
(3) Microspore tetrad
(4) Only tapetum and sporogenous cells
BSR344

31. Which of the following statements are correct?
(1) Pollen grains are rich in nutrients.
(2) In some cereals like rice and wheat pollen grains lose viability within 30 minutes of their release
(3) In some members of rosaceae, Leguminosae and solanaceae, pollen grains maintain viability for months
(4) All of the above
BSR345

32. What would be the ploidy of cells of tetrad ?
(1) n (2) 2n (3) 3n (4) 4n
BSR346

33. Each lobe of a typical anther in angiosperm having two theca it is called :
(1) Monothecous (2) Dithecous
(3) Monosporangiate (4) Tetrasporangiate
BSR347

34. From outer to inner, what is the sequence of wall layers in anther lobes?
(1) Epidermis, middle layers, tapetum, endothecium
(2) Epidermis, endothecium, tapetum, middle layers
(3) Epidermis, endothecium, middle layers, tapetum
(4) Tapetum, middle layers, endothecium, epidermis
BSR348

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35. Which statement is correct.
(1) China rose plant have unisexual flowers
(2) China rose plant have bisexual flowers
(3) Maize have bisexual flowers
(4) Both (1) and (3)

BSR349

36. Sporopollenin is a :-
(1) Polymer of amines acids
(2) Polymer of carotenoids
(3) Polymer of NAG
(4) Polymer of ribose sugar

BSR350

37. Which plant is monoecious?
(1) *Marchantia* (2) Papaya
(3) *Chara* (4) Both (1) and (2)

BSR351

38. Read the following four statements (A-D) :-
(A) Both wind and water pollinated flowers are not very colourful and do not produce nectar.
(B) Wind pollinated flowers often have a single ovule in each ovary
(C) Wind pollinated pollens are light and non sticky
(D) *Vallisneria* is an aquatic plant but it is pollinated by insect
How many of the above statements are correct?
(1) One (2) Two (3) Three (4) Four

BSR352

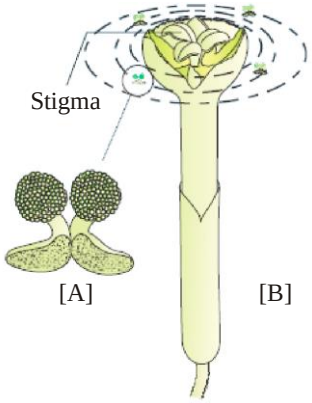
39. If pollen release and stigma receptivity within a flower is non synchronised then it is known as :-
(1) Dichogamy (2) Homogamy
(3) Heterostyly (4) Cleistogamy

BSR353

40. Which of the following statements are correct?
(1) The body of ovule fuses with funicle in the region called chalaza
(2) Polar nuclei are situated in the central cell above the egg apparatus
(3) Cleistogamous flowers are invariably autogamous
(4) Antipodal cells are present towards the micropylar end

BSR354

41. Following diagrams (A) & (B) represent ?



(1) (A) Female flower of *Hydrilla*
(B) Male flower of *Hydrilla*
(2) (A) Male flower of *Hydrilla*
(B) Female flower of *Hydrilla*
(3) (A) Female flower of *Vallisneria*
(B) Male flower of *Vallisneria*
(4) (A) Male flower of *Vallisneria*
(B) Female flower of *Vallisneria*

BSR355

42. In which set of plants an ovary contains one ovule.
(1) Papaya, water melon, orchids
(2) Wheat, paddy, mango
(3) Papaya, water melon, mango
(4) Papaya, water melon, wheat

BSR356

43. The functional megaspore generally undergoes:
(1) 2 meiotic divisions to form mature embryo sac
(2) 3 mitotic divisions to form mature embryo sac
(3) 2 mitotic divisions to form mature embryo sac
(4) 3 meiotic divisions to form mature embryo sac

BSR357

44. The number of ovules in an ovary may be
(1) One (2) Many
(3) Two (4) One to many

BSR358

45. Each ovule has one or two protective envelopes called
(1) Micropyle (2) Integuments
(3) Hilum (4) Chalaza

BSR359

46. Chalaza represents the
(1) Tip of the ovule (2) Base of the ovule
(3) Both (1) & (2) (4) Stalk of the ovule

BSR360

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47. Polar nuclei are situated in the central cell
(1) Below the egg apparatus
(2) Above the egg apparatus
(3) Below the antipodals
(4) Both (2) and (3)
BSR361

48. Pollen tube releases :-
(1) One male gamete in the cytoplasm of synergid & other in the cytoplasm of central cell
(2) Two male gametes in the cytoplasm of central cell
(3) Two male gametes in the cytoplasm of antipodal cell
(4) Two male gametes in the cytoplasm of a synergid
BSR362

49. In a flowering plant, how many division/s is/ are required for the formation of mature male gametophyte from mature pollen grain (having two cells) –
(1) 1 meiotic division is required
(2) 1 meiotic and 1 mitotic divisions are required
(3) 1 mitotic division is required
(4) 1 meiotic and 2 mitotic divisions are required
BSR363

50. The first cell of male gametophyte in flowering plant is ?
(1) Microspore mother cell
(2) Megaspore mother cell
(3) Male gamete
(4) Microspore
BSR364

51. How many meiosis are required for the formation of 80 male gametes in a typical flowering plant?
(1) 20 (2) 10 (3) 80 (4) 64
BSR365

52. How many in the list given below are diploid in a typical angiosperm plant.
Endothecium, Middle layer, Perisperm, Egg cell, Antipodals, Synergids, Nucellus, Integuments, Pollen grain, Endosperm, Embryosac.
(1) four (2) five (3) six (4) seven
BSR366

53. Read the following four statements (A-D) with respect to flowering plants :-
(A) Ploidy of the cells of the tetrad is haploid.
(B) The stigma serves as a landing platform for pollen grains.
(C) Pollen grains are generally spherical measuring about 100-200 micrometers in diameter.
(D) Exine of pollen grain has prominent apertures, called germ pores, where sporopollenin is absent.
How many of the above statements are correct?
(1) One (2) Two (3) Three (4) Four
BSR367

54. In which of the following plants, the nucellar cells start dividing and develop into the embryos ?
(1) *Citrus* and mango
(2) *Parthenium* and grasses
(3) Pea and gram
(4) Wheat and rice
BSR368

55. Fruit production without fertilisation is called?
(1) Parthenogenesis (2) Parthenocarpy
(3) Polyembryony (4) Amphimixis
BSR369

56. The zygote gives rise to the ..... and subsequently to the globular, heart shaped and mature embryo.
(1) Torpedo embryo (2) Endosperm
(3) Proembryo (4) All of the above
BSR370

57. In which of the following, endosperm persists in the mature seed ?
(1) Pea (2) Bean
(3) Groundnut (4) Castor
BSR371

58. Both autogamy and geitonogamy is prevented in :
(1) Papaya and date palm.
(2) Castor and maize
(3) Mango and papaya.
(4) Castor and date palm.
BSR372

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59. An example of syncarpous ovary is :
(1) *Hibiscus*
(2) *Papaver*
(3) *Michelia*;
(4) Both (1) and (2)

BSR373

60. A typical angiosperm anther is :
(1) Bilobed, dithecous and tetrasporangiate.
(2) Bilobed, monothecous and bisporangiate.
(3) Bilobed, monothecous and tetrasporangiate.
(4) Bilobed, dithecous and bisporangiate.

BSR374

61. In which set of plants, an ovary contains many ovule ?
(1) Wheat and paddy.
(2) Papaya and watermelon.
(3) Paddy and mango.
(4) Wheat and orchids.

BSR375

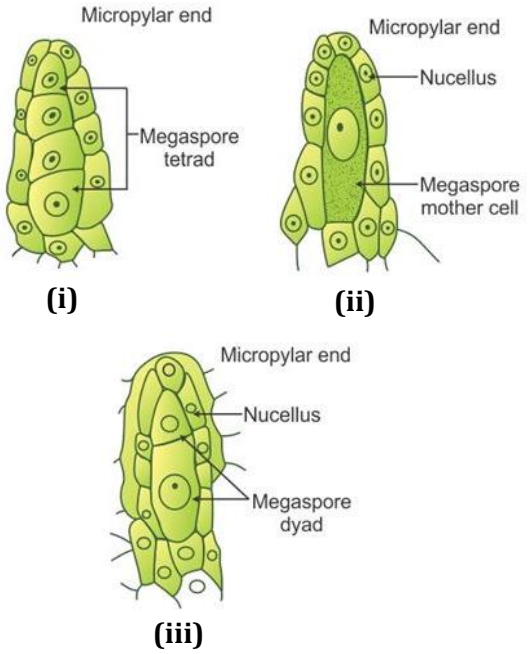
62. Choose the correct set of statements from the option given below:
(A) The microsporangia develop further and become pollen sacs.
(B) Longitudinal groove runs lengthwise separating the theca.
(C) Sporogenous tissue occupies the periphery of each microsporangium.
(D) Cells of the tapetum do not have dense cytoplasm
(1) A and B
(2) B and C
(3) A and C
(4) All of the above

BSR376

63. Which one of the following represents megasporangium ?
(1) Ovule
(1) Embryo sac
(3) Ovary
(4) Nucellus

BSR377

64. Few events of development of megaspore from megaspore mother cell is shown in figure below. Arrange the diagram in progressive order of their occurrence and select the correct order from options given below.



Megasporogenesis

Options :
(1) iii, ii, i
(2) i, ii, iii
(3) ii, iii, i
(4) ii, iii, i

BSR378

65. Choose the correct set of statements from the option given below:
(A) Hilum represents the junction between ovule and funicle.
(B) Chalaza, represents the tip of the of the ovule.
(C) Cells of the nucellus have abundant reserve food materials.
(D) Micropyle is surrounded by integuments.
(1) A, B and C
(2) B and C
(3) A and C
(4) All of the above

BSR379

66. Characteristic feature of angiosperm is :-
(1) Syngamy
(2) Triple fusion
(3) Double fertilization
(4) Seed formation

BSR380

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67. Read the following 4 statements (A-D) with respect to flowering plants :
(A) Vegetative cell of pollen grain is smaller than generative cell.
(B) Exine is made up of sporopollenin that can be degraded.
(C) The period for which pollen grains remain viable is highly variable.
(D) *Parthenium* or carrot grass, causes pollen allergy.
How many of the above statements are correct?
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BSR381

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BSR383

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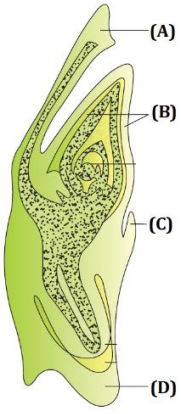
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(3) A–Coleorhiza, B–Scutellum, C–Epiblast, D–Coleoptile
(4) A–Epiblast, B–Coleorhiza, C–Scutellum, D–Coleoptile,
BSR391

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78. Which type of pollination brings genetically different pollen grains on the stigma ?
(1) Autogamy and geitonogamy
(2) Only autogamy
(3) Xenogamy
(4) Both (1) and (3)

BSR392

79. Which of the following is **incorrect** for insect pollinated flower ?
(1) The flowers pollinated by flies and beetles secrete foul odour to attract these animal
(2) Flowers are large and brightly coloured.
(3) Single ovule in each ovary.
(4) Nectar and pollen grains are the usual floral rewards.

BSR393

80. Choose the correct set of statements :
(A) *Zostera* has long ribbon like pollen grains.
(B) Water lily and water hyacinth are pollinated by water.
(C) Anemophilous flowers usually have single ovule and produce large amount of pollen grains.
(D) Pollen grains of *Hydrilla* are covered by mucilaginous covering.
(1) A, C and D
(2) B and C
(3) A and C
(4) All of the above

BSR394

81. Pollination by water in flowering plants is limited to :
(1) 20 genera, mostly monocots.
(2) 30 genera, mostly dicots.
(3) 50 genera, mostly monocots.
(4) 30 genera, mostly monocots.

BSR395

82. The central cell after triple fusion forms :
(1) Primary endosperm cell
(2) Embryo
(3) Zygote
(4) Both (1) and (2)

BSR396

83. Which of the following are endospermic seeds?
(1) Maize
(2) Castor
(3) Gram
(4) Both (1) and (2)

BSR397

84. Choose the correct set of statements :
(A) Artificial hybridisation is one of the major approaches of crop improvement programme.
(B) Removal of anthers from the flower bud before the anther dehisces is called bagging.
(C) Filiform apparatus present at the micropylar part of the synergids guides the entry of pollen tube.
(D) Primary endosperm nucleus (PEN) is diploid in nature.
(1) A, C and D (2) B and C
(3) A and C (4) All of the above

BSR398

85. Which of the following are parthenocarpic fruits ?
(1) Banana (2) Grapes
(3) Apple (4) Both (1) and (2)

BSR399

86. Read the following four statements (A-D):-
(A) Pericarp is fleshy in mango, orange and guava.
(B) Apple and pear are example of parthenocarpic fruits.
(C) In dicot seed, root cap enclosed in an undifferentiated sheath called coleorhiza.
(D) In dicot seed, the zygote gives rise to the proembryo and subsequently to the globular, heart-shaped and mature embryo.
How many of the above statements are correct?
(1) One (2) Three
(3) Two (4) Four

BSR400

87. Polyembryony is seen in :
(1) *Citrus*
(2) Mango
(3) Papaya
(4) Both (1) and (2)

BSR401

88. The arrangement of nuclei in a normal embryo sac in the dicot plant is :
(1) 2+4+3 (2) 3+2+3
(3) 2+3+3 (4) 3+3+2

BSR402

89. In some plants, embryo develops without fertilization. This phenomenon is known as :
(1) Parthenogenesis (2) Parthenocarpy
(3) Apomixis (4) Polyembryony

BSR403

90. Continued self-pollination result in:
(1) High vigour
(2) Variations
(3) Heterozygosity
(4) Inbreeding depression

BSR404

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	1	4	3	3	2	3	4	2	3	3	1	4	2	3	2
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	4	1	4	2	4	4	4	2	4	3	3	2	4	4	3
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	4	1	2	3	2	2	3	3	1	3	2	2	2	4	2
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	2	1	4	3	4	2	2	3	1	2	3	4	1	4	1
Question	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Answer	2	1	1	3	3	3	1	4	2	3	1	2	2	1	3
Question	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
Answer	2	1	3	3	1	4	1	4	3	4	2	4	2	1	4

Principle of Inheritance and Variations

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1. The character which can express only in homozygous condition in case of garden pea is
(A) Constricted pod
(B) Green seed colour
(C) Axial flower
(D) Yellow pod colour
(1) A, B & D (2) A, C & D
(3) B, C & D (4) A, B & C
BPV538

2. Mendel published his work on inheritance of characters in 1865, but it remained unrecognised till 1900 due to the reason (s):
(A) Communication was not easy in those days.
(B) His contemporaries did not accept factors as discrete units of heredity.
(C) Mathematical approach to explain inheritance was totally new.
(D) Mendel could not provide any physical proof for the existence of factors
(1) Only A and B are correct
(2) Only B and C are correct
(3) Only C and D are correct
(4) All are correct
BPV539

3. Which is not true for test cross -
(1) F_2 Dihybrid test cross ratio is 1 : 1 : 1 : 1
(2) Performed to know the alleles of a gene in an individual.
(3) A test cross is a back cross
(4) Dominant offsprings in test cross is always homozygous.
BPV540

4. Total number of genotypes and phenotypes in a dihybrid test cross is -
(1) 9 (2) 8 (3) 16 (4) 13
BPV541

5. What is the frequency of pure round seeded pea plants in F_2 generation of a dihybrid cross
(1) 4/16 (2) 6/16
(3) 1/16 (4) 2/16
BPV542

6. How many types of gametes are produced from individual of AABBCcDdee genotype
(1) 4 (2) 8 (3) 5 (4) 7
BPV543

7. How many kinds of gametes can be produced by an individual of genotype $\frac{A}{a}\frac{B}{b}\frac{C}{c}\frac{D}{d}\frac{E}{e}\frac{F}{f}$
(1) 32 (2) 16 (3) 8 (4) 4
BPV544

8. Proportion of progenies with round seeded and red flowered pea plants if cross is made between WwRr and ww Rr would be :-
(1) 3/8 (2) 1/8 (3) 4/8 (4) 2/8
BPV545

9. Ratio of dominant and recessive for both character plants in F_2 generation of dihybrid cross is :-
(1) 1 : 1 (2) 5 : 3
(3) 9 : 1 (4) 3 : 1
BPV546

10. Which will not be the genotype of pollen grain of a plant with genotype HHbbDdnnMMaa :-
(1) HbdNMa (2) HbDnMa
(3) HbdnMa (4) All the above
BPV547

11. What is the probability of gamete AbC produced by a plant having genotype AaBbCc (If all the genes are situated on different chromosomes) ?
(1) 1/2 (2) 1/4
(3) 1/6 (4) 1/8
BPV548

12. A plant has genotype TtRrGgPP. On selfing of this plant, what is the sum total of phenotypic categories ?
(1) 8 (2) 16 (3) 4 (4) 27
BPV549

13. In a dihybrid cross involving RRTT(red tall) × rrtt(white dwarf), what will be the ratio of red & tall plants in F_2-
(1) 3 : 1 (2) 1 : 1
(3) 1 : 2 (4) 2 : 3
BPV550

14. Offsprings of which cross will be phenotypically & genotypically similar -
(1) Dd × Dd (2) DD × dd
(3) DD × Dd (4) Ww × Ww
BPV551

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15. What will be the phenotypic ratio of offsprings of F_2 generation obtained by selfing of F1 plant having AABbCC genotype
(1) 3 : 1
(2) 1 : 1
(3) 0 : 3 : 3 : 1
(4) 27 : 9 : 9 : 9 : 3 : 3 : 3 : 1

BPV552

16. Which of the following crosses will produce 1 : 1 phenotypic ratio :-
(1) AaBb × aabb
(2) AaBB × aaBB
(3) aabb × aabb
(4) AABB × aabb

BPV553

17. What is the probability of obtaining pure homozygous individuals, if a cross is made between AaBbCcDd × aaBBCCdd individuals ?
(1) $\frac{2}{16}$ (2) $\frac{1}{16}$ (3) $\frac{16}{248}$ (4) $\frac{1}{64}$

BPV554

18. Based on the observations in hybridisation experiment on garden pea plant, Mendel proposed that something was being stably passed down, unchanged, from parent to off spring through the gametes, over successive generations. He called these things as :
(1) Genes (2) Factors
(3) Alleles (4) Multiple gene

BPV555

19. A diploid organism is heterozygous for 3 loci, how many types of gametes can be produced?
(1) 16 (2) 64 (3) 27 (4) 8

BPV556

20. A child has blood group O. If the father has blood group A and mother has blood group B then find out the incorrect statement :-
(1) Child is homozygous for blood group O
(2) Father is homozygous for blood group A
(3) Mother is heterozygous for blood group B
(4) Both (1) & (2)

BPV557

21. When pure green pod colour garden pea plant is crossed with yellow pod colour plant, then yellow pod colour plants obtained in F_2 generation will be :-
(1) 75% (2) 50%
(3) 25% (4) 37.5%

BPV558

22. What is correct about true-breeding pea line?
(1) A true breeding line is one that, having undergone continuous self-pollination, shows the stable trait inheritance and expression for several generations.
(2) Mendel selected 14 true breeding pea plant varieties, as pairs which were similar except for one character with contrasting traits
(3) Mendel selected 7 true-breeding pea plant varieties, as pairs which were similar except for one character with contrasting traits
(4) Both (1) & (2)

BPV559

23. Which of the following law explains the proportion of 3 : 1 obtained at the F_2 in a monohybrid cross ?
(1) Law of dominance
(2) Theory of linkage
(3) Law of independent assortment
(4) Both (2) & (3)

BPV560

24. Which of the following provides a good example of multiple alleles ?
(1) ABO blood group
(2) Sickle-cell anaemia
(3) Flower colour in the dog flower plant
(4) Starch synthesis in garden pea seeds

BPV561

25. Mendel crossed round and yellow seeded pea plant with wrinkled and green seeded plant, then in F_2 generation probability of plants having wrikled seed would be :-
(1) 3/16 (2) 1/16
(3) 12/16 (4) 4/16

BPV562

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26. Which of the following statement can be explained on the basis of law of dominance?
(1) characters are controlled by discrete units called factors
(2) Factors occur in pairs
(3) In a dissimilar pair of factors one member of the pair dominates the other
(4) All of the above

BPV563

27. Genes or alleles which can express independently when they are present together and also follow the law of segregation are -
(1) Multiple genes
(2) Codominant genes
(3) Epistatic genes
(4) Complementary genes

BPV564

28. Maximum height of a plant is 30ft. and minimum height is 10 ft. If plant height is controlled by 3 pairs of genes, then the height of plant with genotype AaBbCc will be -
(1) 13.2 ft. (2) 23.2 ft.
(3) 16.6 ft. (4) 19.9 ft.

BPV565

29. What percentage of pea seeds would have intermediate size of starch grains in F_2 generation?
(1) 0% (2) 25%
(3) 50% (4) 75%

BPV566

30. Select wrong statement with respect to multiple alleles -
(1) It can be detected only in a population.
(2) They are mutant forms of the same gene.
(3) Number of phenotypes in multiple alleles is calculated by formula 2n + 1. Here n is number of alleles.
(4) They occupy same locus on homologous chromosomes.

BPV567

31. In co-dominance the genes of an allelomorphic pair express _____ and follow law of _____.
(1) Equally independent assortment
(2) Unequally, dominance
(3) Equally, segregation
(4) Equally, dominance

BPV568

32. In *Mirabilis jalapa,* the percentage of plants having pink and red flowers would be _____ when pink flower plant is test crossed.
(1) 25% pink & 25% red
(2) 50% pink & 50% red
(3) 50% pink & 0% red
(4) 0% pink & 50% red

BPV569

33. Multiple alleles can be found only when a population is studied because -
(1) An individual has only two alleles.
(2) A single trait is controlled by more than two alleles.
(3) A single gene affects different characters.
(4) It has quantitative inheritance.

BPV570

34. What phenotypic ratio is obtained by selfing of a dihybrid, in which one genes is completely dominant and one gene is incompletely dominant ?
(1) 18 : 6 : 6 : 2 : 9 : 3 : 3 : 1 : 9 : 3 : 3 : 1
(2) 27 : 9 : 9 : 9 : 3 : 1 : 3 : 3
(3) 3 : 6 : 3 : 1 : 2 : 1
(4) 18 : 6 : 6 : 2 : 9 : 3 : 3 : 1 : 1 : 2 : 1

BPV571

35. If all the four children of a family will not have similar blood group, than what will be the blood group genotype of father & mother ?
(1) $I^A I^A$ and $I^B I^B$ (2) $I^A I^O$ and $I^B I^B$
(3) $I^A I^O$ and $I^B I^O$ (4) $I^A I^A$ and $I^B I^O$

BPV572

36. The ratio between carrier, disease free & diseased individual on marriage between two carriers of sickle cell anaemia will be -
(1) 0 : 2 : 1 (2) 0 : 1 : 1
(3) 1 : 2 : 1 (4) 2 : 1 : 1

BPV573

37. A monohybrid cross between two plants, one having 24 cm long internode and other having 12 cm long internode, produces F_1 hybrid, all having 18 cm long internodes. This is a case of
(1) Complete dominance
(2) Incomplete dominance
(3) Co-dominance
(4) Multiple allelism

BPV574

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38. In a polygenic inheritance of skin colours controlled by two polygenes, how many phenotypes, mulattoes and total progenies are formed in F_2.
(1) 3, 3, 16 respectively
(2) 5, 6, 16 respectively
(3) 7, 20, 64 respectively
(4) 7, 6, 16 respectively
BPV575

39. How many genotype are produced in F_2 generation in a dihybrid cross if both genes are pleiotropic -
(1) 3
(2) 4
(3) 9
(4) Cannot be calculated
BPV576

40. If a particular trait is controlled by 2 genes (polygenic), then calculate the total number of phenotype and genotypes in F_2 generation
(1) 13 (2) 8 (3) 14 (4) 27
BPV577

41. In Pea plant, when homozygous round seeded plant is crossed with homozygous wrinkled seeded plant, then the offsprings will be :-
(1) 100% with large starch grains
(2) 75% with medium sized starch grains
(3) 25% with small sized starch grains
(4) 0% with small sized starch grains
BPV578

42. Red eyed female *Drosophila* is crossed with white eyed male. If this cross results in 50% progeny being red eyed, we can say that -
(1) Both parents were hemizygous
(2) Female parent was heterozygous
(3) Female parent was pure
(4) Both parents were pure
BPV579

43. In a linear chromosome, map distance between four loci are ab = 12, bc = 4, ad = 2, ac = 8. The distance between c and d will be -
(1) 10% (2) 6%
(3) 10% or 6% (4) 3% or 6%
BPV580

44. In a certain taxon of insects some have 17 chromosomes and the other have 18 chromosomes. The 17 and 18 chromosome bearing organisms are :-
(1) Male and females, respectively
(2) Females and males, respectively
(3) All males
(4) All females
BPV581

45. What will be the percentage inheritance of a disorder from an X-linked diseased father to his son.
(1) 0% (2) 25% (3) 50% (4) 100%
BPV582

46. A homozygous recessive with cd genes was crossed with dominant (+ +). If their hybrid is test crossed and the following results were obtained

+ +	-	900
c d	-	880
+ d	-	115
+ c	-	105

than what will be the distance between genes c and d.
(1) 5.75 unit (2) 11 unit
(3) 44 unit (4) 90 unit
BPV583

47. Two genes a and b are linked and show 30% recombination. If ++ / ++ individual married with ab / ab then the types and proportion of gametes produced by F1 will be -
(1) ++ 35% : ab : 35% : + a 15% : +b 15%
(2) ++ 15% : ab : 15% : + a 35% : +b 35%
(3) ++ 35% : ab : 15% : + a 35% : +b 15%
(4) ++ 15% : ab : 35% : + a 15% : +b 35%
BPV584

48. How many types of gametes are formed by a plant having AaBbCc genotype, in which A & B genes shows complete linkage ?
(1) 2 (2) 4 (3) 6 (4) 8
BPV585

49. Which of the following gametic combination will not form in plant having AaBbCc genotype, in which A & B genes show complete linkage & cis arrangement ?
(1) ABC (2) ABc (3) abC (4) aBC
BPV586

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50. Which genotype ratio will appear by the trihybrid test cross in a plant having completely linked genes :-
(1) 1 : 1 : 1 : 1 : 1 : 1 : 1 : 1
(2) 1 : 1 : 1 : 1
(3) 1 : 1
(4) 1 : 2 : 1

BPV587

51. A dihybrid plant with genotype PpNn. It produces four types of gametes in following number PN = 200, pn = 200, Pn = 800, pN = 800 then what is the distance between linked genes?
(1) 30 cM (2) 10 cM
(3) 25 cM (4) 20 cM

BPV588

52. Find out the % of plants, which are dominant for both the character by the cross of $\frac{++}{ab}$ with it's recessive form, if 20% recombination is present:-
(1) 10% (2) 20% (3) 45% (4) 40%

BPV589

53. Four genes A, B, C and D located on chromosome and cross over frequencies between different genes are :
A – D ⇒ 20%, A – C ⇒ 4%
C – D ⇒ 16%, B – C ⇒ 9%
A – B = 5%
based on above data, what is the arrangement of linked genes on chromosome
(1) A – B – C – D (2) B A C D
(3) C B A D (4) C A B D

BPV590

54. A woman of normal vision whose father was colourblind, marries a man of normal vision, whose father was also colourblind. The sons of this marriage would be :-
(1) All normal (2) All colourblind
(3) 50% colourblind (4) 100% colourblind

BPV591

55. If a colourblind female marries a normal male (XY) then :-
(1) All the sons are normal
(2) All the sons are colourblind
(3) All the daughter are colourblind
(4) Both sons and daughters are colourblind

BPV592

56. By the cross of F1 linked dihybrid plant with a plant of genotype aabb, we get the 1300 plants out of 2000 of parental type. The distance between A and B gene is :-
(1) 35 map unit (2) 15 map unit
(3) 65 map unit (4) 85 map unit

BPV593

57. Genotype of plant is $\frac{++}{ab}$ and there is 18% recombination present in it then percentage of gametes will be :-
(1) ++ = 9% (2) ab = 18%
(3) +b = 9% (4) a+ = 41%

BPV594

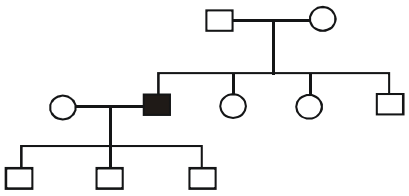
58. Find out the number of dominant individual in a population of 6000, if the frequency of recessive phenotype is 25%
(1) 3000 (2) 4500
(3) 1500 (4) 2000

BPV595

59. The frequency of an autosomal recessive allele is 0.4. Then what will be the frequency of heterozygotes in progeny among 4000 individual ?
(1) 2000 (2) 1920
(3) 2400 (4) 1600

BPV596

60. In this given pedigree what will be the nature of inheritance :



(1) X-linked dominant
(2) Y-linked
(3) Autosomal dominant
(4) Autosomal recessive

BPV597

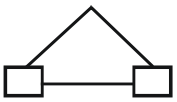
61. Pedigree analysis is useful in human genetics because :
(1) It can trace the inheritance of a particular trait
(2) Control crosses can not be performed in human beings
(3) It helps to trace disease in a family
(4) All of these

BPV598

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62. Find out the number of heterozygous free ear lobed persons in a population of 3000 if the proportion of fused ear lobed persons is 9%:
(1) 1260 (2) 1470
(3) 270 (4) 2730
BPV599

63. In pedigree analysis the symbol represents :-

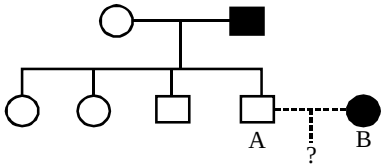


(1) Consanguineous marriage
(2) Sex unspecified
(3) Monozygotic twin
(4) Dizygotic twin
BPV600

64. If in a population, frequency of organisms of recessive trait is 25% then calculate the frequency of recessive allele :
(1) 25% (2) 50%
(3) 75% (4) 5%
BPV601

65. If the frequency of recesive phenotype is 9% then find out the number of homozygous organisms in population of 50,000 individuals:-
(1) 42,000 (2) 29,000
(3) 4500 (4) 24,500
BPV602

66. In the pedigree shown below, individuals with the solid symbols suffers from albinism. You would counsel the couple A & B that the probability that each of their child will have the disease is :-

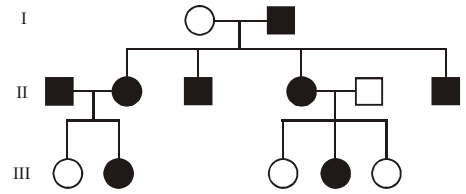


(1) 0% (2) 25% (3) 50% (4) 75%
BPV603

67. Organisms in which the total number of chromosomes are same in both male and female, But two different types of gametes in terms of the sex chromosomes, are produced by female are :-
(1) Human (2) Grasshopper
(3) Birds (4) Honey bee
BPV604

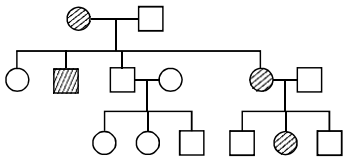
68. Dihybrid cross conducted by Morgan in which yellow body and white eyed female Drosophila is crossed with wild type male then non-parental combinations obtain in F_2-generation were :-
(1) 98.7% (2) 37.2%
(3) 62.8% (4) 1.3%
BPV605

69. Study the given pedigree carefully, the trait indicated is :-



(1) Autosomal recessive
(2) X-linked recessive
(3) Autosomal dominant
(4) Maternal imprinting
BPV606

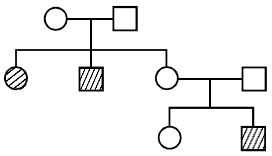
70. Given pedigree show inheritance of autosomal dominant disorder (myotonic dystrophy) then what will be genotype of shaded symbol :-



(1) AA (2) Aa
(3) AA or Aa (4) aa
BPV607

71. The map distance between A and C = 30 cM, B and C = 20 cM, A and B is 50 cM, B and D = 10 cM, C and D = 30 cM. What is the arrangement of ABCD gene on chromosome
(1) ABCD (2) DBCA (3) BACD (4) CABD
BPV608

72. Predict from the following chart :-



(1) Character is X-linked dominant
(2) Character is X-linked recessive
(3) Character is Autosomal dominant
(4) Character is Autosomal recessive
BPV609

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73. Thalassemia is differ from sickle cell anaemia because :-
(1) Thalassemia is a qualitative disease
(2) Thalassemia is a quantitative disease
(3) Thalassemia is an autosomal dominant disease
(4) Thalassemia is a chromosomal disorder
BPV610

74. A normal woman has a haemophilic son besides three perfectly normal daughters. The genotypes of the woman and her husband with respect to haemophilia would respectively be:-
(1) X^+X^+ and X^hY (2) X^+X^h and X^hY
(3) X^hX^h and X^+Y (4) X^+X^h and X^+Y
BPV611

75. A trait controlled by genes at several loci is an example of
(1) Multiple allelism
(2) Incomplete dominance
(3) Pleiotropy
(4) Polygenic inheritance
BPV612

76. A husband and wife have normal vision but fathers of both of them were colour blind. Probability of their first daughter to be colour blind is
(1) 25% (2) 50% (3) 75% (4) 0%
BPV613

77. A dihybrid cross produces 6 phenotype in F2 generation then which statement is correct :-
(1) Both genes show dominance
(2) Both genes show incomplete dominance
(3) One show dominance and another show incomplete dominance
(4) Both genes show codominance
BPV614

78. Which of the following is the incorrect statement-
(1) Artificial selection and domestication are required to make useful breeds of cows.
(2) Humans knew the benefits of sexual reproduction as early as 8000-1000 B.C.
(3) Heredity is the degree by which progeny differs from parents.
(4) Both (1) and (2)
BPV615

79. The major benefit Mendel had by using a large sampling size is-
(1) More variations
(2) Greater credibility of data
(3) More traits of one character
(4) Greater chances of successful pollination
BPV616

80. How many statements are correct about Mendel-
A. Mendel performed his experiments for seven years during 1956-1963.
B. Mendel considered 7 traits of 14 characters for his experiments.
C. His results allowed him to set up a basic framework of rules governing inheritance, which was expanded on by later scientists to account for all the diverse natural observations and the complexity inherent in them.
D. A true-breeding line is one that, having undergone continuous cross-pollination.
E. Mendel selected 14 true-breeding pea plant varieties, as pairs which were different except for one character with contrasting traits.
(1) A, B and C (2) C, D and E
(3) C and D (4) C only
BPV617

81. A cross between plants of exactly the same genotypes can be called
(1) Homozygous (2) Self-pollination
(3) Autosomal cross (4) Allosomic cross
BPV618

82. Which is not true for Mendel's monohybrid cross results?
(1) Factor occurs in pairs.
(2) F_1 progeny resembles with either of the parents.
(3) Inheritance of one character is independent of another character.
(4) None
BPV619

83. The number of genotypes and phenotypes in F_1 generation of Mendel's monohybrid cross, respectively:-
(1) 3 and 2 (2) 9 and 4
(3) 1 and 1 (4) 3 and 1
BPV620

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84. How many of the following do not hold true for Mendel's dihybrid cross-

A. Factors occur in pairs.
B. Characters are controlled by discrete units called factors.
C. In a dissimilar pair of factors one member of the pair dominates (dominant) the other (recessive).
D. The ratio of 9:3:3:1 can be derived as a combination series of 3 yellow: 1 green, with 3 round: 1 wrinkled.
E. F_1 generation individuals produce two types of gametes.

(1) A, B, C and D (2) C, D and E
(3) A, B, C and E (4) E only

BPV621

85. A plant with genotype AaBbCc is crossed with AAbbCc, what would be the phenotypic ratio-

(1) 9:3:3:1
(2) 1:2:1: 2:4:2: 1:1:2:1
(3) 1: 2: 1 :1 :2 :1
(4) 3 : 1 :3 :1

BPV622

86. Which of the following crosses will produce 1 : 1 genotypic ratio :-

(1) AaBb × aabb (2) AaBB × aaBB
(3) aabb × aabb (4) AABB × aabb

BPV623

87. What would be the phenotypic ratio in F_2 generation of a dihybrid cross if one character follows dominance and the other follows incomplete dominance-

(1) 1: 1: 1: 1 (2) 3: 1: 3 :1
(3) 1: 2: 1: 3: 1 (4) 1: 2: 1: 3: 6: 3

BPV624

88. The probability of AbC gamete production by an individual with genotype AaBbCc is-

(1) 50 % (2) 25 %
(3) 37.5 % (4) 12.5 %

BPV625

89. Genotypic and phenotypic ratio in the progeny would be the same if the cross is between-

(1) AaBb × AaBb (2) AaBb × AABB
(3) AaBb × aabb (4) AaBb × AABb

BPV626

90. Select the correct statement(s) regarding the process mentioned in following diagram-



A. It is performed in unisexual flowers only.
B. It is performed before pollination.
C. Not required if the cross is between two dihybrid plants.
D. Mendel used this step for his hybridization experiments.

(1) A and B only (2) B and D only
(3) C and D only (4) A and C only

BPV627

91. A plant has two independently assorting genes, one controlling seed shape (showing complete dominance) and the other controlling size of starch grain (showing incomplete dominance).
An individual plant which has wrinkled seed and small-sized starch grains is crossed with the plant which is dihybrid for these two characters. What would be the phenotypic ratio in the progeny

(1) 1: 1: 1: 1 (2) 3: 1: 3 :1
(3) 1: 2: 1: 3: 1 (4) 1: 2: 1: 3: 6: 3

BPV628

92. Match the list I with list II.

	List I		List II
A.	Punnett square	I.	Used for determining the genotype of F_1 individual
B.	Test cross	II.	Cross between plants with the same genotypes
C.	Selfing	III.	To study the inheritance of one gene
D.	Monohybrid cross	IV.	To determine the genotypes of progeny from a cross

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(1) A-I, B-II, C-III, D-IV
(2) A-IV, B-I, C-II, D-III
(3) A-II, B-IV, C-III, D-I
(4) A-I, B-IV, C-III, D-II

BPV629

93. Match the list I with list II.

	List I		List II
A.	Boveri	I.	Noted that the behaviour of chromosomes was parallel to the behaviour of genes and used chromosome movement to explain Mendel's laws.
B.	Walter Sutton and Theodore Boveri	II.	United the knowledge of chromosomal segregation with Mendelian principles and called it the chromosomal theory of inheritance.
C.	Henking	III.	Used the frequency of recombination between gene pairs on the same chromosome as a measure of the distance between genes and 'mapped' their position on the chromosome.
D.	Alfred Sturtevant	IV.	Gave the name X body

(1) A-II, B-I, C-III, D-IV
(2) A-II, B-I, C-IV, D-III
(3) A-I, B-II, C-IV, D-III
(4) A-III, B-IV, C-I, D-II

BPV630

94. Read the following statements and choose the correct set of statements regarding the behaviour of chromosomes-

A. Occur in pairs.
B. Segregate at the time of gamete formation and only one of each pair is transmitted to a gamete.
C. One pair segregates independently of another pair.
D. Independent pairs segregate independently of each other.

(1) A, B and C (2) A and B
(3) A, B, C and D (4) A, B and D

BPV631

95. Read the following statement and choose the correct set of statements.

A. Henking discovered X body in human spermatids.
B. Only one type of sex chromosome is found in the gametes of male Grasshopper.
C. In *Drosophila* the males have one X and one Y chromosome.
D. Allosomes are responsible for the characters that are common in male and female individuals.

(1) A and C only (2) A, B and C only
(3) B and C only (4) A, B, C and D

BPV632

96. Phenylketonuria is an inborn error of metabolism and is also inherited as the autosomal recessive trait. The affected individual lacks an enzyme that converts-
(1) Alanine into Phenylalanine
(2) Phenylalanine into Tyrosine.
(3) Tyrosine into Phenylalanine
(4) Phenylalanine into phenylketonuria

BPV633

97. Genes for alpha-thalassemia and beta-thalassemia are present on:-
(1) Same chromosome
(2) Two chromosomes of a homologous pair
(3) Two chromatids of one chromosome
(4) Non-homologous chromosomes

BPV634

98. Which of the following is not a symptom of Down's syndrome?
(1) Flat back of head
(2) Small round face
(3) Big wrinkled tongue
(4) Congenital heart disease

BPV635

99. Which of the following statements is/are correct?

(A) Turner's syndrome is due to 44+XXY condition.
(B) Klinefelter's syndrome is due to 44+X0 condition.
(C) Rudimentary ovaries are found in the individuals affected with Klinefelter's syndrome.

(D) Turner's syndrome and Klinefelter's syndrome, both are related to sex chromosomes so somatic characters are not affected in these conditions.

(E) Gynaecomastia can be seen in individuals affected with Klinefelter's syndrome.

(1) A, B and E only

(2) A, B, C and E only

(3) A, B and D only

(4) E only

BPV636

100. A man affected with Haemophilia marries a girl who is heterozygous for Haemophilia. What are the chances that their first male child would be affected by Haemophilia?

(1) 100% (2) 50%

(3) 25% (4) 25%

BPV637

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	1	4	4	2	1	1	1	1	3	1	4	1	2	2	1
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	2	2	2	4	2	3	4	1	1	4	4	2	4	3	3
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	3	3	1	3	3	4	2	2	3	3	4	2	3	1	1
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	2	1	2	4	3	4	4	2	3	2	1	3	2	2	4
Question	61	62	63	64	65	66	67	68	69	70	74	72	73	74	75
Answer	4	1	3	2	2	3	3	4	3	2	2	4	2	4	4
Question	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
Answer	4	3	3	2	4	2	3	3	4	4	2	4	4	3	2
Question	91	92	93	94	95	96	97	98	99	100					
Answer	1	2	2	3	3	2	4	2	4	2					

Molecular Basis of Inheritance

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1. A DNA molecule in which both strands have radioactive thymidine is allowed to duplicate in an environment containing non-radioactive thymidine. What will be the exact number of DNA molecules that contain some radioactive thymidine after 3 duplication ?
(1) One (2) Two
(3) Four (4) Eight

BMI435

2. Enzyme, that is not participates in the process of DNA-replication in prokayotes :-
(1) DNA ligase
(2) DNA topoisomerase
(3) Reverse transcriptase
(4) DNA polymerase

BMI436

3. Which of the following is not a characteristic feature of genetic material ?
(a) It should be chemically and structurally less stable
(b) It should be able to produce its replica
(c) It should be more reactive
(d) It does not follow Mendelian inheritance
(e) It provides scope for slow changes i.e. mutation
(1) only a, b, c and d
(2) only b, c, a, d and e
(3) only c, d and e
(4) only a, c and d

BMI437

4. Viruses having RNA genome evolve faster, because :-
(1) RNA is comparatively unstable, so mutate at a faster rate
(2) RNA is stable, and mutate at a slow rate
(3) RNA is chemically less reactive
(4) RNA is double stranded

BMI438

5. DNA is better genetic material than RNA, because:-
(1) DNA is chemically less reactive and structurally more stable as compared to RNA
(2) Presence of thymine at the place of uracil provides additional stability to DNA
(3) DNA being stable, so mutate at a slow rate
(4) All the above

BMI439

6. Consider the following four statements A, B, C and D and select the right option for two correct statement :-
Statements :-
(A) RNA was first genetic material
(B) 5-methyl uracil is found in RNA
(C) DNA polymerase-III does not require RNA primer for the synthesis of DNA strand
(D) RNA primer is formed during the process of DNA replication
The correct statements are :-
(1) (A) and (B) (2) (B) and (C)
(3) (A) and (D) (4) (A) and (C)

BMI440

7. If *E.coli* containing N15-N15 DNA was allowed to grow for 80 minutes in medium containing N14 then, what would be the percentage of light density and hybrid density DNA molecule ?
(1) 87.5% light density ; 12.5% hybrid density
(2) 75% light density ; 25% hybrid density
(3) 25% light density ; 75% hybrid density
(4) 12.5% light density ; 87.5% hybrid density

BMI441

8. Which is not found in DNA ?
(1) Deoxy ribose sugar
(2) 5-methyl uracil
(3) Demethylated thymine
(4) H_3PO_4

BMI442

9. Viruses having RNA genome evolve faster, because :-
(1) RNA is unstable and mutate at a faster rate
(2) RNA is more stable
(3) RNA is first genetic material
(4) RNA is the main genetic material

BMI443

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10. A segment of DNA molecule contains 20,000 base pairs. What is the number of phosphate molecule in this DNA molecule ?

(1) 20,000 (2) 40,000
(3) 10,000 (4) 5,000

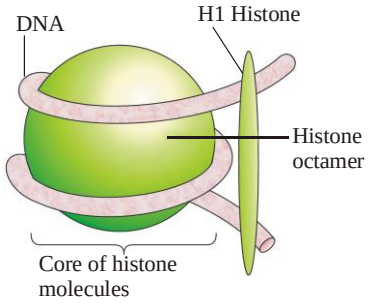
BMI444

11. In a segment of DNA molecule there occurs 200 molecules of adenine nitrogen bases and 200 molecules of cytosine nitrogen bases. What will be the total number of purine nitrogen bases in this segment ?

(1) 800 (2) 400 (3) 100 (4) 200

BMI445

12.



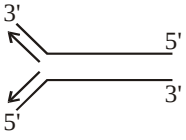
(A) Histone octamer is made up of 5 types of histone protein
(B) A typical nucleosome contains 200 bp of DNA helix
(C) Histones are organised to form a unit of eight molecules called as histone octamer
(D) The negatively charged DNA is wrapped around the positively charged histone octamer to form a structure called nucleosome.

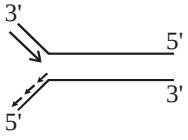
How many statements are correct ?

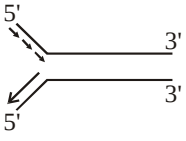
(1) One (2) Two
(3) Three (4) Four

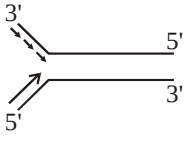
BMI446

13. Which one is true among the following for DNA replication :-

(1) 

(2) 

(3) 

(4) 

BMI447

14. In a phosphodiester bond formation a phosphate group is bonded with

(1) 5'–OH of one sugar and 2'–OH of next sugar
(2) 3'–OH of one sugar and 5'–OH of next sugar
(3) 3'–OH of one sugar and 2'–OH of next sugar
(4) 5'–OH of one sugar and 1'–OH of next sugar

BMI448

15. Which of the following enzyme inhibit transformation of R-strain bacteria to S-strain:

(1) DNAs (2) RNase
(3) DNase (4) Both (1) and (3)

BMI449

16. Okazaki fragments are :–

(1) Segments of DNA capable of translation
(2) Segments of DNA which undergo recombination
(3) Segments of nucleotides formed during transcription
(4) Segments of chain of nucleotides formed during replication of DNA

BMI450

17. In a ds DNA if the sequence of N-Bases in one strand is known then the sequence in other strand can be predicted, this is because of :-

(1) Antiparallel nature
(2) Purine-purine pairing
(3) Pyrimidine-pyrimidine pairing
(4) Complementary nature

BMI451

18. A molecular biologist analysed a nucleic acid sample and found that it contain N_2 bases in following proportion A = 20%, T = 15%, C = 35%, G = 30%. This data indicates that –

(1) Nucleic acid in the sample is single stranded DNA
(2) Nucleic acid in the sample is double stranded DNA
(3) Nucleic acid in the sample is double stranded RNA
(4) Nucleic acid in the sample is single stranded RNA

BMI452

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19. Which one is incorrect statement about genetic material ?

(1) It should be able to express itself in the form of "Mendelian characters".

(2) DNA chemically less reactive and structurally more stable than RNA

(3) RNA can directly code for the synthesis of proteins

(4) Only DNA is able to mutate.

BMI453

20. If *E.coli* with light DNA is allowed to grow for 80 minutes in 15NH4Cl medium then what would be the proportion of hybrid and heavy density DNA molecules respectively :-

(1) 7 : 1 (2) 1 : 7

(3) 2 : 0 (4) 0 : 2

BMI454

21. The following features occur during protein synthesis:

(i) Movement of m-RNA from the nucleus into the cytoplasm, where it binds to the ribosomes

(ii) Formation of amino acyl t-RNA complex

(iii) Transcription of specific segments of DNA into m-RNA molecules in the nucleus

(iv) Binding of N-terminal amino acid, methionine t-RNA to the 'P' site of the ribosome. The t-RNA anticodon "pairs with AUG on m-RNA"

(v) Formation of peptide bond between first and second amino acids at the 'A' site

(vi) Release of the completed polypeptide chain from ribosomes

(vii) Migration of dipeptide to form a tripeptide and then a polypeptide

The correct order of events is

(1) (iii) – (ii) – (i) – (iv) – (v) – (vi) – (vii)

(2) (i) – (ii) – (iii) – (iv) – (v) – (vi) – (vii)

(3) (iii) – (i) – (ii) – (iv) – (v) – (vi) – (vii)

(4) (iii) – (i) – (ii) – (iv) – (v) – (vii) – (vi)

BMI455

22. TMV infects a variety of plants, including *Zinnia* and tobacco. When TMV-RNA is used for infection, the amino acid sequence of the coat protein of the progeny is the same whether the host is *Zinnia* or tobacco. This proves that the genetic code is :-

(1) degenerate (2) universal

(3) unidirectional (4) non-overlapping

BMI456

23. Transfer RNA :-

(1) forms hydrogen bonds between its codon and the anticodon of an m-RNA in the 'A' site of a ribosome

(2) binds specific amino acid by the help of aminoacyl t-RNA synthetase

(3) uses GTP as the energy source to bind its amino acid

(4) is translated from m-RNA

BMI457

24. Several proteins may be produced at the same time from a single m-RNA by :-

(1) the action of several ribosomes in a string called a polyribosome

(2) several RNA polymerase molecules work sequentially

(3) single peptides that associate ribosomes with rough E.R.

(4) the involvement of multiple spliceosomes complex

BMI458

25. Which of the following is true for RNA processing ?

(1) Exons are excised before the m-RNA processing

(2) Assemblies of protein and snRNPs, called spliceosomes, may catalyze splicing

(3) The RNA transcript that leaves the nucleus may be much longer than the original transcript

(4) Large quantities of r-RNA are assembled into ribosomes

BMI459

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26. All of the following would be found in a prokaryotic cell except :-
(1) m-RNA
(2) r-RNA
(3) simultaneous transcription and translation
(4) sn-RNA

BMI460

27. A functional unit of gene which specifie the synthesis of one polypeptide is known as :-
(1) recon (2) clone
(3) codon (4) cistron

BMI461

28. A DNA strand is directly involved in the synthesis of all the following except :
(1) t-RNA molecule
(2) m-RNA molecule
(3) another DNA strand
(4) protein synthesis

BMI462

29. Because most of the amino acids are represented by more than one codons, the genetic code is said to be :-
(1) deaminated (2) comma less
(3) degenerate (4) overlapping

BMI463

30. Which of the following RNA has structural and catalytic role in translation ?
(1) m-RNA (2) t-RNA
(3) r-RNA (4) sn-RNA

BMI464

31. Match the column–I with column–II and select the correct answer :-

Column-I		Column-II	
(A)	Replication	(i)	RNA → DNA
(B)	Transcription	(ii)	m-RNA → protein
(C)	Translation	(iii)	DNA → mRNA
(D)	Reverse transcription	(iv)	DNA → DNA
(E)	Ribosome	(v)	r-RNA + protein

(1) A – i, B – ii, C – iii, D – iv, E–v
(2) A – ii, B – i, C – iii, D – iv, E–v
(3) A – iii, B – iv, C – ii, D – i, E–v
(4) A – iv, B – iii, C – ii, D – i, E–v

BMI465

32. The same amino acid sequence of a polypeptide chain can be represented by more than one type of nucleotide sequence in mRNA. It explains which property of genetic code?
(1) Universal nature
(2) Degeneracy
(3) Un-ambiguous
(4) Comma-less or continuous

BMI466

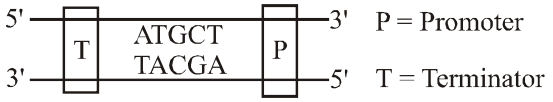
33. The process of translation requires the transfer of genetic information from :-
(1) A polymer of amino acids to a polymer of carbohydrates.
(2) A polymer of deoxy nucleotides to a polymer of nucleotides.
(3) A polymer of ribonucleotides to a polymer of amino acids.
(4) A polymer of amino acids to proteins.

BMI467

34. Which of the following region of transcription unit is located towards 3' end of the coding strand and it usually defines the end of the process of transcription ?
(1) Promoter (2) Structural genes
(3) Terminator (4) Coding strand

BMI468

35.



The m-RNA produced by transcription of above given DNA is :-
(1) 5' AUGCU 3' (2) 5' AGCAU 3'
(3) 3' UCGUA 5' (4) 5' UACGA 3'

BMI469

36. Which statement is true for transcription?
(1) Both strands of DNA are copied during transcription
(2) The total DNA of an organism gets transcribes
(3) Only a segment of DNA and only one of the strands is copied into RNA
(4) RNA dependent DNA polymerase enzyme catalyses process of transcription

BMI470

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37. Which of the following codon has dual functions ?
(1) UUU (2) UGA (3) AUG (4) UGG
BMI471

38. Which of the following RNA act as enzyme for the formation of peptide bond in bacteria :-
(1) 28s rRNA (2) 23s rRNA
(3) 16s rRNA (4) 18s rRNA
BMI472

39. Which of the following enzyme is helpful in polymerising RNA with defined sequences in a template independent manner?
(1) Severo ochoa enzyne
(2) Polynucleotide phosphorylase
(3) DNA dependent RNA polymerase
(4) Both (1) and (2)
BMI473

40. Which of the following statement is **incorrect** with respect to lac operon ?
(1) Single promoter for three structural genes, lac z, lac y, lac a
(2) Polycistronic m-RNA formation
(3) Glucose results in induction of lac operon
(4) Repressor protein shows negative control
BMI474

41. In structure of normal DNA, replacement of one nitrogenous base by another nitrogenous base is called as 'A' and replacement of adenine by guanine is called as 'B'. In the above statement A & B here refers to :-
(1) A $\rightarrow$ Substitution, B $\rightarrow$ Transition
(2) A $\rightarrow$ Point mutation, B $\rightarrow$ Transversion
(3) A $\rightarrow$ Frame shift mutation, B $\rightarrow$ Transversion
(4) Both (2) and (3)
BMI475

42. A classical example of point mutation is a change of single base pair in the gene for beta globin chain of haemoglobin that results in the change of amino acid residue glutamate to valine. It result into a diseased condition called as :-
(1) Sickle cell anemia (2) Phenylketonuria
(3) Haemophila (4) Thalassemia
BMI476

43. Select the **incorrect** statement for RFLP :-
(1) It is the basis of DNA fingerprinting
(2) It is due to variable length of restricted fragment.
(3) It is due to variable number of minisatellite.
(4) It is same for all human beings.
BMI477

44. Any Allelic sequence variation has traditionally been described as DNA polymorphism if more than one allele at a locus show : -
(1) Frequency greater than 0.01
(2) Frequency less than 0.01
(3) Frequency greater than 1.0
(4) Frequency less than 0.01%
BMI478

45. Find an incorrect statement with respect to results of HGP :-
(1) Dystrophin protein gene, is the largest gene of human genome.
(2) Chromosome Y has minimum 231 genes
(3) SNPs are identified at about 2.4 million locations.
(4) The functions are unknown for over 50% of discovered genes.
BMI479

46. If there are four different types of nitrogeneous bases (A, T, G and C) then how many different types of transitions and transversions are possible?
(1) Transition = 8, Transversion = 4
(2) Transition = 4, Transversion = 4
(3) Transition = 8, Transversion = 8
(4) Transition = 4, Transversion = 8
BMI480

47. Choose the incorrect match w.r.t. HGP :
(1) Expressed Sequence Tags - Identifying all the genes that are expressed as RNA
(2) Sequence Annotation - Sequencing both coding and non-coding sequences then sassigning different regions with functions
(3) YAC and BAC - Cloning vectors
(4) Human chromosome number 22 - The last chromosome to be sequenced
BMI481

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48. Which one is correct for variable number tandem repeat (VNTR) :
(1) Is a location in a genome where nucleotide sequence is organised as tandem repeat
(2) The number of repeat varies from chromosome to chromosome in an individual and also between individuals
(3) Its size varies from 0.1 to 20 kbp
(4) All are correct

BMI482

49. The key difference between the structure of thymine and uracil is due to:-
(1) –OH group (2) –COOH group
(3) Methyl group (4) Ethyl group

BMI483

50. The number of purine base(s) present in one base pair of nucleotides is/are:-
(1) One (2) Two
(3) Zero (4) One or two

BMI484

51. All are the correct statements, except:-
(1) One strand of DNA is made up of two types of nucleotides, so a total of four types of nucleotides are present in DNA.
(2) DNA molecules have an equal number of purines and pyrimidines in their structure.
(3) Chargaff's rule does not apply to single-stranded DNA.
(4) A bacteriophage known as $\phi \times 174$ has 5386 nucleotides.

BMI485

52. Select the incorrect statement:-
(1) A nitrogenous base is linked to the OH of 1' C pentose sugar through an N-glycosidic linkage to form a nucleoside.
(2) A phosphate group is linked to OH of 5' C of a nucleoside through phosphoester linkage.
(3) Two nucleotides are linked through 3'-5' phosphodiester linkage to form a dinucleotide.
(4) A DNA polymer (strand) has at one end a free phosphate moiety at 3' -end of sugar.

BMI486

53. Which of the following is a biomacro-molecule?
(1) DNA (2) Sucrose
(3) Lipid (4) Ribose

BMI487

54. Chemical bond(s) present in the structure of ATP is/are:-
(1) Glycosidic bond
(2) Peptide bond
(3) Phosphoester bond
(4) Both (1) and (3)

BMI488

55. The term 'Nuclein' was given by:-
(1) Wilkins and Franklin
(2) Watson and Crick
(3) Friedrich Miescher
(4) Nirenberg

BMI489

56. The ratio between purines and pyrimidines in a DNA molecule is:-
(1) Constant (2) Variable
(3) Always one (4) Both (1) and (3)

BMI490

57. Match the following and select the correct option: -

Column I		Column II	
A.	Nucleotide	I.	Adenine
B.	Nucleoside	II.	Cytosine
C.	Purine	III.	Adenosine
D.	Pyrimidine	IV.	Adenylic acid

(1) A-I B-II C-III D-IV (2) A-IV B-III C-I D-II
(3) A-IV B-III C-II D-I (4) A-III B-IV C-I D-II

BMI491

58. In RNA, every nucleotide residue has an additional –OH group present at _____ in the ribose.
(1) 2' -position (2) 3' -position
(3) 1' -position (4) 5' -position

BMI492

59. If *E. coli* containing N^{14}-N^{14} DNA was allowed to grow for 80 minutes in a medium containing N^{15}, then what would be the percentage of light density and hybrid density DNA molecule?
(1) 0% light density; 12.5% hybrid density
(2) 75% light density; 25% hybrid density
(3) 25% light density; 75% hybrid density
(4) 12.5% light density; 87.5% hybrid density

BMI493

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60. Select the set of correct statements only.

A. Maurice Wilkins and Rosalind Franklin, based on the X-ray diffraction data produced by James Watson and Francis Crick, proposed a very simple but famous Double Helix model for the structure of DNA.

B. The base pairing confers a very unique property to the polynucleotide chains.

C. The backbone of DNA is constituted by sugar-phosphate, and the bases project outside.

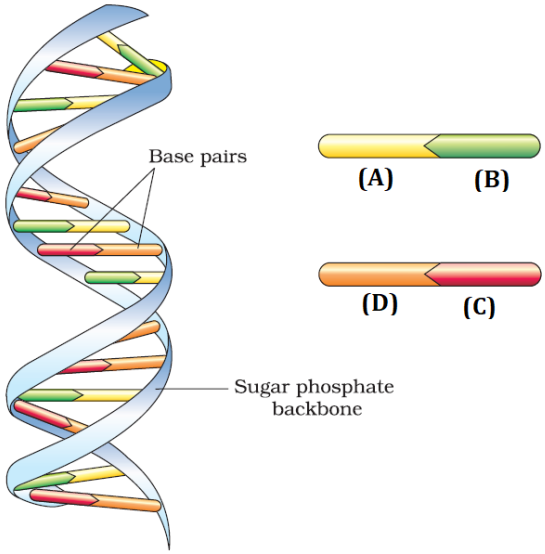
D. The pitch of the helix is 3.4 nm.

(1) B and D only (2) A and C only

(3) A, B, and D only (4) A, B, and C only

BMI494

61. In the following diagram, A, B, C, and D represent four bases of DNA. Select the option that shows the correct base pairing.



(1) A= Adenine B= Thymine

(2) A= Adenine B= Guanine

(3) C= Adenine D= Cytosine

(4) C= Thymine D= Guanine

BMI495

62. Central dogma in molecular biology was proposed by-

(1) Maurice Wilkins

(2) Rosalind Franklin

(3) James Watson

(4) Francis Crick

BMI496

63. Select the correct statement.

(1) In prokaryotes, such as *E. coli*, because they do not have a defined nucleus, the DNA is scattered throughout the cell.

(2) DNA (being negatively charged) is held with some proteins (that have positive charges) in a region termed as 'nucleoid' in eukaryotic cells.

(3) Basic amino acid residues carry positive charges in their side chains.

(4) A typical nucleosome contains 200 bases of DNA helix.

BMI497

64. _____ constitutes the repeating unit of a structure in the nucleus called chromatin.

(1) Nucleosome (2) Chromosome

(3) Chromatid (4) Histone protein

BMI498

65. A segment of DNA that is 2000 bp long, contains 400 Adenine bases. Select the correct option about it:-

(1) The number of Guanine is 800

(2) The number of Cytosine is 400

(3) The number of Cytosine is 600

(4) The number of Thymine is 600

BMI499

66. The packaging of chromatin at a higher level in eukaryotes requires an additional set of proteins that collectively are referred to as:-

(1) Non-chromosomal proteins

(2) Non histone proteins

(3) Polyamines

(4) Both (2) and (3)

BMI500

67. Prior to the work of Oswald Avery, Colin MacLeod, and Maclyn McCarty (1933-44), the genetic material was thought to be-

(1) Cellulose (2) RNA

(3) Protein (4) Lipid

BMI501

68. In the experiment done by Hershey and Chase, viruses grown in the presence of radioactive phosphorus contained-

(1) Radioactive DNA but not radioactive protein

(2) Radioactive protein but not radioactive DNA

(3) Radioactive DNA and protein

(4) Non-radioactive DNA and protein

BMI502

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69. RNA is not preferred as genetic material over DNA because-
(1) It is not able to express itself in the form of 'Mendelian Characters'.
(2) Chemically less stable than DNA.
(3) The mutation rate is very low.
(4) Both (1) and (2)

BMI503

70. Select the incorrect statement.
(1) Both the nucleic acids (DNA and RNA) have the ability to direct their duplications.
(2) The genetic material should be stable enough not to change with different stages of life cycle.
(3) Transcription as one of the properties of genetic material was very evident in Griffith's 'transforming principle'.
(4) The 2'-OH group present at every nucleotide in RNA is a reactive group and makes RNA labile and easily degradable.

BMI504

71. Select the option with the correct statements.
(A) Presence of uracil at the place of thymine also confers additional stability to DNA.
(B) Both DNA and RNA are able to mutate.
(C) RNA can directly code for the synthesis of proteins.
(D) The protein synthesising machinery has evolved around RNA.
(E) For the transmission of genetic information, DNA is better.
(1) B, C, and D only (2) B, C, D and E only
(3) A, B, C and E only (4) C, D and E only

BMI505

72. Which statement is incorrect?
(1) Matthew Meselson and Franklin Stahl performed the experiment in 1958 to prove the semiconservative nature of DNA replication.
(2) DNA has evolved from RNA with chemical modifications that make it more stable.
(3) DNA is double-stranded and having complementary strands further supports changes in its structure easily.
(4) The semi-conservative nature of DNA replication in eukaryotes was proved by an experiment done on *Vicia faba*.

BMI506

73. The energy requirement during DNA replication is fulfilled by
(1) ATPs produced by Ribosome
(2) Deoxyribonucleoside triphosphates
(3) DNA-dependent DNA polymerase
(4) Rho-factor

BMI507

74. For long DNA molecules, the two strands of DNA cannot be separated in its entire length because-
(1) DNA polymerase is not able to replicate the whole DNA.
(2) Energy requirement is very high.
(3) Chances of error are very high if whole DNA is replicated at once.
(4) Cells do not need to replicate the whole DNA.

BMI508

75. DNA replication in *E. coli* takes place during
(1) G_1 phase
(2) S phase
(3) G_0 phase
(4) No such specific phase

BMI509

76. The process of copying genetic information from one strand of the DNA into RNA is termed as transcription. Here also, the principle of _____ governs the process of transcription.
(1) Molecularity (2) Adaptability
(3) Complementarity (4) Accessibility

BMI510

77. A transcription unit in DNA is defined primarily by the three regions in the DNA:
(A) A Promoter
(B) The Structural gene
(C) A Terminator
(D) Rho factor
(1) B only (2) A, B, and C only
(3) A and B only (4) A, B, C and D

BMI511

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78. What are the similarities between DNA-dependent DNA polymerase and DNA-dependent RNA polymerase enzymes?
(A) Substrate used by them.
(B) Direction of polymerization
(C) Initiation factors
(D) Termination factors
(1) A and B only (2) A, C and D only
(3) B only (4) B, C and D only

BMI512

79. The definition of coding and template strands could be reversed by:-
(1) Interchanging the direction of coding and non-coding strands
(2) Switching the position of promoter and terminator
(3) Changing the location of Ori site
(4) Both (1) and (2)

BMI513

80. Select the correct statement.
(1) The DNA sequence coding for tRNA or rRNA molecule is not defined as a gene.
(2) Defining a gene in terms of DNA sequence is easy.
(3) Cistron is a segment of DNA coding for a polypeptide.
(4) Mostly, monocistronic genes are present in prokaryotes.

BMI514

81. Match the following and select the correct option.

Column I	Column II
A. Cistron	(I) Transcription unit
B. Exon	(II) Non-coding part
C. Intron	(III) Coding part
D. Monocistronic	(IV) Prokaryotes
E. Polycistronic	(V) Eukaryotes

(1) A-I, B-II, C-III, D-IV, E-V
(2) A-I, B-III, C-II, D-V E-IV
(3) A-I, B-II, C-III, D-V, E-VI
(4) A-V, B-III, C-I, D-IV, E-II

BMI515

82. Inheritance of a character is affected by:-
(1) Structural gene
(2) Regulatory sequence
(3) Personal belief
(4) Both (1) and (2)

BMI516

83. Which of the RNAs are involved in protein synthesis?
(1) rRNA and mRNA only
(2) mRNA and tRNA only
(3) rRNA and tRNA only
(4) mRNA, rRNA and tRNA

BMI517

84. How many types or DNA-dependent RNA polymerase are required for synthesis of mRNA, rRNA, and tRNA in *E. coli*?
(1) One
(2) Two
(3) Three
(4) RNA is not synthesized by RNA polymerase enzyme.

BMI518

85. Read the following statements and select the correct option.
(A) DNA is more stable and less reactive than RNA due to the presence of deoxyribose sugar.
(B) Uridine consists of uracil and ribose sugar.
(C) RNA is the genetic material in most of bacteria.
(D) The base sequence of only one strand of DNA can be determined if the base sequence of RNA is known.
(1) A and B only (2) A, B, and C only
(3) B, C, and D only (4) A and D only

BMI519

86. The essential life processes such as metabolism, splicing, and translation evolved around-
(1) DNA (2) RNA
(3) Protein (4) Lipid

BMI520

87. Match the columns and select the correct option:

Process	Product of the process
A. Replication	(I) DNA
B. Transcription	(II) RNA
C. Translation	(III) Protein

(1) A-I, B-II, C-III (2) A-II, B-I, C-III
(3) A-I, B-III, C-II (4) A-III, C-II, C-I

BMI521

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88. How is the RNA polymerase of prokaryotes able to catalyse all three steps, which are initiation, elongation, and termination?
(1) RNA polymerase changes its structure during different steps.
(2) Other enzymes help RNA polymerase.
(3) Some factors get associated with RNA polymerase to complete different steps.
(4) RNA polymerase is not able to complete these three steps; different enzymes are involved.

BMI522

89. Which of the following is not related to transcription in prokaryotes?
(1) There are at least three RNA polymerases.
(2) The direction of transcription is 5' to 3'.
(3) The transcription and translation can be coupled.
(4) Both (1) and (3)

BMI523

90. Select the pair of incorrect matches.
(1) Ribozyme- nucleic acid
(2) tRNA- Adapter molecule
(3) Amino acid acceptor end- mRNA
(4) Stop codon- no tRNA

BMI524

91. Beta-galactosidase enzyme is primarily responsible for:-
(1) Synthesis of glycosidic bond
(2) Breakdown of glycosidic bond
(3) Synthesis of galactose
(4) Breakdown of galactose

BMI525

92. Which of the *lac* operon products is not required for the metabolism of lactose?
(1) Beta-galactosidase
(2) Transacetylase
(3) Permease
(4) None

BMI526

93. Select the incorrect statement.
(1) The preferred carbon source in bacteria, that have *lac*-operon, is lactose.
(2) The repressor of the operon is synthesised (all-the-time – constitutively) from the i gene.
(3) Regulation of *lac* operon can also be visualised as regulation of enzyme synthesis by its substrate.
(4) Glucose or Galactose cannot act as inducers for *lac* operon.

BMI527

94. *Lac* operon is under control of
(1) Negative regulation
(2) Positive regulation
(3) DNA polymerase accessibility to the gene
(4) Both (1) and (2)

BMI528

95. Which of the following is not true about the goals of the Human Genome Project?
(A) Identify all the approximately 2,000-2,500 genes in human DNA.
(B) Determine the sequences of the 3 billion chemical base pairs that make up human DNA.
(C) Store this information in databases.
(D) To elucidate the mechanism of gamete formation
(E) To calculate the number of chromosomes in the human cell.
(1) D and E only
(2) B, D and E only
(3) A, D and E only
(4) A, B, D, and E only

BMI529

96. Genomes of how many of the following organisms are also sequenced apart from the human genome?
Caenorhabditis elegans, *Drosophila*, Rice, *Arabidopsis*
(1) One (2) Two
(3) Three (4) Four

BMI530

97. Fredrick Sanger is known for:-
(1) Developing the method for sequencing of DNA
(2) Developing the method for sequencing of lipid
(3) Developing the method for sequencing of carbohydrates
(4) Both (1) and (3)

BMI531

98. Allelic sequence variation has traditionally been described as a DNA polymorphism if more than one variant (allele) at a locus occurs in human population with a frequency greater than ____
(1) 1% (2) 0.1%
(3) 0.01% (4) 0.001%

BMI532

99. The probability of inheritable mutation responsible for DNA polymorphism to be observed is-
(1) Higher in coding DNA
(2) Higher in non-coding DNA
(3) Equal in coding and non-coding DNA
(4) Random in coding and lower non-coding DNA

BMI533

100. Select the correct sequence of steps involved in DNA fingerprinting-
(A) Digestion of DNA by restriction endonucleases
(B) Isolation of DNA
(C) Hybridization using labelled VNTR probe.
(D) Transferring (blotting) of separated DNA fragments to synthetic membranes, such as nitrocellulose or nylon.
(E) Separation of DNA fragments by electrophoresis.
(1) B, A, E, D, C (2) A, B, C, D, E
(3) B, C, E, D, A (4) B, C, D, E, A

BMI534

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	3	4	1	4	3	1	3	1	2	2	3	2	2	3
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	4	4	1	4	2	4	2	2	1	2	4	4	4	3	3
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	4	2	3	3	2	3	3	2	4	3	1	1	4	1	3
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	2	4	4	3	1	1	4	1	4	3	4	2	1	1	1
Question	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Answer	1	4	3	1	3	2	3	1	2	3	1	3	2	2	4
Question	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
Answer	3	2	3	4	3	2	4	4	1	1	2	1	3	1	3
Question	91	92	93	94	95	96	97	98	99	100					
Answer	2	4	1	4	3	4	1	1	2	1					

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Microbes in Human Welfare

1. Which product are synthesized from microbes?
(1) Antibiotics and bread
(2) Fermented beverages and citric acid
(3) Enzymes and acetic acid
(4) Bioactive molecules and curd
(1) A, C, D (2) A, B, C, D
(3) A, C (4) A, B, C

BMH135

2. Which one of the following is not a biopesticide?
(1) *Bacillus thuringiensis*
(2) *Nucleopolyhedrovirus*
(3) *Trichoderma*
(4) *Agrobacterium*

BMH136

3. Biogas is the mixture of gases produced by the microbial activity. The type of the gas produced depends upon :-
(1) Type of microbes
(2) Type of organic substrate/waste
(3) Size of digester
(4) Both (1) and (2)

BMH137

4. Which of the following bioactive molecule has been commercialised as blood cholesterol lowering agent ?
(1) Cyclosporin-A (2) Streptokinase
(3) Statin (4) All of the above

BMH138

5. Which of the following alcoholic drinks are produced by distillation of the fermented broth?
(1) Wine (2) Whisky
(3) Rum (4) (2) & (3) Both

BMH139

6. Biogas is a mixture of gases produces by the microbial activity in which predominant gas is:
(1) CO_2 (2) CO
(3) Methane (4) H_2

BMH140

7. Identify the correct match from the Column-I and Column-II :-

	Column-I		Column-II
(a)	*Aspergillus niger*	(i)	Butyric acid
(b)	*Acetobacter aceti*	(ii)	Citric acid
(c)	*Clostridium butylicum*	(iii)	Lactic acid
(d)	*Lactobacillus*	(iv)	Acetic acid

(1) a-ii, b-iv, c-i, d-iii (2) a-iii, b-iv, c-i, d-ii
(3) a-ii, b-i, c-iv, d-iii (4) a-iv, b-ii, c-i, d-iii

BMH141

8. The dough, which is used for making bread is fermented using :-
(1) *Saccharomyces cerevisiae*
(2) *Propionibacterium sharmanii*
(3) *Acetobacter aceti*
(4) *Streptococcus*

BMH142

9. Enzyme which is produced by the bacterium *Streptococcus* and modified by genetic engineering is used as a "clot buster" for removing clots from the blood vessels of patients who have undergone myocardial infarction leading to heart attack is :-
(1) Pectinases
(2) Streptokinase
(3) Polygalacturonase
(4) Adenosine deaminase

BMH143

10. Which of the following organism serve as an important biofertiliser in paddy fields ?
(1) Fungi (2) Cyanobacteria
(3) *Trichoderma* (4) Ladybird

BMH144

11. Which of the following statements is not incorrect?
(1) LAB produces acids that coagulate and completely digest the milk proteins.
(2) A small amount of cheese is added to the fresh milk as inoculum or starter contain millions of LAB,
(3) LAB multiply at a suitable temperature, thus converting curd to milk, which also improves its nutritional quality by increasing vitamin B_{12} in our stomach.
(4) LAB plays very beneficial role in checking disease causing microbes.

BMH145

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12 How many of the following statements are correct?

I. The dough which is used for making food such as dosa and idli is also fermented by yeast.

II. The puffed-up appearance of dough is due to the production of CO_2 gas.

III. The dough which is used for making bread, is fermented using brewer's yeast

IV. 'Toddy', a traditional drink of some parts of Southern India is made by fermenting fruits from palms

V. Microbes are also used to ferment fishes, soyabean and bamboo-roots to make foods.

(1) 4 (2) 3

(3) 2 (4) 1

BMH146

13. Yeast is used in the production of

(1) citric acid and lactic acid

(2) lipase and pectinase

(3) bread and beer

(4) cheese and butter

BMH147

14. How many of the following statements are not incorrect?

I. Anti is a Greek word that means 'against', and bio means 'life', together they mean 'against life'.

II. Antibiotics are chemical substances, which are produced by some microbes and can kill or retard the growth of other (disease-causing) microbes.

III. Penicillin is the first antibiotic to be discovered.

IV. Alexander Fleming discovered Penicillin while working on a *Staphylococci bacteria*.

V. The full potential as an effective antibiotic was established much later by Ernest Chain and Howard Florey.

(1) 5 (2) 4

(3) 3 (4) 1

BMH148

15. Which of the following statements is correct?

(1) Penicillin was used to treat American soldiers wounded in World War I.

(2) Fleming, Chain and Florey were awarded the Nobel Prize in 1954, for this discovery.

(3) Before Penicillin, antibiotics were also purified from other microbes.

(4) Antibiotics have greatly improved our capacity to treat deadly diseases such as plague, whooping cough, diphtheria and leprosy.

BMH149

16. Microbes are used for commercial and industrial production of certain chemicals like organic acids, alcohols and enzymes.

Which of the following is incorrect regarding this?

(1) Aspergillus – fungus – citric acid

(2) Acetobacter aceti – bacteria – vinegar

(3) Clostridium butylicum – protozoan - butyric acid

(4) Lactobacillus - bacteria - lactic acid.

BMH150

17. Which of the following statements is incorrect with respect to waste water treatment?

(1) The primary effluent is passed into large aeration tanks where it is constantly agitated mechanically and air is pumped into it. This allows vigorous growth of useful aerobic microbes into flocs.

(2) Flocs are masses of bacteria associated with fungal filaments to form mesh like structures.

(3) While growing, microbes consume the major part of the organic matter in the effluent. This significantly elevates the BOD (biochemical oxygen demand) of the effluent.

(4) BOD refers to the amount of the oxygen that would be consumed if all the organic matter in one liter of water were oxidised by bacteria.

BMH151

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18. Which of the following statements is wrong with respect to waste water treatment?
(1) The BOD test measures the rate of uptake of oxygen by micro-organisms in the sample of water.
(2) BOD is a direct measure of the organic matter present in the water.
(3) The greater the BOD of waste water, more is its polluting potential.
(4) Once the BOD of sewage or waste water is reduced significantly, the effluent is then passed into a settling tank where the bacterial 'flocs' are allowed to sediment.

BMH152

19. The gases which evolve from the anaerobic sludge digester constituting biogas are-
(1) Methane, sulphur dioxide, carbon sulphide
(2) Carbon dioxide, hydrogen chloride, methane
(3) Methane, hydrogen chloride and carbon dioxide
(4) Methane, carbon dioxide, hydrogen sulphide

BMH153

20. Which of the following statement is wrong?
(1) In settling tank, the sedimented bacterial flocs are called activated sludge.
(2) A small part of the activated sludge is pumped back into the anaerobic sludge digester to serve as the inoculum.
(3) In anaerobic sludge digesters, other kinds of bacteria which grow anaerobically, digest the bacteria and the fungi in the sludge.
(4) The biogas can be used as source of energy as it is inflammable.

BMH154

21. The high value of BOD (Biochemical Oxygen Demand) indicates that
(1) Water is pure.
(2) Water is highly polluted.
(3) Water is less polluted.
(4) Consumption of organic matter in the water is higher by microbes.

BMH155

22. If the activated sludge flocs do not get enough oxygen supply,
(1) It will increase the rate of the treatment.
(2) The center of flocs will become anoxic, which would cause the death of bacteria and eventually breakage of flocs.
(3) It will increase the size of flocs.
(4) Protozoa would grow in large numbers.

BMH156

23. Methanogenic bacteria are not present in
(1) gobar gas plant
(2) stomach of ruminants
(3) bottom of water-logged paddy fields
(4) activated sludge

BMH157

24. Activated sludge should settle quickly so that
(1) it is rapidly pumped back from sedimentation tank to aeration tank
(2) absorb pathogenic bacteria present in wastewater
(3) it is anaerobically digested
(4) absorbs inorganic matter

BMH158

25. The sludge generated by wastewater treatment is treated by
(1) anaerobic digesters
(2) floc
(3) chemicals
(4) oxidation pond

BMH159

26. How many of the following are correct with respect to mycorrhiza?
I. Fungi are also known to form symbiotic associations with plants (mycorrhiza).
II. Many members of the genus Glomus form mycorrhiza.
III. It provide resistance to root-borne pathogens.
IV. It provide tolerance to salinity and drought.
V. It helps in overall increase in plant growth and development.
(1) 5 (2) 4 (3) 3 (4) 1

BMH160

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27. Which of the following is incorrect?
(1) *Rhizobium* fixes atmospheric nitrogen into inorganic forms which are used by plants as nutrients.
(2) Cyanobacteria are autotrophic microbes widely distributed in aquatic and terrestrial environments many of which can fix atmospheric nitrogen.
(3) Blue green algae add organic matter to the soil and increase its fertility.
(4) In paddy fields, cyanobacteria serve as an important biofertiliser.

BMH161

28. The bacteria which can fix atmospheric nitrogen in it's free-living form in the soil are–
(1) *Azospirillum* and *Azorhizobium*
(2) *Azospirillum* and *Azotobacter*
(3) *Azotobacter* and *Rhizobium*
(4) *Frankia* and *Azospirillum*

BMH162

29. The technology of biogas production was developed in India mainly due to the efforts of:
(1) Indian Agricultural Research Institute (IARI) and Khadi and Village Industries Commission (KVIC).
(2) National Botanical Research Institute (NBR1).
(3) Indian Council of Medical Research (ICMR).
(4) Indian Council of Agricultural Research (ICAR).

BMH163

30. Insect resistant transgenic cotton has been produced by inserting a piece of DNA from
(1) an insect
(2) a wild relative of cotton
(3) a bacterium
(4) a virus

BMH164

31. Microbe used for biocontrol of pest butterfly caterpillars is
(1) *Trichoderma sp.*
(2) *Saccharomyces cerevisiae*
(3) *Bacillus thuringiensis*
(4) *Streptococcus sp.*

BMH165

32. Organic farming includes
(1) use of fertilizers and pesticides of biological origin.
(2) IPM (integrated pest management programme).
(3) locally developed pest resistant varieties.
(4) all of the above

BMH166

33. Which of the following statements is not true for stirred tank fermentaion ?
(1) Buffer needed to control pH.
(2) Batch and feed possible.
(3) Control dissolved oxygen.
(4) Easy in process sampling.

BMH167

34. Which of the following statement is correct ?
(1) *Methanobacterium* is an aerobic bacterium found in rumen of cattle.
(2) Biogas (commonly called gobar gas) is pure methane.
(3) Activated sludge sediment in settlement tanks of sewage treatment plants is rich source of aerobic bacteria.
(4) Biogas is produced by the activity of aerobic bacteria on animal waste.

BMH168

35. Which one of the following is the most important role of microorganism for the well-being of humans?
(1) Sewage treatment
(2) Biological control of plant disease
(3) Conversion of milk to curd
(4) All of these

BMH169

36. Consider the following statements about organic farming.
(i) It utilizes genetically modified crops like Bt cotton.
(ii) It uses only naturally produced inputs like compost.
(iii) It does not use pesticides and urea.
(iv) It produces vegetables rich in vitamins and minerals.
Which of the above statements are correct?

(1) (ii), (iii) and (iv)
(2) (iii) and (iv) only
(3) (ii) and (iii) only
(4) (i) and (ii) only

BMH170

37. Microbes are used in the
I. primary treatment of sewage.
II. secondary treatment of sewage.
III. anaerobic sludge digester.
IV. production of bioactive molecules.
Choose the correct option showing the uses of microbes.
(1) I, III, IV (2) I, II, III, IV
(3) II, III, IV (4) IV, I, II

BMH171

38. A lake with an inflow of domestic sewage rich in organic waste may result in
(1) drying of the lake very soon due to algal bloom.
(2) an increased production of fish due to lot of nutrients.
(3) death of fish due to lack of oxygen.
(4) increased population of aquatic food web organization.

BMH172

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	4	4	3	4	3	1	1	2	2	4	3	3	1	4
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	3	3	2	4	2	2	2	4	1	1	1	1	2	1	3
Question	31	32	33	34	35	36	37	38							
Answer	3	4	3	3	4	3	3	3							

Organisms and Populations

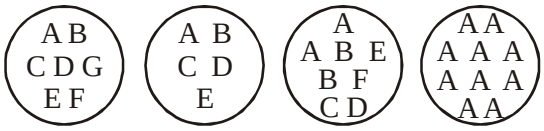
1. Physiological ecology tries to understand :-
(1) How different type of organisms interact among themselves.
(2) How different type of ecosystems interact with local abiotic factors.
(3) How different organisms are adapted to environments in forms of not only survival but also reproduction
(4) How different type of abiotic factors interact among themselves and one factor remains limiting

BOP202

2. Fig trees can maintain community structure during food scarcity in tropical deciduous forest. These act as
(1) Exotic species (2) Pioneer species
(3) Edge species (4) Key stone species

BOP203

3.



In above diagram four areas are given. In which area has maximum and minimum species diversity respectively.
(1) Maximum = B, Minimum = D
(2) Maximum = A, Minimum = D
(3) Maximum = A, Minimum = B
(4) Maximum = C, Minimum = D

BOP204

4. Ecological succession usually focuses on which changes ?
(1) type of animals
(2) type of decomposers
(3) vegetation
(4) humus content

BOP205

5. Which species have high value of biotic potential and short generation time?
(1) Pioneer species (2) Climax species
(3) Seral species (4) Sere

BOP206

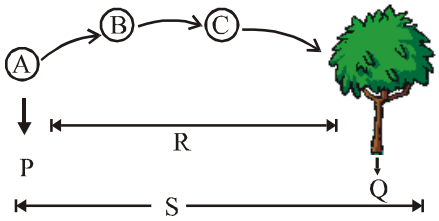
6. Which of the following parameter decreases as succession proceeds?
(1) Humus content
(2) Species diversity
(3) Net community productivity
(4) Biomass

BOP207

7. Primary succession is development of biotic communities on :-
(a) Newly exposed habitat
(b) Cleared forest area
(c) Freshly harvested crop field
(d) Pond formed after a rainy season
(1) only a (2) a & d
(3) a, c, d (4) b, c, d

BOP208

8.



In above given diagram Identify P, Q, R, S respectively and give **correct** option:-
(1) P → pioneer community,
Q → Climax community,
R → Sere, S → Seral community
(2) P → Climax community,
Q → Sere, R → Seral,
S → Pioneer community
(3) P → Pioneer community,
Q → Climax community,
R → Seral community, S → Sere
(4) P → Climax community, Q → Seral,
R → Sere, S → Pioneer community

BOP209

9. Which of the following zone of the lake also known as ecotone and has maximum species diversity ?
(1) Limnetic zone
(2) Profundal zone
(3) Benthic zone
(4) Littoral zone

BOP210

10. In rocky intertidal communities removal of which of the following predator became the cause of extinction of 10 species of invertebrates?
(1) Abingdon tortoise
(2) Monarch butterfly
(3) *Pisaster* starfish
(4) *Calotropis*

BOP211

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11. Select the correct match :

	A Key stone species	B Critical link species	C Endemic species
(1)	Metasequioa in China valley	Lion in forest	Mycorrhizal fungi
(2)	Kangaroo in Australia	Fig tree in tropical forest	Pollinator insect
(3)	Kangaroo rat in desert	Mycorrhizal fungi	Metasequioa in China valley
(4)	Lion in forest	Kangaroo in Australia	Pollinator insect

BOP212

12. Select the correct match :-

	A Mutualism	B Parasitism	C Commensalism
(1)	Barnacle & Whale	*Loranthus* & mango	*Yucca* plant & *pronuba* moth
(2)	Leech on cattle	Orchid & Bee	Pilot fish & shark
(3)	Fig tree & wasp	*Viscum* & Oak	Orchid & mango
(4)	Sucker fish & shark	Lichen	Ticks on dog

BOP213

13. Select the correct match in the following :-
(1) Brood parasitism – Partial stem parasite
(2) Holoparasite – Cuckoo & crow
(3) Hemiparasite – Parasite which totally dependent upon the host
(4) Hyper parasite – A parasite living on another parasite

BOP214

14. The Prickly pear cactus introduced into Australia in the early 1920's caused havoc by spreading rapidly in to millions of hectares of rangeland because -
(1) There is no competitor
(2) Due to less intra specific competition
(3) Absence of natural predator
(4) Availability of high nutrient

BOP215

15. Mark the important defence mechanism in plants against herbivory :-
(1) Spines (2) Toxic chemical
(3) Both (1) & (2) (4) Green leaves

BOP216

16. The Abingdon tortoise in Galapagos Island became extinct within decade after goats were introduced on the island is the example of :
(1) Competitive release
(2) Resource partitioning
(3) Interference competition
(4) Competitive exclusion

BOP217

17. Identify the correct match from the column I, II and III.

Column-I	Column-II	Column-III
1. Mutualism	a. (+, –)	i. Visiting flamingoes and resident fishes in south american lakes
2. Competition	b. (+, 0)	ii. Starfish *Pisaster* in american pacific coast
3. Predation	c. (+, +)	iii. Cattle egret birds and cattle
4. Commensalism	d. (–, –)	iv. Lichens

(1) 1 - d - iv, 2 - c - i, 3 - a - ii, 4 - b - iii
(2) 1 - c - iv, 2 - d - i, 3 - a - ii, 4 - b - iii
(3) 1 - c - iv, 2 - d - iii, 3 - a - ii, 4 - b - i
(4) 1 - d - iv, 2 - c - i, 3 - b - ii, 4 - a - iii

BOP218

18. What benefits does fig tree offers to wasp in return for its pollination ?
(1) Delicious fruits
(2) Nectar
(3) Pollens
(4) Oviposition and developing seeds

BOP219

19.



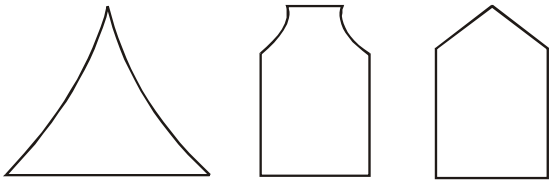
Which of the following statement is incorrect with respect to the above given figure ?
(1) One petal of the flower bears an uncanny resemblance to the female bumble bee in size, colour and markings.

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(2) The female bumble bee is a critical link species which helps in pollination of the flower.
(3) The flower employs 'Sexual deceit' to get pollination done by a particular species of bumble bee.
(4) Male bumble pseudocopulates with the flower.

BOP220

20. Age structure diagrams (A, B and C) for the populations are shown below. They represent:-



(1) A-Declining populations
B-Stable population
C-Growing population
(2) A-Growing population
B-Stationary population
C-Declining populations
(3) A-Exponential growth
B-Indeterminate growth
C-Stationary population
(4) A-Growing population
B- and C-Stable population

BOP221

21. The logistic population growth is expressed by the equation :

(1) $dN/dt = rN\left(\frac{N-K}{N}\right)$

(2) $dt/dN = Nr\left(\frac{K-N}{K}\right)$

(3) $dN/dt = rN\left(\frac{K-N}{K}\right)$

(4) $dN/dt = rN$

BOP222

22. In a population birth rate is 0.15 and death rate is 0.08 during a unit time period. What is the value of r (intrinsic rate of natural increase) for given population ?

(1) 0.23 (2) 0.07
(3) 0.05 (4) 0.25

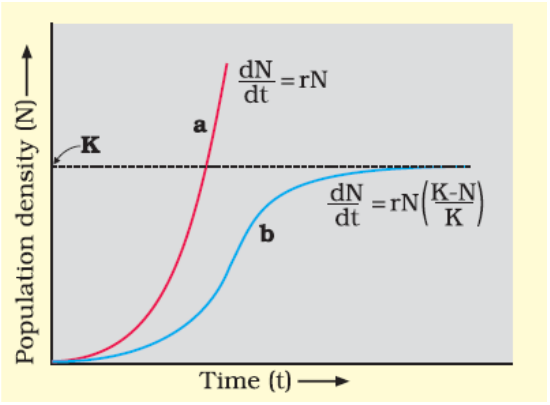
BOP223

23. What is the reproductive strategy adopted by bamboo to maximise its fitness ?

(1) Breed multiple times in a life time
(2) Breed only once in a life time
(3) Produce small number of small sized offsprings
(4) Produce small number of large sized offsprings

BOP224

24.



For the given population growth curve which statement is correct .

(1) Curve 'a' does not show saturation because rate of population growth is very low.
(2) 'K' is carrying capacity and is specific for a given population.
(3) Curve 'a' is obtained when responses are limiting.
(4) Curve 'b' is depicted by a population of small organisms.

BOP225

25. Which of the following is not an attribute of population ?

(1) Biotic potential (2) Dominance
(3) Natality (4) Age pyramid

BOP226

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26. Which among the following represents commensalism?
(1) Plasmodium living in human RBCs
(2) Tick birds helping rhino to get rid of leeches.
(3) Orchid plant living on a mango tree
(4) Hermit crab and sea anemone

BOP227

27. Observe the following table

A	B	Interaction
+	–	a = ?
–	–	b = ?
–	0	c = ?
+	0	d = ?

The correct order of a, b, c, d would be:
(1) Commensalism, Parasitism, Predation, Mutualism
(2) Predation, Competition, Amensalism, Commensalism
(3) Amensalism, predation, Commeusalism, Parasitism
(4) Parasitism, Predation, Amensalism, Commensalism

BOP228

28. Which of the following is a conduit for energy transfer across trophic levels?
(1) Mutualism
(2) Parasitism
(3) Predator
(4) Proto-cooperation

BOP229

29. What is the ecological principle behind the biological control method of managing with pest?
(1) Mutualism (2) Predation
(3) Parasitism (4) Commensalism

BOP230

30. Given below are two statements :
Statement-I : Under a particular set of selection pressures organisms evolve towards the most efficient reproductive strategy.
Statement-II : Some organisms breed only once in their lifetime (pacific salmon fish, bamboo) while others breed many times during their life time.
(1) Both statement-I and statement-II are incorrect.
(2) Statement-I is correct but statement-II is incorrect.
(3) Statement-I is incorrect but statement-II is correct.
(4) Both Statement-I and statement-II are correct.

BOP231

31. What is true for individuals of same species :-
(1) Live in same niche
(2) Live in same habitat
(3) Interbreeding
(4) Live in different habitat

BOP232

32. Which of the following means was used by warblers to avoid competition?
(1) Difference in foraging activities
(2) Habitat fragmentation
(3) Competitive release
(4) All of these

BOP233

33. Which one of the following interaction shares a characteristic where none of the organism can complete its life cycle without the other participating organism?
(1) *E.coli* and Human
(2) Cattle egret bird and Cattle
(3) Hermit Crab and Sea Anemone
(4) Fig and Wasp

BOP234

34. Which of the following organism is highly distasteful to its predator because of a special chemical present in its body: -
(1) *Pisaster* starfish
(2) Prickly pear cactus
(3) Monarch butterfly
(4) Parthenium

BOP235

35. Which is the correct order of ecological hierarchy?
(1) Biome → Populations → Community → Organism
(2) Organism → Biome → Population → Community
(3) Population → Community → Biome → Organism
(4) Organism → Population → Community → Biome

BOP236

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36. A population has certain attributes that an individual organism does not, like :-
(a) birth (b) birth rate
(c) death rate (d) sex ratio
(1) a and d (2) a and b
(3) b, c and d (4) c and d

BOP237

37. Under most favourable environment conditions the maximum reproductive capacity of an species is known as :-
(1) Biotic potential (2) Birth rate
(3) Death rate (4) Natality

BOP238

38. How many statements are incorrect ?
(a) Predators also help in maintaining species diversity in a community.
(b) Parasite provides benefits to host.
(c) *Balanus* & *Chathamalus* are motile crustaceans.
(d) *Pisaster* maintains species diversity of it's prey species.
(1) Three (2) One
(3) Two (4) None

BOP239

39. **Assertion :** Predators never overexploit their prey.
Reason : Predators are prudent in nature.
(1) Both assertion and reason are true and the reason is a correct explanation of Assertion.
(2) Both assertion and reason are true but the reason is not the correct explanation of Assertion.
(3) Assertion is true but reason is false.
(4) Both assertion and reason are false.

BOP240

40. **Assertion :-** *Cuscuta* is a parasite commonly found on hedge plants.
Reason :- *Cuscuta* has lost its chlorophyll and leaves in the course of evolution.
(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

BOP241

41. **Statement-I :** The size of a population for any species is a static parameter.
Statement-II : Natality and emigration contribute to increase in population density.
(1) Statement-I & Statement-II both are correct.
(2) Statement-I is correct & Statement-II are incorrect.
(3) Statement-I is incorrect & Statement-II are correct.
(4) Both Statement-I & Statement-II are incorrect.

BOP242

42. **Statement-I :** A population at any given time is composed of individuals of same species.
Statement-II : The shape of age pyramid reflects the growth status of the population.
(1) Statement-I & Statement-II both are correct.
(2) Statement-I is correct & Statement-II is incorrect.
(3) Statement-I is incorrect & Statement-II is correct.
(4) Both Statement-I & Statement-II is incorrect.

BOP243

43. Identify the organism on the basis of following characters :-
(a) have simple digestive system
(b) have lost its unnecessary sense organ during evolution
(c) have high reproduction potential
(d) presence of suckers
(1) Prey (2) Predator
(3) Parasite (4) Host

BOP244

44. Choose the correct match :-
(a) Co-evolution – parasite and host
(b) Commensalism – relation of Egret birds and cattle
(c) Sexual deciet – shown by *opuntia* plant
(d) Connell's elegant field experiment – competition
(1) a, b (2) b, c
(3) c, d (4) a, b, d

BOP245

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45. Which of the following statements are true for predator ?
(a) Conduit of energy
(b) Over exploits prey
(c) Helps in maintaining species diversity
(d) Show (+/–) interaction with it's prey.
(1) a, b, c, d (2) a, b, c
(3) a, c, d (4) a, b, d

BOP246

46. Given below are two statements :-
Statement I : An organism is the smallest unit of ecological hierarchy and basic unit of ecological study.
Statement II : Ecological niche is the functional role of any species in an ecosystem or community.
In the light of above statements, choose the correct answer from the options given below
(1) Statement I is true but statement II is false
(2) Statement I is false but statement II is true
(3) Both statement I and statement II is true
(4) Both statement I and statement II is false

BOP247

47. **Assertion :** The species which have great influence on the community characteristics relative to their low abundance or biomass are called as key stone species.
Reason : The activities of key stone species determine the structure of the community.
(1) Both **A** and **R** are true, and **R** is correct explanation of the **A**.
(2) Both **A** and **R** are true, but **R** is not a correct explanation of **A**.
(3) **A** is true but **R** is false.
(4) **A** is false but **R** is true.

BOP248

48. Sum total of biotic and abiotic factors that do not allow a population to achieve its maximum biotic potential is :-
(1) Represented by $\frac{dN}{dt} = rN$
(2) Called carrying capacity
(3) Known as environmental resistance
(4) Represented as $\frac{N-K}{N}$

BOP249

49. Match the columns I and II, and choose the correct combination from the options given :-

Column-I 'r' Values		**Column-II Organism**	
a.	0.12	1.	Human population in India in 1981
b.	0.015	2.	Norway rat
c.	0.0205	3.	Flour beetle

	a	b	c
(1)	1	2	3
(2)	3	1	2
(3)	2	3	1
(4)	3	2	1

BOP250

50. In equation $\frac{dN}{dt} = rN\,\frac{N-K}{N}$; here K is defined by :-
(1) Maximum population density achievable under available resources.
(2) Carrying capacity of habitat
(3) Both (1) & (2)
(4) Change in population density

BOP251

51. Which of the following factors can cause change in population size?
(1) Food
(2) Predation pressure
(3) Adverse weather
(4) All of these

BOP252

52. ___ is the number of individuals of population who left the habitat during the time period.
(1) Mortality (2) Emigration
(3) Natality (4) Immigration

BOP253

53. **Assertion :** In an area, if there are 200 *Parthenium* plants and one huge banyan tree with large canopy, biomass is more meaningful measurement of Population Density.
Reason : Pug marks and fecal pellets are used to estimate tiger census in our national parks and tiger reserves.

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(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

BOP254

54. In the following expression -

$N_{t+1} = N_t + [(B + I) - (D + E)]$

where I, B and D stand for:
(1) Emigration, natality, mortality
(2) Mortality, natality, immigration
(3) Mortality, immigration, natality
(4) Immigration, natality, mortality

BOP255

55. When environmental conditions are favourable then population growth curve will be :-
(1) Logistic (2) J-Shaped
(3) 'S' shaped (4) Both 1 and 2

BOP256

56. A biologist studied a population of rats in barn. He found that average natality was 250, average mortality 240, immigration 20 and emigration 30. The net increase in population is :
(1) 10 (2) 15 (3) 5 (4) Zero

BOP257

57. When does the growth rate of a population following the logistic model equal zero ? The logistic model is given as $\frac{dN}{dt} = rN\left(1 - \frac{N}{K}\right)$:-
(1) when N/K is exactly one.
(2) when N nears the carrying capacity of the habitat.
(3) when N/K equals zero.
(4) when death rate is greater than birth rate.

BOP258

58. Which of the following is a feature of population NOT of organism?
(1) Birth (2) Age
(3) Growth rate (4) Death

BOP259

59. If N = 1000, b = 0.7, d = 0.2, find dN/dt?
(1) 700 (2) 500 (3) 200 (4) 100

BOP260

60. Match the columns–I and II, and choose the correct combination from the options given.

Column-I		Column-II	
a.	Breed only once in life time	1.	Bird and mammals
b.	Breed many times in their lifetime	2.	Oysters and pelagic fishes
c.	Produce large number of small sized offspring	3.	Bamboo
d.	Produce small number of large-sized offspring	4.	Mammals

(1) a – 2, b – 1, c – 3, d – 4
(2) a – 4, b – 1, c – 2, d – 3
(3) a – 3, b – 2, c – 4, d – 1
(4) a – 3, b – 4, c – 2, d – 1

BOP261

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	4	2	3	1	3	2	3	4	3	3	3	4	3	3
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	4	2	4	2	4	3	2	2	2	2	-	3	3	2	4
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	3	1	4	3	4	3	1	3	1	1	4	3	3	4	3
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	3	1	3	4	3	4	2	2	4	2	4	1	3	2	4

Ecosystem

1. Which of the following ecosystem has the highest gross primary productivity :-
(1) mangrove (2) rain forest
(3) grassland (4) coral reef
BES190

2. Mass of living matter at a trophic level in an area at anytime is called :-
(1) Standing state (2) Standing crop
(3) Detritus (4) Humus
BES191

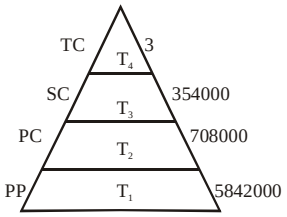
3. Which of the following group is essential component for survival of an ecosystem :-
(1) Producers
(2) Producers and herbivores
(3) Producers and decomposers
(4) Detritivores
BES192

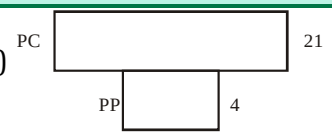
4. In an ecosystem, which one shows one-way passage :-
(1) energy (2) carbon
(3) nitrogen (4) potassium
BES193

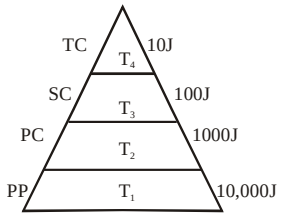
5. Concept of ecological pyramids was proposed by:-
(1) Odum (2) Tansley
(3) Reiter (4) Charles Elton
BES194

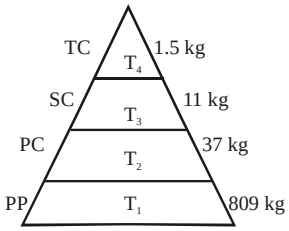
6. Guild is -
(1) Organism which have similar trophic level
(2) Organism which have different trophic level
(3) Producer's and consumer's of ecosystem
(4) Consumer's of ecosystem
BES195

7. The four pyramids (A, B, C and D) given below represent four different type of ecosystem ecological pyramids, which one of these is correctly identified in the options given, along with its correct ecosystem and type of pyramid:-

(A)


(B)


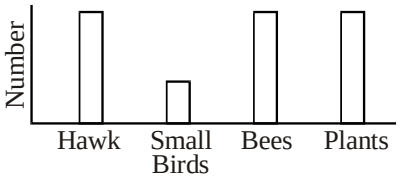
(C)


(D)


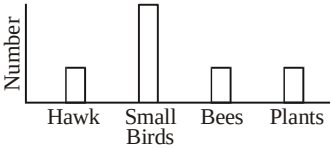
		Type of pyramid	Ecosystem
(1)	A	Pyramid of number	Tree ecosystem
(2)	B	Pyramid of biomass	Grassland
(3)	C	Pyramid of Energy	Grassland Ecosystem
(4)	D	Pyramid of number	Tree Ecosystem

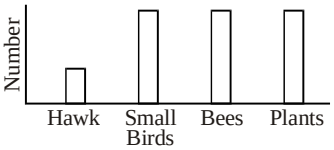
BES196

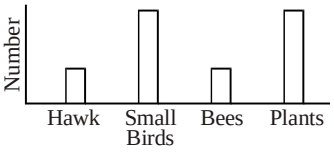
8. In an ecosystem, four major interdependent components were in the given proportions :



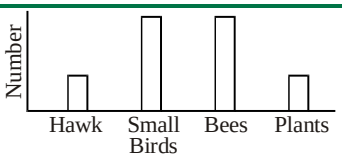
What would be the new composition of the same ecosystem when majority of the Hawk were killed -

(1)


(2)


(3)


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(4) 

BES197

9. Plant → Insect → P → Snake → Hawk
In above given food chain identify P:-
(1) Frog (2) Cobra
(3) Fox (4) Lion

BES198

10. Plant → Rabbit → Wolf → Lion
In above food chain if plant produce 1000 KCal energy during photosynthesis, then How much amount of energy available for wolf in given food chain ?
(1) 100 KCal (2) 1 KCal
(3) 10 KCal (4) 0.1 KCal

BES199

11. Which of the following is an example of artifical ecosystem ?
(1) Crop land ecosystem
(2) Tree ecosystem
(3) Grass land ecosystem
(4) desert ecosystem

BES200

12. Which one is **incorrect** about ecological pyramids ?

		Number	Biomass	Energy
(1)	Lake			
(2)	Grassland			
(3)	Ocean			
(4)	Tree			

BES201

13. Choose the **incorrect** one regarding ecological pyramids :-
(1) No place for decomposers
(2) Represent unidirectional flow of energy
(3) Can represent food web
(4) Is erect for number of organisms in grassland ecosystem.

BES202

14. Which statement is true concerning this food chain :–
Grass → Rabbits → Snakes → Hawk :-
(1) Each population is omnivores
(2) Each predator population has a greater biomass than its prey population
(3) Both (1) and (2)
(4) Each population has a greater biomass than its predator population

BES203

15. (I) Plant → Insect → Frog → Eagle
(II) Plant → Rat → Snake → Peacock
When food chain I will connected with food chain-II and snake eats frog than snake is :-
(1) Primary consumer
(2) Secondary consumer
(3) Tertiary consumer
(4) Top consumer

BES204

16. Which of the following is not an abiotic component of an ecosystem :-
(1) water (2) minerals
(3) litter (4) micro organisms

BES205

17. How much amount of incident solar radiation (ISR) is used in the photosynthesis process by plants :-
(1) 1-5% (2) 5-10%
(3) 20-40% (4) 50-70%

BES206

18. Which of the following is most ecologically relevant environmental abiotic factor :-
(1) light (2) temperature
(3) soil (4) water

BES207

19. Which of the following is not a physiological adaptation :-
(1) Blubber in whale and seal in Antarctica region
(2) Antifreezing protein
(3) Increase number of RBC in blood at high altitude
(4) In Cactus leaves modified into spines

BES208

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20. Under certain conditions soil nutrients get tied up with the biomass of microbes and becomes temporarily unavailable to other organism such incorporation of nutrients is called –

(1) Nutrient immobilisation

(2) Acclimatisation

(3) Mineralisation

(4) Leaching

BES209

21. Which one of the following processes during decomposition is **correctly** described ?

(1) Leaching – Water soluble inorganic nutrients rise to the top layers of soil

(2) Fragmentation – Carried out by organisms such as earthworm

(3) Humification – Leads to the accumulation of a dark coloured substance humus which undergoes microbial action at a very fast rate

(4) Catabolism – Last step in the decomposition under fully anaerobic condition

BES210

22. Identify the correct match from the following column I, II and III :-

	Column-I		Column-II		Column-III
(1)	Eury-thermal	(a)	Able to tolerate narrow range of temperature	(i)	99% Animals
(2)	Steno-thermal	(b)	A stage of suspended development	(ii)	Thermo-regulation
(3)	Conformers	(c)	Body temp. changes with ambient temp.	(iii)	Zooplanktons
(4)	Diapause	(d)	Able to tolerate wide range of temp.	(iv)	Poikilothermal

(1) 1–b–ii, 2–c–iii, 3–a–i, 4–d–iv

(2) 1–d–ii, 2–a–i, 3–b–iv, 4–c–iii

(3) 1–d–ii, 2–a–iv, 3–c–i, 4–b–iii

(4) 1–a–iii, 2–b–ii, 3–d–iv, 4–c–i

BES211

23. How many of the following are **correct** ?

(A) On planet earth life exists in just few favourable habitats not in extreme and harsh habitats

(B) Our intestine is a unique habitat for hundreds of species of microbes

(C) Physio-chemical components along do not characterise the habitat of an organism completely

(D) Habitate includes biotic components also with which organism interact constantly

(E) Over a period of time, the organism had through natural selection evolved adaptations to optimise it's survival and reproduction in it's habitat

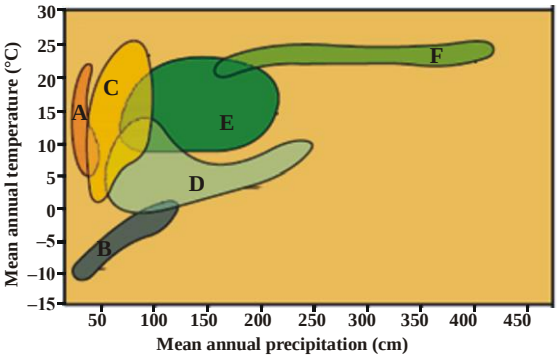
(F) Mango tree do not and can not grow in temperate countries, Tuna fish are rarely caught beyond tropical latitudes

(1) 6 (2) 5

(3) 4 (4) 3

BES212

24. Which of the following statement is correct with respect to different biomes represented in the given figure :-



(1) Biome 'A' and 'C' show high annual temp and they are highly rich in vegetation

(2) Biome 'C' and 'D' have same temperature as well as rainfall and their vegetation type is also similar

(3) Biome 'A' and 'B' differ in their mean annual temperature but both have very low vegetation

(4) Biome 'A' and 'C' have low precipitation and thus vegetation type is similar

BES213

25. Which of the following pairs is a sedimentary type of biogeochemical cycle :-

(1) phosphorus and nitrogen

(2) phosphorus and sulphur

(3) oxygen and nitrogen

(4) phosphorus and carbon dioxide

BES214

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26. About 70% of total global carbon is found in:
(1) forest (2) grassland
(3) agroecosystem (4) oceans

BES215

27. The correct ascending order of temperature is:
(1) Polar areas → Temperate areas → sub tropical areas → Tropical areas
(2) Equator → Timber line → Tropical deserts
(3) Deep sea hydrothermal vents → Tropical deserts → mountain top
(4) Polar areas → Tropical deserts → Mountain top → Timber line

BES216

28. Which of the following characters are changed with altitude and latitude ?
(a) Temperature (b) Rain fall
(c) Vegetation (d) Species diversity
Options :
(1) only b (2) only a and b
(3) only a, b and c (4) a, b, c & d

BES217

29. Alpine vegetation is found in :
(1) Tundra region
(2) Arctic desert
(3) Above 60°N latitude
(4) All of above

BES218

30. Which of the following place have sub zero temperature :
(a) High altitude place
(b) Hydrothermal vent
(c) Polar area
(d) Tropical desert
(1) a & c (2) a, b & c
(3) only c (4) a, b, c & d

BES219

31. Which one of the following biome experiences minimum average annual rain fall and minimum average annual temperature :-
(1) Desert
(2) Arctic and alpine tundra
(3) Coniferous forest
(4) Temperate forest

BES220

32. Earth is an :
(1) Open system for energy and open system for minerals
(2) Closed system for minerals and open system for energy
(3) Closed system for energy and open system for minerals
(4) Closed system for energy and closed system for minerals

BES221

33. Species diversity increase :-
(1) Low altitude to high altitude and from low latitude to high latitude
(2) High altitude to low altitude, and from high latitude to low latitude
(3) High altitude to low altitude, and from low latitude to high latitude
(4) Low altitude to high altitude, and from high latitude to low latitude

BES222

34. Holozoic mode of nutrition is shown by :
(1) Producers
(2) Macroconsumers
(3) Micro-consumers
(4) BGA

BES223

35. Peacock eats snake and snake eats insect while insect eats green plant. Then the position of peacock is :-
(1) Primary producer
(2) Secondary producer
(3) Decomposer
(4) Top level consumer at the apex of ecological pyramid

BES224

36. Read the following statements :-
(i) .....A.... is defined as the amount of biomass or organic matter produced per unit area over a time period by total plants during photosynthesis
(ii) Gross primary productivity minus respiration losses (R) is the .....B.....
(iii) .....C..... is defined as the rate of formation of new organic matter by consumers.

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Identify the A, B & C :-

	A	B	C
(1)	Secondary productivity	Net primary productivity	Primary productivity
(2)	Gross primary productivity	Net primary productivity	Secondary productivity
(3)	Net primary productivity	Secondary productivity	Primary productivity
(4)	Primary productivity	Net secondary productivity	Secondary productivity

BES225

37. The annual net primary productivity of whole hydrosphere is :
(1) 170 million tons (2) 55 billion tons
(3) 115 billion tons (4) 170 billion tons

BES226

38. Most resistant and resilient biome is -
(1) Desert (2) Grassland
(3) Tropical rainforest (4) Tundra

BES227

39. The reason(s) for low productivity of oceans is that:
(1) Light decreases with depth
(2) Nitrogen becomes limiting
(3) Both (1) and (2)
(4) Nutrients become unlimiting

BES228

40. **Statement-I :** Temperature and soil moisture are the important factors that regulate decomposition.
Statement-II : Decomposition rate is quicker if detritus is rich in lignin and chitin.
(1) Statement-I & Statement-II both are correct.
(2) Statement-I is correct & Statement-II is incorrect.
(3) Statement-I is incorrect & Statement-II is correct.
(4) Both Statement-I & Statement-II is incorrect.

BES229

41. NPP means
(1) Available biomass for the consumption of heterotrophs
(2) Biomass produced by producers
(3) Gross primary product of respiratory losses
(4) All of the above

BES230

42. Which one of the following is odd:
(1) Fragmentation (2) Humification
(3) Stratification (4) Mineralisation

BES231

43. Net community productivity (NCP) is given by the formulae :
(1) NCP = GPP + NPP
(2) NCP = $\frac{GPP}{NPP} \times 100$
(3) NCP = GPP – R
(4) NCP = NPP – Heterotrophic Consumption

BES232

44. In deep sea hydrothermal ecosystems, producers are:-
(1) Chemoautotrophs
(2) Chemoheterotrophs
(3) Photoautotrophs
(4) Photoheterotrophs

BES233

45. In a food chain, level of energy across successive trophic levels :
(1) Increase up to secondary consumers, then decreases
(2) Decreases up to secondary consumers, then increases
(3) Increases continuously
(4) Decreases continuously

BES234

46. In arrangement of trophic level system a human baby that feeds on mother's milk is considered as which consumer :
(1) Primary (2) Secondary
(3) Tertiary (4) Top

BES235

47. Match list-I with list-II and select correct answer:

	List I		List II
(A)	Primary consumer	(i)	Phytoplankton
(B)	Primary Producer	(ii)	Small fish
(C)	Secondary consumer	(iii)	Zooplankton
(D)	Tertitary consumer	(iv)	Fish eating bird

(1) **A**-ii **B**-iii **C**-i **D**-iv
(2) **A**-iii **B**-i **C**-ii **D**-iv
(3) **A**-i **B**-ii **C**-iii **D**-v
(4) **A**-iii **B**-ii **C**-i **D**-iv

BES236

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48. During decomposition humification leads to accumulation of a dark coloured amorphous substance called humus. About humus which of the following is correct?
(1) Susceptible to microbial action
(2) Undergoes decomposition at an extremely high rate
(3) Colloidal in nature
(4) It promotes compaction of soil

BES237

49. How many statements are correct ?
(A) Predatory food chain begins with producers and in successive order it goes from smaller organism to bigger organism
(B) Parasitic food chain also starts from producers but in successive order it goes from bigger organism to the smaller organism.
(C) Detritus food chain begins with decomposition of dead organic matter by decomposers.
(D) In aquatic ecosystem, DFC is the major conduit for energy flow.
(1) One (2) Two
(3) Three (4) Four

BES238

50. Leaching is :-
(1) Break down of detritus into smaller particles
(2) Downward movement of water-soluble inorganic nutrients
(3) Degradation of detritus into simpler inorganic nutrients
(4) Releasing of inorganic nutrients from humus

BES239

51. **Assertion :-** Detritus food chain may be connected with grazing food chain at some levels.
Reason :- Some of the organisms of DFC are prey to the GFC animals and some of the organisms are of omnivorous nature.
(1) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.
(2) Assertion is true but Reason is false.
(3) Assertion is false but Reason is true.
(4) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

BES240

52. **Assertion:** Complexity of food web is dependent over length of food chains
Reason: Complexity of food webs determine the stability of an ecosystem
(1) Both Assertion & Reason are true, and the reason is the correct explanation of the Assertion.
(2) Both Assertion & Reason are true, but the Reason is not the correct explanation of the Assertion.
(3) Assertion is true statement, but Reason is false.
(4) Both Assertion and Reason are false statements.

BES241

53. **Assertion :** Saprotrophs are not given any place in ecological pyramids.
Reason : Saprotrophs play a vital role in the ecosystem.
(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

BES242

54. Choose the incorrect match :-

Column-I	**Column-II**
(i) Grass	Primary producer
(ii) Deer	Primary consumer
(iii) Lion	Top consumer
(iv) Fox	Secondary consumer
(v) Insectivorous plants	Primary consumer

(1) (iii) and (iv) (2) (iii) and (v)
(3) (iv) and (v) (4) (v) only

BES243

55. Which of the following option correctly represent functions of an ecosystem ?
(A) Energy flow
(B) Decomposition
(C) Productivity
(D) Nutrient cycling
(1) A and D (2) A, C and D
(3) B, C and D (4) A, B, C and D

BES244

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56. Match the column-I with column-II ?

	List I		List II
(A)	Pyramid of number represents	(i)	Standing crop
(B)	Pyramid of biomass represents	(ii)	Productivity
(C)	Pyramid of energy represents	(iii)	Biotic potentia

(1) (A) - (ii) ; (B) - (iii) ; (C) - (i)
(2) (A) - (iii) ; (B) - (i) ; (C) - (ii)
(3) (A) - (iii) ; (B) - (ii) ; (C) - (i)
(4) (A) - (ii) ; (B) - (i) ; (C) - (iii)

BES245

57. Primary productivity is affected by
(i) Temperature
(ii) Sunlight
(iii) Moisture
(iv) Availability of nutrients
(1) (i), (ii), (iii) and (iv)
(2) (i) and (ii)
(3) (i), (ii) and (iii)
(4) (ii), (iii) and (iv)

BES246

58. **Statement-I :** Of the incident solar radiation more than 50% of it is photosynthetically active radiation (P.A.R.)
Statement-II : Plants capture only 15-20% of the P.A.R.
(1) Both Statement-I & II are correct.
(2) Statement-I is correct & Statement-II is incorrect.
(3) Statement-I is incorrect & Statement-II is correct.
(4) Both Statement-I & II are incorrect.

BES247

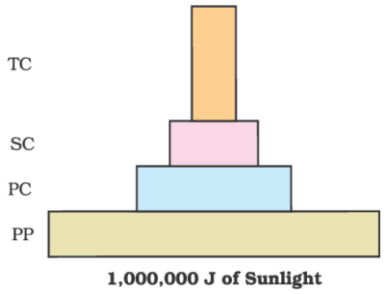
59. **Statement-I :** Vertical distribution of different species occupying different levels is called stratification.
Statement-II : Secondary productivity is defined as the amount of biomass produced per unit area over a time period by plants.
(1) Statement-I & Statement-II both are correct.
(2) Statement-I is correct & Statement-II is incorrect.
(3) Statement-I is incorrect & Statement-II is correct.
(4) Both Statement-I & Statement-II are incorrect.

BES248

60. **Statement-I :** Detritivores break down detritus into smaller particles this process is called catabolism.
Statement-II : Bacterial and fungal enzymes degrade detritus into simpler inorganic substances, this process is called fragmentation.
(1) Statement-I & Statement-II both are correct.
(2) Statement-I & Statement-II both are incorrect.
(3) Only Statement-I is correct.
(4) Only Statement-II is correct.

BES249

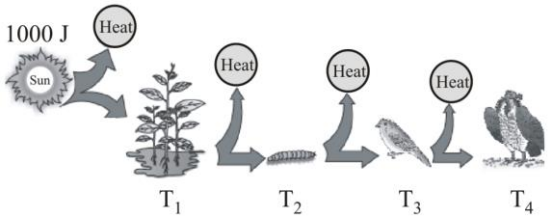
61. In the given pyramid calculate the amount of energy available to secondary consumer.



(1) 100 (2) 1000
(3) 5000 (4) 10

BES250

62. What is the magnitude of energy at top carnivores (T) level in given figure ?



(1) 0.1 J (2) 0.01 J
(3) 1 J (4) 10 J

BES251

63. **Assertion :** A sparrow is a primary consumer when it eats seed and a secondary consumer when it eats insects.

Reason : A given species may occupy more than one trophic level in the same ecosystem at the same time.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

BES252

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	2	3	1	4	1	3	1	1	1	1	3	3	4	3
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	4	1	2	4	1	2	3	2	3	2	4	1	4	4	1
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	2	2	2	2	4	2	2	3	3	2	1	3	4	1	4
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	2	2	3	3	2	4	2	2	4	4	2	1	4	2	2
Question	61	62	63												
Answer	2	2	1												

Biodiversity and Conservation

1. Which of the following character is not related to biodiversity hot spot?
(1) High degree of endemism
(2) Habitat loss
(3) Lesser interspecific interaction
(4) High species richness

BAC166

2. In "rivet popper hypothesis" rivet indicates?
(1) Community (2) Biosphere
(3) Ecosystem (4) Species

BAC167

3. Western Ghat's of India is hot spot :
(1) It is a ecotone in two community
(2) It is a mega diversity zone
(3) It is a warm places
(4) It is a Hydrothermal vents

BAC168

4. Social forestry is :
(1) Planting tree on public places and common land's
(2) Wood species are grown in combination with herbaceous crop
(3) Growing ornamental and fruit tree in urban areas
(4) Growing tree's needed for industries in specific area

BAC169

5. Which of the following is **correctly** matched
(a) Carrot grass → *Lantana*
(b) Farmers friend → Fungi
(c) Amazon rain forest → Lungs of planet
(d) World summit → Johannesburg
(1) only c, d (2) b, c, d
(3) only d (4) a, b, d

BAC170

6. How many example given below are the example of *in-situ* conservation? Tissue culture, pollen bank, national park, Biosphere reserve, Botanical garden
(1) Three (2) Two
(3) One (4) Five

BAC171

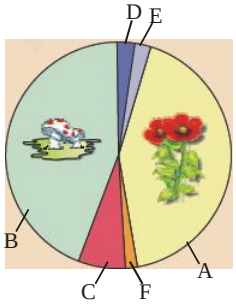
7. Extinction of stellar's sea cow is due to which of the following reasons :-
(1) Habitat loss
(2) Over exploitation
(3) Alien species invasion
(4) Co-extinction

BAC172

8. Kajiranga National Park is famous for :-
(1) One horned rhino (2) Hangul
(3) Wild Indian ass (4) Snow Leopard

BAC173

9. Identify the correct match regarding plant diversity ?



	A	B	C	D	E	F
(1)	Angio-sperms	Fungi	Mosses	Algae	Ferns	Lichens
(2)	Fungi	Angio-sperms	Mosses	Algae	Lichens	Ferns
(3)	Angio-sperms	Algae	Fungi	Mosses	Lichens	Ferns
(4)	Angio-sperms	Fungi	Algae	Mosses	Ferns	Lichens

BAC174

10. India has more than 50,000 strains of rice and 1000 varieties of mango. It is related to :-
(1) Species diversity
(2) Genetic diversity
(3) Ecological diversity
(4) Temperate climate in India

BAC175

11. India's share in the global species diversity is about:-
(1) 2% (2) 4% (3) 6% (4) 8.1%

BAC176

12. Choose the correct option :
(i) Sacred groves
(ii) Cryopreservation
(iii) Germplasm bank
(iv) Wildlife safari park
(v) Home gardens
(1) All are example of in-situ conservation
(2) All are example of ex-situ conservation
(3) Only (i), (iv), (v) are example of in-situ conservation
(4) All are examples of ex-situ conservation except one

BAC177

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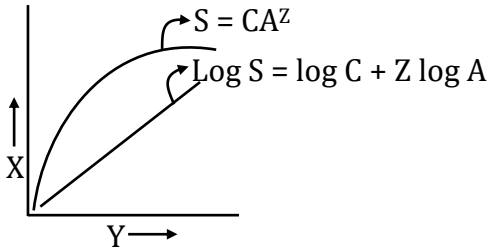
13. How many of the following are considered as threat to ecological security?
Habitat loss, Afforestation, Alien species invasion, Over-exploitation, Co-existance, Sacred grove
(1) Four (2) Two
(3) Five (4) Three

BAC178

14. Which of the following characters is not related to biodiversity hot spot?
(1) High species richness
(2) High endemism
(3) Lesser interspecific interaction
(4) Accelerated habitat loss

BAC179

15. For the given graph identify X, Y and Z :



(1) X = Area, Y = Slope of curve,
Z = Species richness
(2) X = Area, Y = Species richness,
Z = Intercept on y axis
(3) X = Species richness,
Y= Slope of curve,
Z = Area
(4) X = Species richness, Y = Area,
Z = Slope of line

BAC180

16. Which of the following uses of biodiversity are related to the narrow utilitarian arguments for conserving biodiversity?
(1) It is source of firewood.
(2) It is source of industrial products.
(3) It is source of medicines.
(4) All of the above

BAC181

17. Species facing high risk of extinction in the wild in the near future are called as :-
(1) Critically endangered
(2) Endangered
(3) Vulnerable
(4) Rare species

BAC182

18. Main reason of disturbance of biological diversity :-
(1) Green House effect
(2) Hunting
(3) Soil erosion
(4) Destruction of Natural habitats

BAC183

19. According to tilman how species rich biological community show more stability:-
(A) Plots with more species show less year to year variation in total biomass
(B) Increased diversity contributed to higher productivity
(C) Provide resistance to invasions by alien species
(1) A only (2) A, B only
(3) A, B, C (4) C only

BAC184

20. In any island ecosystem, introduction of exotic or alien species may cause :-
(1) Disappearance of native species
(2) Ecological imbalance
(3) Increase biodiversity
(4) Both (1) & (2)

BAC185

21. Which of the following statements are correct?
(a) Passenger pigeon extinct due to over exploitation
(b) Earth summit was occurred in 1992 at Rio-de-Janeiro.
(c) World summit on sustainable development was held at Johannesburg in 2002.
(d) Amazonian rain forest in south America has the greatest biodiversity on earth.
(1) Only a, b, c (2) Only b, c
(3) Only b (4) a, b, c, d

BAC186

22. Which one of the following cause of biodiversity loss, is because of a positive ecological interaction:-
(1) Over-exploitation
(2) Habitat fragmentation
(3) Co-extinction
(4) Alien species invasion

BAC187

23. Which one is not an example of Alien species?
(1) *Parthenium*
(2) *Eichhornia*
(3) *Clarias gariepinus*
(4) *Rauwolfia vomitoria*

BAC188

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24. The species - area relationship studied by Alexander von Humboldt is a straight line on a logarithmic scale and is represented by equation Log S = Log C + Z Log A in this equation 'S' stands for :-
(1) Species richness
(2) Area
(3) Slope of the line
(4) Regression coefficient

BAC189

25. Which of the following is not under broadly utilitarian reason for conserving biodiversity:
(1) Pollination
(2) Asthetic pleasures
(3) Controlling soil erosion
(4) 25% drugs sold in market are plant based

BAC190

26. The Amazon rain forest 'lungs of Planet" harbouring probably millions of species is being cut & cleared for which purpose–
(1) For cultivation of soyabeans
(2) For conservation of grasslands for rawsing beef cattles
(3) For cultivation of Medicinal plants
(4) Both (1) and (2)

BAC191

27. In India how many genetically different strains of rice and mango varieties are present :
(1) 50,000 and 1,0000 respectively
(2) 1000 and 50000 respectively
(3) 50,000 and 1,000 respectively
(4) 50,000 and 5,000 respectively

BAC192

28. In species area relationship, on a logarithmic scale. The relationship is–
(1) Rectangular hyperbola
(2) Rectangular parabola
(3) Straight line
(4) Sigmoid

BAC193

29. Match the animals given in column A with their extinct place in column B :-

Column-A		Column-B	
(i)	Dodo	(a)	Russia
(ii)	Thylacine	(b)	Mauritius
(iii)	Quagga	(c)	Australia
(iv)	Steller's sea cow	(d)	Africa

Choose the correct match from the following:
(1) i-a, ii-c, iii-d, iv-b (2) i-b, ii-d, iii-c, iv-a
(3) i-d, ii-c, iii-b, iv-a (4) i-b, ii-c, iii-d, iv-a

BAC194

30. Which of the following belong to critically endangered animals ?
(1) Red Panda
(2) Asiatic Wild Ass
(3) Hangul Kashmir Stag
(4) Siberian crane

BAC195

31. Loss of Biodiversity in a region may not lead to :-
(1) Decline in plant production
(2) Increase in Resistance to flood condition
(3) More year to year variation in productivity
(4) Variable pest and disease cycle

BAC196

32. (A) More than 70 percent of all the species recorded are animals.
(B) Out of every 10 animals on this planet, 7 are insects.
(C) The number of fungi species in the world is more than the combined total of the species of fishes, amphibians, reptiles and mammals.
(D) Number of fishes is very less than that of mammals.
(1) All correct
(2) All incorrect
(3) A, B and C are correct
(4) Only D is is correct

BAC197

33. All given are included among 34 hot spots of world, except :-
(1) Indo-Burma
(2) Western Ghats and Sri Lanka
(3) Himalaya
(4) Sarguja, Chanda and Bastar

BAC198

34. Species diversity increases as we go :-
(1) from pole to equator
(2) from equator to pole
(3) from mountain to mountain
(4) from sea level to mountain

BAC199

35. Deserts, rain forests, mangroves, coral reefs, wet lands estuaries & Alpine meadows can be employed to describe the :-
(1) Genetic diversity
(2) Species diversity
(3) Ecological diversity
(4) All of the above

BAC200

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36. When we conserve and protect the whole ecosystem, it's biodiversity at all levels is conserved called :-
(1) In-situ conservation
(2) Ex-situ conservation
(3) On-site conservation
(4) Both (1) and (3)

BAC201

37. Genetic Diversity is shown by :-
(1) Rice
(2) Mango
(3) *Rauwolfia vomitoria*
(4) All of the above

BAC202

38. In 2012 Rio-20 summit was held at Rio de Janeiro it is meant for :-
(1) Biodiversity conservation
(2) Kyoto protocol
(3) Climate change
(4) All of the above

BAC203

39. **Assertion :** A stable community should not show too much variation in productivity from year to year.
Reason : Increased diversity in community contribute to higher productivity.
(1) Both Assertion and Reason are true and the Reason is a correct explanation of Assertion.
(2) Both Assertion and Reason are true but the Reason is not the correct explanation of Assertion.
(3) Assertion is true but Reason is false.
(4) Both Assertion and Reason are false.

BAC204

40. **Assertion:** Over exploitation is one of the important cause driving organisms to extinct.
Reason: Steller's sea cow and passenger pigeon were extinct due to fragmentation.
(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

BAC205

41. **Assertion :** Tropics harbour more species than temperate and polar areas.
Reason : Tropical environments are less seasonal and more constant which promote niche-specialization and greater species diversity.
(1) Both Assertion & Reason are true & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are true but Reason is not a correct explanation of the Assertion.
(3) Assertion is true but the Reason is false.
(4) Both Assertion & Reason are false.

BAC206

42. **Assertion:** Many species extinct in the last 500 years (steller's sea cow, passenger pigeon)
Reason: Habitat loss and fragmentation is the most important cause driving animals and plants to extinction.
(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

BAC207

43. **Assertion :** Biodiversity hotspots are regions with very high level of species richess and high degree of endemism.
Reason : These hotspots are also regions of accelerated habital loss.
(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

BAC208

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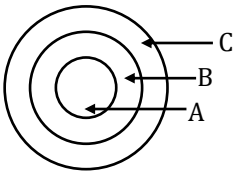
44. According to 'rivet popper hypothesis' rivet on wings is

(1) An ecosystem (2) Key stone species

(3) The whole biome (4) A community

BAC209

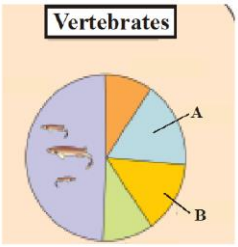
45. Biosphere Reserve is a specified protected area in which multiple use of the land is permitted. Three zones of biosphere reserve are found identify the correct zone (A, B, C) in given biosphere reserve zones.



	A	B	C
(1)	Transition zone	Buffer zone	Core zone
(2)	Buffer zone	Transition zone	Core zone
(3)	Core zone	Buffer zone	Transition zone
(4)	Core zone	Transition zone	Buffer zone

BAC210

46. Identify A & B in given pie chart, (representing global biodiversity of vertebrates).



(1) A = Amphibian, B = Birds

(2) A = Reptiles, B = Mammals

(3) A = Reptiles, B = Birds

(4) A = Birds, B = Reptiles

BAC211

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	4	2	1	1	2	2	1	4	2	4	4	4	3	4
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	4	2	4	3	4	4	3	4	1	4	4	3	3	4	4
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	2	3	4	1	3	4	4	1	2	3	1	2	2	2	3
Question	46														
Answer	4														

Important Notes

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